

Weak and difference ascent sequences (OEIS A336070)

Factorial growth, the logarithmic-square correction, weighted levels,
sharp remainders and bounded-error inverses

Three research reports of one external research session
(bundle Reports 105, 107 and 108), merged into one ProveIt report

Parts I–III: 2 October 2026 merged: 5 October 2026

Abstract

A weak ascent sequence (Bényi–Claesson–Dukes) starts at 0 and never exceeds one more than the number of its weak ascents $x_i \leq x_{i+1}$; its count a_n is OEIS A336070. More generally, a difference ascent sequence with parameter d (Dukes–Sagan) counts the indices with $x_{i+1} > x_i - d$; $d = 0$ gives the Fishburn numbers and $d = 1$ gives A336070. Put $\mu = 6/\pi^2$. This report prints three manuscripts as Parts I–III. Part I proves, for every fixed $d \geq 0$, that $(a_n^{(d)}/n!)^{1/n} \rightarrow \mu$ and $a_{n+1}^{(d)}/((n+1)a_n^{(d)}) \rightarrow \mu$, with exponential concentration of the ascent count, and for $d = 1$ that $\log(a_n/(n!\mu^n)) = \frac{1}{2}(\log n)^2 + O((\log n)^2/\log \log n)$, with a signed Lambert- W threshold inverse. Part II weights each equal adjacent pair (level) by $w > 0$, proves $\log H_n(w) = \log n! + n \log \mu + \frac{w}{2}(\log n)^2 + O((\log n)^2/\log \log n)$ uniformly on compact weight intervals, and derives a large-deviation principle at speed $(\log n)^2$ with rate $x \log(2x/w) - x + w/2$ for the number of levels, the exact zero-level rate and a threshold inverse. Part III reduces both remainders to $O(\log n)$, proves the correction $\frac{d}{2}(\log n)^2$ with remainder $O_d(\log n)$ for every fixed integer $d \geq 1$, and deduces threshold inverses with bounded additive error and sharper moment bounds for the level count. A relative-error equivalent, an amplitude, the coefficient of a possible $\log n$ term, a central limit theorem and uniformity in a growing d remain open. The report is unrefereed, and nothing in it is formalized.

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Guide to this report

The three Parts are three manuscripts of the session bundle of Reports 1–243 (arrival commit 60f54ea06), placed in this directory by commit 6ab1f1979 (batch 102). They form one chain, printed in the order in which they were written; each later manuscript is self-contained but sharpens the earlier ones:

Part	Bundle report	Delivered title	Labels
I	105 (base)	Weak Ascent Sequences and Fixed Difference Ascent Growth	wasc:gr:
II	107	Weighted Weak Ascent Levels and Their Probability Law	wasc:lv:
III	108	Sharp Logarithmic Remainders and Bounded Error Inverses	wasc:sh:

Each Part prints its manuscript’s text, with every result, proof, remark, open question and limitation; statements keep their delivered wording. Sections, statements and equations are numbered continuously, and statements and equations are numbered within their sections. Part I keeps exactly the section, statement and equation numbers of Report 105 (Sections 1–12). In Part II a number is Report 107’s plus 12 in its section part: Report 107’s Section k is Section $k + 12$ here, and its Theorem $k.j$ and equation $(k.j)$ are Theorem $(k + 12).j$ and $(k + 12.j)$. Part II ends with a section added in the write, Section 23. In Part III the offset is 23: Report 108’s Section k is Section $k + 23$. Report 108’s Sections 3 and 4 repeat Report 107’s Sections 3 and 4 word for word; they are printed once, in Part II, and Sections 26 and 27 of Part III keep only their titles and a pointer. Report 108’s references to equations of those two sections point to Part II.

Notes added when the manuscripts were merged are set in small type and tagged [**write**], with their date. They give cross-references between the Parts, mark results proved twice, mark statements that a later Part supersedes, and credit prior work that a manuscript used without credit. Superseded statements are printed as delivered, with a dated note pointing forward, as in the merged reports **a202058-ascent-000-growth** and **a202061-ascent-120-deficit** of the same collection. No delivered statement was changed, and no symbol was renamed. Each Part ends with a section “Further questions and research”, which collects, under Vladimir’s standing rule of 4 October 2026, every claim of that Part that is stated without proof, with its source, its sketch and what is missing.

What is proved, and where

Here $a_n^{(d)}$ counts difference ascent sequences of length n with parameter d (the d -ascent condition is $x_{i+1} > x_i - d$), $a_n = a_n^{(1)}$ (A336070), $H_n(w) = \sum_x w^{E(x)}$ is the level-weighted count of weak ascent sequences ($E(x)$ the number of equal adjacent pairs), $\mu = 6/\pi^2$, and $T = \log n$.

Statement	Where
$(a_n^{(d)}/n!)^{1/n} \rightarrow \mu$, $a_{n+1}^{(d)}/((n+1)a_n^{(d)}) \rightarrow \mu$, exponential concentration of the ascent count at μn , tilted upper bound; every fixed $d \geq 0$ $\log(a_n/(n!\mu^n)) = \frac{1}{2}T^2 + O(T^2/\log T)$	Part I, Theorem 1.1
$\log H_n(w) = \log n! + n \log \mu + \frac{w}{2}T^2 + O_{\mathcal{W}}(T^2/\log T)$, w in a compact $\mathcal{W} \subset (0, \infty)$ The same with remainder $O_{\mathcal{W}}(T)$; $\log a_n^{(d)} = \log n! + n \log \mu + \frac{d}{2}T^2 + O_d(T)$ for every fixed integer $d \geq 1$	Part I, Theorem 1.2; again (second route) as the case $w = 1$ of Part II, Theorem 13.1 Part II, Theorem 13.1
Threshold inverses: error $O(\log x / \log \log x)$ ($d = 1$; weighted); $o(L/(\log L)^2)$ (every fixed d); $O(1)$ (weighted; every fixed $d \geq 1$) Level law: log-MGF limit $\frac{w}{2}(e^t - 1)$ at speed T^2 , concentration and L^p convergence of E_n/T^2 to $w/2$, full large-deviation principle Mean $\frac{w}{2}T^2 + O(T^2/\sqrt{\log T})$, improved to $\frac{w}{2}T^2 + O(T^{3/2})$; central moments $O(T^{3p/2})$, variance $O(T^3)$	Part III, Theorem 24.1; the case $d = 0$ from the Fishburn asymptotic, [write] Remark 33.1
$H_n(0) = P_n$ (primitive ascent sequences); $\log \mathbb{P}_{n,w}(E_n = 0) = -\frac{w}{2}T^2 + O(T^2/\log T)$, improved to $O(T)$	Part I, (1.7) and (10.4); Part II, Theorem 13.4; Part III, Corollary 24.2 ($d = 0$: [write] Remark 33.1) Part II, Theorem 13.2
Exact state recurrences, the $K + 2$ children identity, the catalytic equation (Auli–Elizalde) and its derivative recurrence, the level-marked kernel, the binomial transform between Fishburn and primitive counts (Jelinek’s Fact 5.1), exact counts for $d \leq 3$, $n \leq 10$	Part II, Theorem 13.3; [write] Remark 20.1
<i>Open in all three Parts</i> A relative-error equivalent and an amplitude; whether the residual is $c' \log n + O(1)$; effective constants and a certified finite inverse bracket; the variance order, a central or local limit theorem and Poisson approximation for the level count; the full tilted pressure, a large-deviation principle and a central limit theorem for the ascent count; uniformity in $d = d(n)$ and as $w \rightarrow 0$; the singular type of the Borel transform; the unrestricted generating function for general d (Dukes–Sagan, Problem 8.7). See Sections 12.3, 23 and 37.1.	

Finite computations in all three Parts (exact rational certificates, corruption campaigns, floating diagnostics, the table of r_n and B_n) are checks of formulas and implementations; no asymptotic statement rests on them. The literature searches recorded in the Parts are bounded and are not priority certificates.

Notation across the three Parts

Common to all Parts: $\mu = 6/\pi^2$; $g(v) = \log v/(v-1)$ and $\chi(v) = \int_v^\infty \log t/(t(t-1)) dt$ with $\chi(1) = \pi^2/6$; the state $(K, \text{last letter})$ with alphabet size $M = K + 2$; the frozen matrix with eigenvector $e^{-qI/M}$, its eigenvalue $\lambda = \lambda_M(e^q)$ and the geometric Doob kernel; the calibration $s = n\mu\chi(e^{q(s)})$, $S(s) = n\mu g(e^{q(s)})$ (Part I from Section 6 on; Parts II and III); the corrector b_l , c , h and its stable form $H = Ax(1-x) + J_0x$; $Q = 1+q$, $Q_0(z) = z/(e^z - 1)$; $\Psi(r) = \log r - r + 1$; the cutoff ζ with slope four; W or W_0 the principal Lambert function in the inverses. The symbols below change meaning between the Parts or inside one; each Part’s own text fixes its meaning there.

Symbol	Meanings
$H_n(\cdot)$	Part I: $H_n(v) = \sum v^{\text{wasc}(x)}$, the mark v counts <i>weak ascents</i> . Parts II and III: $H_n(w) = \sum w^{E(x)}$, the mark w counts <i>levels</i> (equal adjacent pairs). <i>Tempting false reading</i> : Part III's $H_3(2) = 12$ is not Part I's $H_3(2)$. Only at argument 1 do both equal a_n .
v	Part I: the ascent mark of (1.3); $v(\tau)$ (Section 4) and $v(s)$ (Section 6) on its calibration curves; in Section 12 the variable marking the last entry. Parts II and III: $v = e^q$, the frozen auxiliary ascent mark (Part II says so explicitly).
w, d	w : level weight (Parts II, III); Part I has no level weight. d : the difference parameter in all Parts; also a forward integer increment in the proof of (8.12), of (16.8) and of Lemma 16.1 (summation index). <i>False reading</i> : substituting $w = d$ does not turn levels into d -ascents (Part III, Section 24).
P	Part I: pressure $P(q) = -\log \chi(e^q)$ (Section 5); Doob kernel $P(l, j)$ ((7.1)); the series $P(t) = 1 + \mathcal{W}_0(t, 1)$ of primitive objects ((12.3)). Parts II and III: Doob kernels P , P_w , P_d ; Part II: P_n the number of primitive ascent sequences, and $P(t)$ their ordinary series. Part I's $P(t)$ and Part II's $P(t)$ are the same series.
g, G	$g(v)$ as above in all Parts. Part I also: $g_n(u) = u^2 H_n(u)$ (Section 2) and $g(x) = f(x) + \frac{1}{2}(\log x)^2$ (Section 10), Part II's A_w at $w = 1$ and Part III's G_c at $c = 1$. $G(u, z)$ is Part I's catalytic series.
L, l	Part I: L the last letter of the state (K, L) , and $L = \log X$ in Section 10. Parts II and III: l the last letter, L_i the last letter on the sampled path, $L = \log X$ or $\log Y$ in the inverses.
$W, V, \mathcal{W}$	Part I: W the Lambert function, and $W = ((j - l) \bmod M)/M$ (also W_i), the circular increment. Parts II and III write that increment V (also V_i) and the Lambert function W_0 ; $\mathcal{W} = [w_-, w_+]$ is a compact weight interval. Part I's V_j , V_i (Sections 4, 6) are potentials, and $\mathcal{W}_\ell$ is its level-marked series.
α	Part I: $\alpha = 3q/(2S)$, α_i (Section 6); Parts II and III: $\alpha(q) = S'(s)$, the tilted mean, which Part I calls $m(q)$ in Section 8; Part II's α_* is Part I's α with $3/2$ replaced by D_* .
h	The corrector $h(q, M, l)$ in all Parts. Also Part I: $h(s)$ in the proof of Lemma 3.1, $h_\alpha(r)$; Part II: the secant step $h > 0$ in Section 19; Part III: $h = T^{-1/2}$ in the proof of Corollary 34.1.
T	The frozen matrix T_v (Part I), T , T_w , T_d (Parts II, III), and $T = \log n$ (Part I from Section 6; Parts II, III).
U	$U = \log T = \log \log n$ (Parts I, II; U_n in Part II, Section 19); Part I also: the potential $U_i(K, L)$ of (8.1).
b, B	b_l the corrector means (all). Part I: b, b_j block lengths. Part II: $b_n = (\log n)^2$. Part III: $b = (w_+ - 1)_+$ (Section 29) and $b = 2C_1\delta$ (proof of Corollary 34.1). $B = r/(1 - r)$ (all). Part I: $B_n = n(r_n/\mu - 1)$ in the table of Section 11, $B(u, t)$ the Borel transform. Part II: B_* a cutoff constant. Part III: $B_n = H_n(w)$ or $a_n^{(d)}$ (Corollary 24.2), B_i an indicator, $B_{\mathcal{W}, p}$, $B_{d, p}$ constants. <i>False reading</i> : Part I's B_n is a ratio diagnostic, Part III's B_n is a count.
c	The corrector mean c (all); c_η , $c_{w, \eta}$ concentration constants; Part III: $c \in \{w, d\}$ the log-square coefficient (Corollary 24.2, Section 33).
$E, \mathbb{E}$	Parts II and III: $E(x)$, E_n the number of levels. $\mathbb{E}$ is always an expectation ($\mathbb{E}_n$, $\mathbb{E}_{n, w}$, $\mathbb{E}_*$, $\mathbb{E}_w^*$, $\mathbb{E}_d^*$).
D	Part I: $D = 1 - r^M$. Part II: $D_0 = 1 - r^M$, $D_w = 1 + (w - 1)t$, the constant D_* . Part III: D_n the normalization (28.4).
R	Part I: R_i the likelihood ratio. Part II: $R = M/S$, R_i^w , and $R = 1/\mu = \pi^2/6$ in Section 20. Part III: $R = M/S$, R_i^w , $R_i^{(d)}$.
F	Part I: $F(n, k, l)$ state counts, F_n^{Fish} Fishburn numbers, $\mathcal{F}$ the level-marked series, $\mathcal{F}_i$ the path filtration. Part II: F_i potentials, F_n Fishburn numbers, $F(z)$ their series. Part III: F_i the capped potential, $\tilde{F}_i$ the lower potential.
A	Part I: A in (7.7), A_0 , the bracket constant A of Section 10, A_F the Fishburn series. Part II: $A_i(K, l; w)$ state weights, $A_w(t)$ the smooth inverse function. Part III: A the cutoff constant in $k = \lceil AT \rceil$ and the coefficient in (16.4); A_1 state weights.

Symbol	Meanings
k	The prefix cutoff: $\lceil 100T^2/U^2 \rceil$ in Part I and in Part II's lower bound, $\lceil B_*T^2/U^2 \rceil$ in Part II's upper bound, $\lceil AT \rceil$ in Part III.
N	Part I: $N(X) = \min\{n \geq 0 : a_n \geq X\}$, $N_d(X)$. Part II: $N_w(X)$. Part III: $N(Y) = \min\{n \geq 1 : B_n \geq Y\}$, and $N_{k,r}$, $N_{k,r}^{(d)}$ prefix counts.
Q	$Q = 1 + q$ (all); Part I: Q_j the block operator of Section 4; Part II: Q_n the tilted law in Section 19.
t, ℓ, u, z	Part I: t the Borel or length variable, ℓ the level mark, u the ascent mark of $G(u, z)$. Part II: $t = v/\lambda$ (Section 15) and the tilt t of the log-MGF.

Part I's ℓ (level mark) and Part II's w (level weight) mark the same statistic: $\mathcal{W}_\ell(t, 1) = \mathcal{F}(t, 1, 1, \ell) = \sum_{n \geq 1} H_n(\ell) t^n$ with Part II's H_n .

Provenance and merge decisions

Three manuscripts of the session bundle of Reports 1–243, all dated 2 October 2026, with the delivered author line “Research report N / Prepared for Vladimir”; none names an author or a tool, and none carries “prepared for private review” wording. They arrived in commit 60f54ea06 and were placed by 6ab1f1979 (batch 102). None pins a ProveIt commit or names a repository path.

- **Part I**, bundle Report 105, archive `A336070_Weak_Ascent_Logarithmic_Correction_Source.zip` (`report105.tex`, 24 pages). The base: its `report105.tex` became this `article.tex`. Contributes the root, ratio and concentration theorem for every fixed d , the logarithmic-square correction for $d = 1$, the inverses, the exact recurrences and formal generating functions, the level-marked kernel and the floating diagnostics.
- **Part II**, bundle Report 107, archive `A336070_Weighted_Levels_and_Large_Deviations_Source.zip` (`report107.tex`, 18 pages). It pins Report 105 by SHA-256 (TeX, PDF and archive; all three match the delivered files) and calls itself a separate continuation of it. Contributes the level-weighted theorem, the level law and its large-deviation principle, the zero-level atom and the weighted inverse.
- **Part III**, bundle Report 108, archive `Ascent_Sequences_Sharp_Remainders_and_Bounded_Inverses_Source.zip` (`report108.tex`, 24 pages, the deterministic concatenation of its eight `source/*.tex` files). Its input-provenance record pins Report 107's TeX and PDF (both match) and nine internal notes that were not delivered. Contributes the $O(\log n)$ remainders, the fixed- d theorem for every $d \geq 1$, the bounded-error inverses, the moment bounds and the exact table.

Where the merge had to choose:

1. *Order and base*. Chain order, which is the order of writing and of logical dependence; Report 105 is the base. Later Parts improve earlier statements; the earlier statements stay, with dated notes.
2. *Verbatim repetition*. Report 108's Sections 3–4 are word for word Report 107's Sections 3–4 (checked by `diff`); they are printed once, in Part II, and Part III keeps their titles as Sections 26 and 27. Report 108's Section 8 repeats Report 107's Section 6 except for the cutoff, two sentences and the closing paragraph; it is printed in full because Part III cites its equations, and a note marks the changes. Parts I and II prove the same theorem at $w = 1$ by different arguments; both are printed (second route).

3. *Attribution.* Part I's binomial transform (12.3) is Jelínek's Fact 5.1 [8]; Report 105 printed it without credit. A dated note supplies the credit. Part II credits it. All three manuscripts cite Mansour [9]; that citation was not checked against the publication at placement or at this write.
4. *Front matter and bibliography.* The three title blocks and abstracts are printed at the head of their Parts; one table of contents replaces two. The three bibliographies are merged into one, keeping every detail of every entry; the write adds the reference [5].
5. *Write additions.* Two remarks with proofs (Remarks 20.1 and 33.1), the three "Further questions and research" sections, the dated notes and this front matter. Everything else is delivered text.

Relation to neighbouring reports

Two reports in the same collection directory (`SetTheory/Cardinals/docs/reports/generating-functions-and-asymptotics/oeis-sequence-asymptotics/`) treat ordinary ascent sequences with restricted multiplicities, by the same toolkit (a frozen geometric kernel, a calibration curve, a Lambert inverse); they are siblings of this report, not hosts. `a202058-ascent-000-growth` treats A202058 (no value three times): factorial growth constant $8/(3\pi^2)$, and a conjecture that $\log(a_n/(n!\mu^n)) \sim \frac{2}{3}(\log n)^2$ pointwise, proved there only in limsup and cumulative form. Parts I–III prove the analogous pointwise law with coefficient $\frac{1}{2}$ for the unrestricted weak family ($d = 1$), and $\frac{d}{2}$ for every fixed $d \geq 1$; they are about different objects and say nothing about the coefficient $\frac{2}{3}$ of that conjecture. `a294220-ascent-multiplicity-caps` treats ascent sequences with every value at most b times; its growth constants μ_b increase to $6/\pi^2$ as $b \rightarrow \infty$ and it asks about the transition to unrestricted ordinary ascent sequences, the Fishburn case $d = 0$ of Part I. The unrestricted families here have $\mu = 6/\pi^2$ for every fixed d (Part I), with a correction $\frac{d}{2}(\log n)^2$ for $d \geq 1$ (Part III) that is absent at $d = 0$, where the Fishburn asymptotic has a factor $\sqrt{n}$. No report other than these treats A336070, weak ascents or difference ascents (searched 5 October 2026).

Part I

Weak ascent sequences and fixed difference ascent growth

Bundle Report 105, archive A336070_Weak_Ascent_Logarithmic_Correction_Source.zip, delivered as report105.tex (24 pages), dated 2 October 2026. Delivered title “Weak Ascent Sequences and Fixed Difference Ascent Growth: Factorial growth, concentration, and a sharp logarithmic square correction”; author line “Research report 105 / Prepared for Vladimir”. Printed as delivered, with [w^{rite}] notes; section, statement and equation numbers are those of the delivery.

Reading conventions of Part I	Elsewhere in the report
$H_n(v)$: v marks weak ascents	Parts II, III: $H_n(w)$, w marks levels
P : pressure (Section 5), Doob kernel (Section 7), primitive series (Section 12)	Parts II, III: P , P_w , P_d kernels; Part II: P_n primitive counts
W : Lambert function and circular increment W_i	Parts II, III: W_0 and V_i
L : last letter; $L = \log X$ in Section 10	Parts II, III: last letter l
$\alpha = 3q/(2S)$; $m(q) = S'$	Parts II, III: $\alpha(q) = S'$
$B_n = n(r_n/\mu - 1)$ (Section 11)	Part III: B_n a count

Abstract of this Part (as delivered). Let a_n count the weak ascent sequences of Bényi, Claesson and Dukes, recorded as OEIS A336070. We prove that both the factorial-normalized root and the scaled consecutive ratio tend to $\mu = 6/\pi^2$. The weak-ascent count divided by the length is exponentially concentrated at μ . The same conclusions hold for every fixed nonnegative integer difference-ascent parameter d , under the convention that a d -ascent satisfies $x_{i+1} > x_i - d$.

For A336070, a quadratic positive corrector and an exact path change of measure give

$$\log \frac{a_n}{n! \mu^n} = \frac{1}{2}(\log n)^2 + O\left(\frac{(\log n)^2}{\log \log n}\right).$$

This also yields a signed correction to the Lambert threshold inverse with a smaller-order error. These are substantial partial asymptotic results: a relative-error coefficient equivalent, smaller correction coefficients, and an amplitude remain unproved. The only external asymptotic input is the established ordinary Fishburn-number asymptotic; all new comparison, calibration, and inverse steps are given in detail.

1 Scope and principal conclusions

A weak ascent of a finite sequence $x = (x_1, \dots, x_n)$ of nonnegative integers is an index $i < n$ with $x_i \leq x_{i+1}$. A weak ascent sequence has $x_1 = 0$ and

$$0 \leq x_{i+1} \leq 1 + \text{wasc}(x_1, \dots, x_i) \quad (1 \leq i < n). \quad (1.1)$$

We count the empty sequence once, so $a_0 = 1$. These are the sequences in A336070 [10, 2], beginning

$$1, 1, 2, 6, 23, 106, 567, 3440, 23286, 173704, 1414102, \dots$$

The definition matters: other papers have used the informal name “weak ascent sequence” for different globally constrained objects. All assertions below concern (1.1).

Put

$$g(v) = \frac{\log v}{v-1}, \quad g(1) = 1, \quad \chi(v) = \int_v^\infty \frac{\log t}{t(t-1)} dt, \quad \mu = \frac{6}{\pi^2}. \quad (1.2)$$

The integrand has its removable value at $t = 1$. For a uniformly chosen length- n weak ascent sequence, let K_n be its weak-ascent count. Write

$$H_n(v) = \sum_{|x|=n} v^{\text{wasc}(x)} \quad (n \geq 1).$$

Theorem 1.1. *For every fixed $v > 0$,*

$$\limsup_{n \rightarrow \infty} \frac{1}{n} \log \frac{H_n(v)}{n!} \leq -\log \chi(v). \quad (1.3)$$

Moreover,

$$\lim_{n \rightarrow \infty} \left(\frac{a_n}{n!} \right)^{1/n} = \mu, \quad \lim_{n \rightarrow \infty} \frac{a_{n+1}}{(n+1)a_n} = \mu. \quad (1.4)$$

For every $\eta > 0$ there are $c_\eta > 0$ and n_η such that

$$\mathbb{P} \left(\left| \frac{K_n}{n} - \mu \right| \geq \eta \right) \leq e^{-c_\eta n} \quad (n \geq n_\eta). \quad (1.5)$$

The same statements hold for every fixed nonnegative integer d when weak ascents and their sequence constraint are replaced by d -ascents and the corresponding constraint of Section 9.

Theorem 1.2. *For weak ascent sequences, namely $d = 1$,*

$$\log \frac{a_n}{n! \mu^n} = \frac{1}{2} (\log n)^2 + O \left(\frac{(\log n)^2}{\log \log n} \right). \quad (1.6)$$

In particular the quotient of this logarithm by $(\log n)^2$ tends to $1/2$. For $N(X) = \min\{n \geq 0 : a_n \geq X\}$, $L = \log X$, and $x = L/W(\mu L/e)$,

$$N(X) = x - \frac{(\log x)^2}{2 \log(\mu x)} + O \left(\frac{\log x}{\log \log x} \right) \quad (X \rightarrow \infty). \quad (1.7)$$

Here W is the principal Lambert function. The logarithmic correction theorem is specific to $d = 1$.

What remains open. Equation (1.6) is a logarithmic asymptotic, not a multiplicative equivalent. The error inside the exponential need not tend to zero. No amplitude, coefficient of a smaller correction, full tilted pressure, central limit theorem, or transseries is established. The upper bound (1.3) need not be an equality away from $v = 1$. The inverse error in (1.7) diverges and has an unspecified asymptotic constant; it is not a bounded-error inverse or a certified finite numerical error bar.

[write] **5 October 2026.** Superseded in part. Part III, Theorem 24.1, improves the remainder in (1.6) to $O(\log n)$ (its case $w = 1$, equivalently $d = 1$), so the error inside the exponential is now known to be $O(\log n)$: a polynomial-factor uncertainty, still not a relative-error equivalent. Corollary 24.2 there replaces the error $O(\log x / \log \log x)$ of (1.7) by $O(1)$, so the inverse is now a bounded-error inverse, with unspecified constants and no certified finite bracket. The other items of this paragraph remain open in all three Parts; see Section 12.3. The counting assertion of Theorem 1.2 is proved a second time, by a weighted kernel and a signed mixture, as the case $w = 1$ of Part II, Theorem 13.1, with the same remainder. The last sentence of Theorem 1.2 remains true of this Part’s proof; Part III proves the correction $\frac{d}{2}(\log n)^2$ for every fixed integer $d \geq 1$.

Relation to the literature. The definition, bijections, and enumerative refinements are due to Bényi–Claesson–Dukes [2]; the catalytic equation used below comes from Auli–Elizalde [1]. Mansour’s 2026 paper [9] provides further statistical generating functions. The fixed- d convention follows Dukes–Sagan [6]. A bounded literature check through 2 October 2026 did not locate the asymptotic conclusions proved here or a full unrestricted coefficient equivalent. This is not a claim of an exhaustive novelty search. Ordinary Fishburn asymptotics are prior work [7]; their use as a lower comparison is explicit throughout.

[write] 5 October 2026. The Mansour citation, made by all three Parts, was not checked against the publication at placement or at this write; see the bibliography.

2 Exact state recurrence and formal generating functions

2.1 Positive recurrence and extension identity

At length $n \geq 1$, write (K, L) for the number of weak ascents and the final letter. Starting from $(0, 0)$, appending j gives

$$(K, L) \mapsto (K + \mathbf{1}_{\{j \geq L\}}, j), \quad 0 \leq j \leq K + 1. \quad (2.1)$$

Induction gives $0 \leq L \leq K \leq n - 1$: without an ascent the new letter is below L , and with an ascent it is at most $K + 1$, the new count. Consequently every such sequence is an inversion sequence and $a_n \leq n!$.

Let $F(n, k, l)$ count sequences in state (k, l) , with $F(1, 0, 0) = 1$ and every out-of-range entry zero. For $0 \leq j \leq k \leq n$,

$$F(n + 1, k, j) = \sum_{l=j+1}^k F(n, k, l) + \sum_{l=0}^{\min(j, k-1)} F(n, k - 1, l). \quad (2.2)$$

The first sum appends a strict descent and preserves the count; the second appends a weak ascent and increases it. Empty sums are zero. Thus

$$a_n = \sum_{k=0}^{n-1} \sum_{l=0}^k F(n, k, l), \quad H_n(v) = \sum_{k,l} F(n, k, l) v^k. \quad (2.3)$$

Prefix and suffix sums compute a new layer in $O(n^2)$ integer operations and all layers through N in $O(N^3)$ integer operations; this is an arithmetic-operation count, not a bit-complexity bound.

There are exactly $K + 2$ children of each state. Counting them in two ways gives the crucial identity

$$\frac{a_{n+1}}{a_n} = \mathbb{E}_n(K_n + 2) \quad (n \geq 1). \quad (2.4)$$

The ratio limit will follow from this identity and concentration, not from a root-limit inference.

2.2 Catalytic equation and its polynomial meaning

Define $g_0(u) = u$ and, for $n \geq 1$,

$$g_n(u) = u^2 H_n(u) = \sum_{k,l} F(n, k, l) u^{k+2}, \quad G(u, z) = \sum_{n \geq 0} g_n(u) z^n.$$

The exact equation of Auli–Elizalde, translated to this refinement, is

$$G(u, z) = u(1 - u) + uG(u + zu(1 - u), z), \quad a_n = g_n(1). \quad (2.5)$$

This is an identity in $\mathbb{Q}[u][[z]]$, the ring of formal power series in z whose coefficients are polynomials in u . It is not an analytic assertion about convergence at positive z . At z^0 it yields $g_0 = u$ after cancellation of $1 - u$ in $\mathbb{Q}[u]$. For $n \geq 1$, formal Taylor expansion gives

$$(1 - u)g_n(u) = \sum_{j=1}^n \frac{u^{j+1}(1 - u)^j}{j!} g_{n-j}^{(j)}(u).$$

Every term is divisible by $1 - u$, so division in the polynomial ring is legitimate and yields the unique recursion

$$g_n(u) = \sum_{j=1}^n \frac{u^{j+1}(1 - u)^{j-1}}{j!} g_{n-j}^{(j)}(u). \quad (2.6)$$

For example,

$$g_1 = u^2, \quad g_2 = 2u^3, \quad g_3 = u^3 + 5u^4, \quad g_4 = 9u^4 + 14u^5.$$

Only after polynomial cancellation do we set $u = 1$. This gives

$$a_n = g'_{n-1}(1), \quad n \geq 1, \quad (2.7)$$

consistent with (2.4) for $n \geq 2$. Setting $u = 1$ directly in (2.5) would merely give a tautology and lose the useful information.

The factorial growth proved below implies that the ordinary series $G(1, z)$ has radius zero. Its Borel transform

$$B(u, t) = \sum_{n \geq 0} g_n(u) \frac{t^n}{n!}$$

obeys the formal identity

$$B(u, t) = u + \sum_{j \geq 1} \frac{u^{j+1}(1 - u)^{j-1}}{j!} \mathcal{I}^j [\partial_u^j B(u, t)], \quad (2.8)$$

where $\mathcal{I}$ integrates formally from 0 to t . Each coefficient receives only finitely many contributions. Equation (2.7) gives $\partial_t B(1, t) = \partial_u B(1, t)$ formally. The root theorem establishes the radius $\pi^2/6$ of $B(1, t)$, but does not identify its singular type or justify analytic continuation of (2.8).

3 The frozen positive matrix

Fix $M \geq 1$ and $v > 0$. On the alphabet $\{0, \dots, M - 1\}$ define

$$T_v(l, j) = v^{\mathbf{1}_{\{j \geq l\}}}, \quad 0 \leq l, j < M. \quad (3.1)$$

For $v \neq 1$ set $r = v^{-1/M}$ and

$$f_l = r^l, \quad \lambda_M(v) = \frac{v - 1}{1 - r}. \quad (3.2)$$

At $v = 1$, take $f_l = 1$ and $\lambda_M(1) = M$.

Lemma 3.1. *The positive vector f satisfies $T_v f = \lambda_M(v) f$. For every integer $b \geq 0$,*

$$T_v^b \mathbf{1} \leq \max(v, v^{-1}) \lambda_M(v)^b \mathbf{1} \quad (3.3)$$

entrywise. Uniformly for v in any fixed compact subset of $(0, \infty)$,

$$\lambda_M(v) \leq \frac{M}{g(v)} + C \quad (3.4)$$

with a constant independent of M .

Proof. The first row has value

$$(T_v f)_0 = v \sum_{j=0}^{M-1} r^j = \frac{v-1}{1-r} = \lambda_M(v),$$

and consecutive rows differ by $(1-v)r^l = \lambda_M(v)(r-1)r^l$. This proves the identity. It remains valid for $v < 1$, when both numerator and denominator of $\lambda_M(v)$ are negative. At $v = 1$, T_v is the all-ones matrix.

If $m_f = \min_l f_l$ and $M_f = \max_l f_l$, positivity implies

$$T_v^b \mathbf{1} \leq \lambda_M(v)^b f / m_f \leq (M_f / m_f) \lambda_M(v)^b \mathbf{1}.$$

Here $M_f / m_f = \max(v, v^{-1})^{(M-1)/M} \leq \max(v, v^{-1})$. Finally let $q = \log v$ and define the continuous removable function

$$h(s) = \frac{1}{1 - e^{-s}} - \frac{1}{s}, \quad h(0) = \frac{1}{2}.$$

Then $\lambda_M(v) - M/g(v) = (v-1)h(q/M)$. For marks in a fixed compact set, q/M stays in a fixed compact interval and this difference is bounded. $\square$

Lemma 3.2. *In a block of b extensions starting from (K, L) , let K' be the final weak-ascent count. Then*

$$\sum_{\text{actual block paths}} v^{K'-K} \leq \max(v, v^{-1}) \lambda_{K+b+2}(v)^b. \quad (3.5)$$

On a fixed compact mark range the eigenvalue on the right is bounded by

$$\lambda_{K+b+2}(v) \leq K/g(v) + C(b+1). \quad (3.6)$$

Proof. The ascent count increases by at most b . Every actual letter lies in the alphabet $\{0, \dots, K+b+1\}$. Embedding the actual paths into all paths of that frozen alphabet preserves each weight $v^{K'-K}$ exactly and adds only nonnegative weights. Apply Lemma 3.1 with $M = K+b+2$. This reasoning also works when $v < 1$: it adds positive terms, rather than relying on monotonicity of v^K in K . The compact bound follows from (3.4) and boundedness of $1/g$. $\square$

4 Calibrated block proof of the tilted upper bound

4.1 A statewise calibration

Fix the terminal mark $v_0 > 0$ and put $\nu = 1/\chi(v_0)$. The function χ is positive, strictly decreasing, and smooth on $(0, \infty)$, with

$$\chi'(v) = -g(v)/v, \quad \chi(v) \longrightarrow 0 \quad (v \rightarrow \infty).$$

For $0 < \tau \leq 1$, define $v(\tau) \geq v_0$ uniquely by

$$\tau = \nu \chi(v(\tau)), \quad \theta(\tau) = \log v(\tau). \quad (4.1)$$

Differentiation gives

$$\theta'(\tau) = -\frac{1}{\nu g(v(\tau))}, \quad \frac{y}{\nu g(v(\tau))} = -\theta'(\tau)y \quad (y \geq 0). \quad (4.2)$$

This is a global identity for every possible scaled state y , not an equation presumed to hold along a typical trajectory. For fixed $\varepsilon > 0$, θ and its first two derivatives are bounded on $[\varepsilon, 1]$. The terminal value $\theta(1) = \log v_0$ can be negative.

For $v \geq 2$,

$$\chi(v) \leq 2 \int_v^\infty \frac{\log t}{t^2} dt = \frac{2(\log v + 1)}{v} = O(v^{-1/2}).$$

As $\varepsilon \downarrow 0$, (4.1) therefore implies

$$|\theta(\varepsilon)| = O(\log(1/\varepsilon)), \quad \varepsilon|\theta(\varepsilon)| \rightarrow 0. \quad (4.3)$$

4.2 One block and cancellation

Fix $\varepsilon \in (0, 1)$, take $k = \lceil \varepsilon n \rceil$, and divide the extensions from length k to length n into $J = O(1/\delta)$ blocks of lengths $b_j \leq \delta n$, allowing harmless integer endpoint adjustments. On a block starting at scaled time τ , put $b = b_j$, $\theta = \theta(\tau)$, and

$$\Delta\theta = \theta(\tau + b/n) - \theta(\tau), \quad v = e^\theta.$$

Since $0 \leq K' - K \leq b$,

$$\sum_{\text{paths}} e^{\theta(\tau+b/n)K'} \leq e^{\theta K + \Delta\theta K + b|\Delta\theta|} \max(v, v^{-1}) \lambda_{K+b+2}(v)^b. \quad (4.4)$$

The constants below may depend on ε, v_0 , but are independent of n , the state, and the subsequently chosen small δ .

Apply $\log x \leq x - 1$ to $x = \lambda/(n\nu)$ and use (3.6) and (4.2):

$$\lambda^b \leq (n\nu/e)^b \exp\left(-\frac{b}{n}\theta'(\tau)K + C\frac{b(b+1)}{n}\right). \quad (4.5)$$

Taylor's theorem gives

$$\Delta\theta = \frac{b}{n}\theta'(\tau) + O(b^2/n^2), \quad |\Delta\theta| = O(b/n).$$

Every reachable state has $K \leq n$. Thus the terms linear in K cancel between (4.4) and (4.5), and the remaining state term is $O(b^2/n)$. The eigenvector condition number contributes $O(1)$ to the logarithm per block. We obtain

$$\sum_{\text{paths}} e^{\theta(\tau+b/n)K'} \leq e^{\theta(\tau)K} (n\nu/e)^b \exp(C(b^2/n + 1)). \quad (4.6)$$

4.3 Positive telescope and ordered limits

Let Q_j denote the positive operator that sums over actual paths in block j , and let $V_j(K, L) = \exp(\theta(\tau_j)K)$. Equation (4.6) says $Q_j V_{j+1} \leq c_j V_j$ for a scalar c_j independent of the state. Positivity allows backward composition of these inequalities. This does not require a shared alphabet or a shared eigenvector between successive blocks.

Using $\sum b_j = n - k$ and $\sum b_j^2/n \leq \delta n$, and then summing over length- k prefixes, yields

$$H_n(v_0) \leq k! \exp(k|\theta(k/n)|) (n\nu/e)^{n-k} \exp(C(n\delta + J)). \quad (4.7)$$

Here $a_k \leq k!$ and $K \leq k - 1$ give the prefix bound for either sign of $\theta(k/n)$. Stirling's formula reduces (4.7) to

$$\limsup_{n \rightarrow \infty} \frac{1}{n} \log \frac{H_n(v_0)}{n!} \leq \varepsilon \log \varepsilon + (1 - \varepsilon) \log \nu + \varepsilon |\theta(\varepsilon)| + C\delta. \quad (4.8)$$

First fix v_0, ε, δ and take the indicated limit in n . Next send $\delta \downarrow 0$ with ε fixed. Finally send $\varepsilon \downarrow 0$ and apply (4.3). The result is (1.3). No uniformity in ε of the block error constant is needed because that error has already been removed before the last limit.

5 The exact constant and the ratio limit

5.1 Ordinary ascent sequences supply the lower bound

Let F_n^{Fish} be the ordinary Fishburn number, counting ordinary ascent sequences with strict ascents. Every ordinary ascent sequence satisfies (1.1), because each prefix has at least as many weak ascents as strict ascents. Hence

$$F_n^{\text{Fish}} \leq a_n \leq n!. \quad (5.1)$$

We use the established result recorded by Hwang–Jin [7, Eq. (1.1)]:

$$F_n^{\text{Fish}} = C_F n! \mu^n \sqrt{n} (1 + O(n^{-1})), \quad C_F > 0. \quad (5.2)$$

Their displayed form uses a positive constant times $(6/(e\pi^2))^n n^{n+1}$; Stirling gives (5.2). The actual value of C_F is not needed here. This is an external theorem about ordinary Fishburn numbers, not about the weak-ascent sequence.

[write] 5 October 2026. Part II records the value $C_F = 12\sqrt{3}\pi^{-5/2}e^{\pi^2/12}$ in (20.4), with the same references, and the corresponding primitive constant $C_P = C_F e^{-\pi^2/6}$ in (20.7).

At $v = 1$, substitute $x = 1/t$ and expand a nonnegative geometric series:

$$\chi(1) = \int_0^1 \frac{-\log x}{1-x} dx = \sum_{m \geq 1} \int_0^1 x^{m-1} (-\log x) dx = \sum_{m \geq 1} \frac{1}{m^2} = \frac{\pi^2}{6}. \quad (5.3)$$

Monotone convergence justifies the exchange. The upper bound (1.3) at $v = 1$ and the lower inclusion (5.2) coincide at μ , proving the first limit in (1.4).

5.2 Two-sided exponential concentration

Set

$$P(q) = -\log \chi(e^q).$$

This is smooth near zero and satisfies

$$P(0) = \log \mu, \quad P'(q) = \frac{g(e^q)}{\chi(e^q)}, \quad P'(0) = \mu. \quad (5.4)$$

Combining (1.3) with the unweighted root limit gives, for every fixed real q ,

$$\limsup_{n \rightarrow \infty} \frac{1}{n} \log \mathbb{E}_n e^{qK_n} \leq P(q) - P(0). \quad (5.5)$$

A lower tilted bound at $q \neq 0$ has not been used or proved.

Given $\eta > 0$, differentiability permits a sufficiently small $q > 0$ with

$$P(q) - P(0) - q(\mu + \eta) < 0.$$

Exponential Markov inequality then makes $\mathbb{P}(K_n/n \geq \mu + \eta)$ exponentially small. A sufficiently small $q < 0$ satisfies

$$P(q) - P(0) - q(\mu - \eta) < 0$$

and gives the lower tail, since $e^{qK_n} \geq e^{qn(\mu - \eta)}$ on that event. Combining the two bounds and slightly reducing the exponent proves (1.5). Empty tail events cause no difficulty.

Since $0 \leq K_n/n \leq 1$, convergence in probability implies convergence of the first moment: for instance split $\mathbb{E}|K_n/n - \mu|$ at any fixed $\eta > 0$ and use the tail bound. Thus $\mathbb{E}K_n/n \rightarrow \mu$. Equation (2.4) now gives

$$\frac{a_{n+1}}{(n+1)a_n} = \frac{n}{n+1} \frac{\mathbb{E}K_n}{n} + \frac{2}{n+1} \rightarrow \mu.$$

This completes the weak-ascent part of Theorem 1.1.

6 A preliminary logarithmic square upper bound

This section gives the simpler $3/4$ upper bound that supplies the calibration and small-state estimates for the sharp proof in Sections 7 and 8. It uses special reachability geometry for weak ascents. All calibrations now depend on the target length n ; the uniform estimates are derived afresh.

6.1 Reachability and the moving calibration

Lemma 6.1. *At length i with K weak ascents, put $M = K + 2$. Then*

$$i \leq \frac{(K+1)(K+2)}{2}, \quad M \geq \sqrt{2i}. \quad (6.1)$$

Proof. Split the sequence at every weak ascent into $K+1$ strictly decreasing runs. Number the runs $r = 0, \dots, K$. The first letter of run r is at most r by the state support, so that run has at most $r+1$ nonnegative letters. Sum these bounds. Since $M(M-1) \geq 2i$, the displayed weaker lower bound on M follows. $\square$

[write] **5 October 2026.** Part II reproves this lemma as (14.3); Part III, (25.4), proves it for every fixed $d \geq 1$ (the runs between d -ascents are strictly decreasing).

For $0 < s \leq n$, define $v(s) \geq 1$, $q(s)$, and $S(s)$ by

$$s = n\mu\chi(v(s)), \quad q(s) = \log v(s), \quad S(s) = n\mu g(v(s)). \quad (6.2)$$

Differentiation gives

$$q'(s) = -1/S(s), \quad S'(s) = \frac{v}{v-1} - \frac{1}{\log v} \in [1/2, 1]. \quad (6.3)$$

To see the interval assertion, put $q = \log v \geq 0$. The expression is the mean of $x \in [0, 1]$ under density proportional to e^{qx} . It equals $1/2$ at $q = 0$, increases with q because its derivative is a variance, and is at most one. Furthermore $S(0+) = 0$, $S(n) = n\mu$, and $q(n) = 0$. Hence

$$\frac{s}{2} \leq S(s) \leq s, \quad 0 \leq q(s) \leq 2 \log(n/s). \quad (6.4)$$

Expanding the integrand for large v and integrating also gives

$$\chi(v) = \frac{\log v + 1}{v} + O\left(\frac{\log v}{v^2}\right), \quad v\chi(v) = \log v + 1 + o(1). \quad (6.5)$$

6.2 An exact moving positive potential

For every reachable state at time i set

$$V_i(K, L) = \exp\left(q(i)K - \frac{q(i)L}{K+2}\right). \quad (6.6)$$

Then $V_n = 1$. In a single extension let $M = K + 2$, $a = \mathbf{1}_{\{j \geq L\}}$, $q = q(i+1)$, $S = S(i+1)$, and $\Delta q = q(i+1) - q(i) \leq 0$. Since $0 \leq j < M$ and $q \geq 0$,

$$\exp\left(-\frac{qj}{M+a}\right) \leq \exp(q/M) \exp(-qj/M). \quad (6.7)$$

Indeed the exponent difference is $qja/[M(M+a)] \leq q/M$. Summing the exact frozen eigenvector identity of Lemma 3.1, then dividing by V_i , yields

$$\frac{\sum_{j=0}^{M-1} V_{i+1}(K+a, j)}{V_i(K, L)} \leq \lambda_M(e^q) \exp(\Delta q K + q/M + |\Delta q|). \quad (6.8)$$

The old-versus-new comparison contributes $-\Delta q L/M \leq |\Delta q|$; this is why the moving denominator cannot simply be ignored.

Because S is increasing,

$$\Delta q = -\int_i^{i+1} \frac{ds}{S(s)} \leq -1/S(i+1), \quad |\Delta q| \leq 1/S(i) \leq 2/i. \quad (6.9)$$

The elementary inequality $2 \sinh(x/2) \geq x$ for $x \geq 0$ gives

$$\frac{x}{1 - e^{-x}} \leq e^{x/2}, \quad \lambda_M(e^q) \leq \frac{M}{g(e^q)} \exp(q/(2M)),$$

with removable interpretations at zero. Let

$$r = M/S, \quad \alpha = 3q/(2S), \quad \Psi(r) = \log r - r + 1. \quad (6.10)$$

Taking logarithms of (6.8), dividing its bound by $n\mu/e$, and using $K = M - 2$ gives

$$\begin{aligned} \log \left(\frac{\sum_j V_{i+1}(K+a, j)}{(n\mu/e)V_i(K, L)} \right) &\leq \log r + 1 - \frac{M-2}{S} + \frac{3q}{2M} + \frac{2}{i} \\ &\leq \Psi(r) + \frac{\alpha}{r} + \frac{6}{i}. \end{aligned} \quad (6.11)$$

This estimate holds for every reachable state, including states with very small K relative to i .

6.3 Uniform treatment of small reachable states

Put $T = \log n$, $U = \log T$, and choose

$$k = \left\lceil \frac{100T^2}{U^2} \right\rceil. \quad (6.12)$$

Only sufficiently large n is considered, so $U > 0$ and $1 \leq k < n$. For $i \geq k$, (6.4) gives $\alpha \leq 6T/i = o(1)$ uniformly, hence eventually $\alpha \leq 1/8$.

For $0 \leq \alpha \leq 1/8$ define $h_\alpha(r) = \Psi(r) + \alpha/r$. Its derivative is

$$h'_\alpha(r) = \frac{r - r^2 - \alpha}{r^2}.$$

Its maximum over $[m, \infty)$ occurs at the endpoint or is no larger than its value at

$$r_+(\alpha) = \frac{1 + \sqrt{1 - 4\alpha}}{2}.$$

The latter point has

$$h_\alpha(r_+(\alpha)) \leq \alpha + 2\alpha^2. \quad (6.13)$$

For a proof, the derivative of this value in α is $1/r_+(\alpha)$. Since $r_+ \geq 1 - 2\alpha$, this derivative is at most $1 + 4\alpha$ on the stated interval, and the value at zero is zero.

The geometry gives $r \geq m = \sqrt{2i}/S(i+1)$. Using $S(i+1) \geq (i+1)/2$, $q(i+1) \leq 2T$, and dropping the nonpositive term $-m$, we obtain

$$h_\alpha(m) \leq -\frac{1}{2} \log i + 1 + \frac{3}{2} \log 2 + \frac{3T}{\sqrt{2i}}. \quad (6.14)$$

The right side decreases with i . At $i = k$ it is bounded by

$$-\left(1 - \frac{3}{\sqrt{200}}\right)U + \log U + O(1),$$

which tends to $-\infty$. Thus the endpoint value is nonpositive for every $i \geq k$, once n is sufficiently large. Equations (6.13) and (6.14) prove the uniform statewise bound

$$\Psi(r) + \alpha/r \leq \alpha + 2\alpha^2. \quad (6.15)$$

The cutoff is used to control all reachable states; no typical-state assumption has entered.

6.4 Accumulation and the prefix cost

Combine (6.11) and (6.15), and compose the positive one-step inequalities backward from $V_n = 1$. Summing the initial states at length k yields

$$a_n \leq k! e^{kq(k)} (n\mu/e)^{n-k} \exp\left(\sum_{i=k}^{n-1} (\alpha_i + 2\alpha_i^2 + 6/i)\right), \quad \alpha_i = \frac{3q(i+1)}{2S(i+1)}. \quad (6.16)$$

The prefix estimate follows from $a_k \leq k!$ and $V_k(K, L) \leq e^{kq(k)}$.

Since $(q/S)' = -(1 + qS')/S^2 < 0$, a right-endpoint integral estimate gives

$$\sum_{i=k}^{n-1} \alpha_i \leq \frac{3}{2} \int_k^n \frac{q(s)}{S(s)} ds = \frac{3}{4} q(k)^2. \quad (6.17)$$

Also,

$$\sum_{i=k}^{n-1} \alpha_i^2 = O(T^2/k) = O(U^2) = o(T^2), \quad \sum_{i=k}^{n-1} 1/i = O(T) = o(T^2). \quad (6.18)$$

As $k/n \rightarrow 0$, the calibration implies $v(k) \rightarrow \infty$. Using (6.5) and $q(k) \leq 2T$,

$$q(k) = T - \log k + \log \mu + \log(q(k) + 1) + o(1) = T + O(\log T).$$

For the lower estimate in the last equality use $\log(q(k) + 1) \geq 0$; the upper estimate uses $q(k) \leq 2T$. Hence $q(k)/T \rightarrow 1$.

Stirling's formula and $k/(n\mu) = \chi(v(k))$ give the precise prefix cancellation

$$\begin{aligned} \log \frac{k! e^{kq(k)} (n\mu/e)^{n-k}}{n! \mu^n} &= k \log \frac{k e^{q(k)}}{n\mu} + \frac{1}{2} \log(k/n) + O(1/k) \\ &= k \log(v(k) \chi(v(k))) + \frac{1}{2} \log(k/n) + O(1/k) \\ &= O(k \log T) + O(T) = O(T^2/U) + O(T) = o(T^2). \end{aligned} \quad (6.19)$$

In particular an apparent $k \log n$ cost has canceled; it cannot be discarded before this calculation. Equations (6.16)–(6.19) prove the preliminary upper limit $3/4$.

Finally (5.2) and inclusion give

$$\log \frac{a_n}{n! \mu^n} \geq \frac{1}{2} \log n + \log C_F + O(n^{-1}),$$

so the normalized lower limit is at least zero. The next two sections sharpen both sides to one half.

7 The sharp upper logarithmic correction

The preceding $3/4$ bound can be improved to $1/2$ by a quadratic corrector. We give the finite identity and all uniform estimates, because replacing a worst-case state by a stationary average without these estimates would not be valid.

7.1 A circulant chain and an exact Poisson identity

For $q > 0$ and $M \geq 1$ set

$$r = e^{-q/M}, \quad B = \frac{r}{1-r}, \quad D = 1 - r^M.$$

The Doob transform of the frozen matrix, defined by $P(l, j) = T_{e^q}(l, j)f_j/(\lambda_M(e^q)f_l)$, is

$$P(l, j) = \frac{1-r}{1-r^M} r^{(j-l) \bmod M}. \quad (7.1)$$

It is a stochastic circulant matrix, so the uniform law on the alphabet is stationary. The forward circular increment $W = ((j-l) \bmod M)/M$ has a truncated geometric distribution independent of l . Write $a(l, j) = \mathbf{1}_{\{j \geq l\}}$ and

$$b_l = \mathbb{E}_l[q a(l, J)J/M], \quad c = \frac{1}{M} \sum_{l=0}^{M-1} b_l. \quad (7.2)$$

Stationarity gives

$$0 \leq c \leq q(M-1)/(2M) \leq q/2. \quad (7.3)$$

Direct geometric summation gives

$$\begin{aligned} b_l &= \frac{q}{MD} \left(l + B - (M+B)r^{M-l} \right), \\ c &= \frac{q}{M^2 D} \left(\frac{M(M-1)}{2} + MB - (M+B)BD \right). \end{aligned} \quad (7.4)$$

Define, also at the auxiliary endpoint $l = M$,

$$h(q, M, l) = \left(\frac{c}{B} + \frac{q}{2MDB} \right) l - \frac{ql^2}{2MDB}. \quad (7.5)$$

Then the exact finite Poisson equation is

$$(Ph)_l - h_l = c - b_l \quad (0 \leq l < M). \quad (7.6)$$

Here h_l denotes (7.5); its value at M need not equal the value at zero.

To verify (7.6) algebraically, set $y_l = h_l + c - b_l$. For $0 \leq l < M-1$, subtraction in (7.4)–(7.5) gives

$$y_{l+1} - y_l = (c - b_l)/B = (y_l - h_l)/B.$$

The same relation holds at the cyclic boundary, because the sum of the right side is zero and the sum of cyclic differences is zero. Thus $y_l - ry_{l+1} = (1-r)h_l$ with cyclic indices. If C is the forward cyclic shift, $I - rC$ is invertible and

$$y = \frac{1-r}{1-r^M} \sum_{s=0}^{M-1} r^s C^s h = Ph,$$

which proves the identity. For $M=1$ it is immediate. All formulas have continuous limits at $q=0$, with $b=c=h=0$ and P the uniform matrix.

[write] 5 October 2026. Part II restates the formulas (7.4)–(7.6) as (16.1)–(16.3) and Lemma 16.1, with the same proof, and (7.7)–(7.8) as (16.4)–(16.5); Part III uses them from there.

7.2 Uniform corrector and moment estimates

Put $Q = 1+q$, and view M as real when differentiating the displayed formulas. For $H(q, M, x) = h(q, M, Mx)$, elementary simplification gives the stable form

$$\begin{aligned} H(q, M, x) &= Ax(1-x) + J_0x, \\ A &= \frac{q^2}{2(1-e^{-q})} \beta(q/M), \quad \beta(z) = \frac{e^z - 1}{z}, \\ J_0 &= Q_0(q) - Q_0(q/M), \quad Q_0(z) = \frac{z}{e^z - 1}. \end{aligned} \quad (7.7)$$

Removable values are used at zero. Choose a sufficiently large absolute constant C_0 . Uniformly for $M \geq C_0 Q$ and $0 \leq x \leq 1$,

$$|H| + |H_x| \leq CQ^2, \quad |H_q| \leq CQ, \quad |H_M| \leq CQ^3/M^2. \quad (7.8)$$

Indeed, β and its first two derivatives are bounded on $[0, 1]$. The function $q^2/[2(1 - e^{-q})]$ is $O(Q^2)$ with derivative $O(Q)$, while Q_0 and Q'_0 are bounded on $[0, \infty)$. Differentiating (7.7) proves the estimates. One may first choose these bounds on $q/M \leq 1$ and increase C_0 afterward, so the constants do not depend circularly on the final choice of C_0 .

The seam jump is particularly small:

$$H(q, M, 1) - H(q, M, 0) = J_0 \in [-1, 0]. \quad (7.9)$$

Integrating H_x along a forward circular arc and crossing this seam at most once gives

$$|h(q, M, J) - h(q, M, l)| \leq CQ^2W + 1. \quad (7.10)$$

For each fixed integer $j \geq 0$ and fixed $A_0 > 0$, the geometric law (7.1) gives

$$\mathbb{E}[W^j \exp(A_0 Q^2 W/M)] \leq C_j Q^{-j}, \quad (7.11)$$

provided C_0 is large enough for that A_0 . For $q \geq 1$, the added exponent is at most $qW/2$, so the sum is bounded using $e^{-qW/2}$. Its normalization is $O(q/M)$ and its j th moment sum is $O(M/q^{j+1})$, uniformly for M/q bounded below. These estimates follow, for example, by comparing the geometric sums with the integrals of $x^j e^{-qx/2}$ plus the endpoint term. For $q \leq 1$, use $W \leq 1$ and boundedness of Q . The law is independent of l , so all bounds are uniform in the current letter.

For $0 \leq \zeta \leq 1$, set

$$Z = qa(l, J)J/M + \zeta[h(q, M, J) - h(q, M, l)].$$

Equation (7.6) says $\mathbb{E}_l Z = (1 - \zeta)b_l + \zeta c$. By (7.10) and (7.11),

$$\mathbb{E}_l[Z^2 e^{|Z|/M}] \leq CQ^2.$$

Apply $|e^x - 1 - x| \leq x^2 e^{|x|}/2$ and then $\log(1 + x) \leq x$ to obtain

$$\log \mathbb{E}_l e^{Z/M} \leq \frac{(1 - \zeta)b_l + \zeta c}{M} + \frac{CQ^2}{M^2}. \quad (7.12)$$

This controls the random fluctuation around the stationary mean, instead of merely substituting that mean in a pointwise expression.

7.3 The corrected moving potential and its three state ranges

Use the calibration $q(s), S(s)$ and cutoff k of Section 6. Let $\zeta(r)$ be zero for $r \leq 1/2$, equal to $4r - 2$ on $[1/2, 3/4]$, and one for $r \geq 3/4$. It is bounded by one and has Lipschitz constant four. Define

$$\tilde{V}_i(K, L) = \exp\left(q(i)K - \frac{q(i)L}{M} + \zeta\left(\frac{M}{S(i)}\right) \frac{h(q(i), M, L)}{M}\right), \quad M = K + 2. \quad (7.13)$$

The terminal potential is exactly one. For a step at time i , freeze $q = q(i + 1)$, $S = S(i + 1)$, and $r_0 = M/S$. When $M \geq 0.49S$, we have $M \geq ci$ and $i/Q \rightarrow \infty$ uniformly for $i \geq k$. Thus (7.8) applies.

Replacing the actual next corrector

$$\zeta((M+a)/S)h(q, M+a, j)/(M+a)$$

by its frozen version $\zeta(M/S)h(q, M, j)/M$ costs $O(Q^2/i^2)$ in the exponent. At fixed j , changing M changes $x = j/M$ by $O(1/M)$, so (7.8), the derivative of $1/M$, and the Lipschitz estimate for ζ give this bound. Replacing the current corrector by its frozen q, S values has the same cost: $|\Delta q| \leq 2/i$, $|\Delta S| \leq 1$, and

$$|\Delta \zeta| \leq 4M|\Delta S|/(S(i)S(i+1)).$$

These estimates also handle the corners of the piecewise linear cutoff.

The eigenvector denominator discrepancy is bounded above by qaj/M^2 . Factoring out the exact frozen eigenvector and applying (7.12), together with the eigenvalue correction $q/(2M)$, gives, if $r_0 \geq 3/4$,

$$\log \frac{\sum_j \tilde{V}_{i+1}(K+a, j)}{(n\mu/e)\tilde{V}_i(K, L)} \leq \Psi(r_0) + \frac{q}{Sr_0} + C \left(\frac{Q^2}{i^2} + \frac{1}{i} \right). \quad (7.14)$$

Here $\zeta = 1$ and $c \leq q/2$; the calibration terms are those in (6.11). In the intermediate range $0.49 \leq r_0 < 3/4$, the same argument with $b_l \leq q$, $c \leq q/2$ gives $3q/(2Sr_0)$ in place of $q/(Sr_0)$. Since $\Psi(r_0) \leq \Psi(3/4) < 0$ there and all other terms are uniformly $o(1)$, this entire range has a negative normalized logarithmic bound for large n .

If $r_0 < 0.49$, both cutoff terms vanish for large i :

$$M/S(i) \leq 0.49(1 + 2/i) < 1/2, \quad (M+a)/S(i+1) \leq 0.49 + 2/(i+1) < 1/2.$$

Use the uncorrected estimate (6.11). Put $\alpha = 3q/(2S) = o(1)$ and $m = \sqrt{2i}/S$. If $m > 0.49$ the range is empty. Otherwise $h_\alpha(r) = \Psi(r) + \alpha/r$ has no local maximum on $[m, 0.49]$: its smaller critical point is a minimum and its larger one tends to one. Its supremum is therefore at an endpoint. Equation (6.14) bounds $h_\alpha(m)$ by a quantity tending uniformly to $-\infty$; the other endpoint satisfies $h_\alpha(0.49) \rightarrow \Psi(0.49) < 0$ uniformly. Both are eventually bounded above by a fixed negative constant, absorbing the additional $6/i$. This proves negativity in the full small-state range without applying a corrector estimate where M might be too small.

Finally for $\beta = q/S = o(1)$, the same critical-point calculation as (6.13) gives

$$\sup_{r \geq 3/4} [\Psi(r) + \beta/r] \leq \beta + 2\beta^2.$$

Since $\beta^2 = O(Q^2/i^2)$, the three ranges combine into the uniform one-step bound

$$\frac{\sum_j \tilde{V}_{i+1}(K+a, j)}{\tilde{V}_i(K, L)} \leq (n\mu/e) \exp \left(\frac{q(i+1)}{S(i+1)} + C \left[\frac{(1+q(i+1))^2}{i^2} + \frac{1}{i} \right] \right). \quad (7.15)$$

7.4 Sharp upper accumulation

At time k , a nonzero cutoff implies $M > S(k)/2 \geq k/4$. Hence the added corrector exponent is $O((1+q(k))^2/k)$, and

$$\sum_{|x|=k} \tilde{V}_k \leq k! \exp \left(kq(k) + O((1+q(k))^2/k) \right).$$

The exact prefix cancellation (6.19) remains unchanged at order $o(T^2)$. Backward iteration of (7.15), using the decreasing function q/S , gives

$$\sum_{i=k}^{n-1} \frac{q(i+1)}{S(i+1)} \leq \int_k^n \frac{q}{S} ds = \frac{1}{2}q(k)^2.$$

The other errors sum to $O(T^2/k + T) = o(T^2)$, and the new prefix error is $O(U^2)$. Since $q(k)/T \rightarrow 1$, we have proved

$$\limsup_{n \rightarrow \infty} \frac{\log(a_n/(n!\mu^n))}{(\log n)^2} \leq \frac{1}{2}. \quad (7.16)$$

8 The matching lower logarithmic correction

The lower bound uses an exact change of measure on legal extension paths. The quadratic corrector controls their expected gain, while a martingale estimate keeps the alphabet size close enough to the calibration. No counting principle for presumed typical paths is needed.

8.1 A legal path measure and Jensen bound

Retain $T = \log n$, $U = \log T$, and $k = \lceil 100T^2/U^2 \rceil$. Choose the single legal prefix of k zeros; its state is $K_k = k - 1$, $L_k = 0$. For each $i = k, \dots, n - 1$, sample the next letter from the geometric Doob matrix (7.1) with $q = q(i + 1)$ and $M = K_i + 2$. All letters $0, \dots, M - 1$ are legal, and every legal completion has positive probability. Let $\mathbb{E}_*$ and $\mathbb{P}_*$ refer to this path measure, which is distinct from the uniform law on all weak ascent sequences.

Since K never decreases, $M \geq k + 1$ along every sampled path. Thus $M/(1 + q) \geq k/(1 + 2T) \rightarrow \infty$ uniformly. We may use all corrector estimates without a state cutoff. Define

$$U_i(K, L) = \exp\left(q(i)K - \frac{q(i)L}{M} + \frac{h(q(i), M, L)}{M}\right). \quad (8.1)$$

Its terminal value is one. For a step to $(K + a, j)$, put

$$R_i = \frac{U_{i+1}(K + a, j)}{U_i(K, L)P_i(L, j)}. \quad (8.2)$$

Multiplication along a path telescopes the potential ratios. Since its probability is the product of its transition probabilities, the exact number of completions of the selected prefix is

$$U_k(k - 1, 0) \mathbb{E}_* \exp\left(\sum_{i=k}^{n-1} \log R_i\right).$$

Here $h(q, M, 0) = 0$, so Jensen's inequality and inclusion of these completions among all objects give

$$\log a_n \geq (k - 1)q(k) + \sum_{i=k}^{n-1} \mathbb{E}_* \log R_i. \quad (8.3)$$

8.2 Tracking the random alphabet size

Set $M_i = K_i + 2$, $X_i = L_i/M_i$, and $Y_i = K_i - X_i$. If $W_i = ((j - L_i) \bmod M_i)/M_i$ and $a = \mathbf{1}_{\{j \geq L_i\}}$, then exactly

$$Y_{i+1} - Y_i = 1 - W_i + aX_{i+1}/M_i. \quad (8.4)$$

Indeed $j/M_i - X_i = W_i - (1 - a)$ and $j/M_i = X_{i+1}(1 + a/M_i)$. The increment in (8.4) belongs to $[0, 2]$. Its conditional mean is at least $1/2$, since the decreasing geometric weights on $0, \dots, M_i - 1$ imply $\mathbb{E}_*(W_i \mid \mathcal{F}_i) \leq 1/2$.

The centered increments therefore form a martingale with differences bounded in absolute value by two. For completeness, the elementary bounded-difference estimate follows by applying the conditional exponential bound $\mathbb{E}_*(e^{tD_i} \mid \mathcal{F}_i) \leq e^{2t^2}$ to each centered difference D_i , iterating, and optimizing exponential Markov inequality. It gives a tail at most $\exp(-z^2/(8m))$ for a

downward displacement z over m increments. Since $Y_k = k - 1$, the event $M_i < i/4$ requires a downward displacement at least $i/4 + k/2 - 1$ from the accumulated lower drift. It is empty at $i = k$. Thus for absolute $c, C > 0$,

$$\mathbb{P}_*(M_i < i/4) \leq Ce^{-ci} \quad (k \leq i \leq n) \quad (8.5)$$

for all sufficiently large n .

The exact conditional geometric mean is

$$\mathbb{E}_*(W_i | \mathcal{F}_i) = \frac{1}{M_i(e^{q/M_i} - 1)} - \frac{1}{e^q - 1}, \quad q = q(i+1). \quad (8.6)$$

Let $m(q) = 1 - 1/q + 1/(e^q - 1)$, with its removable value $1/2$ at zero, so $S'(s) = m(q(s))$. Expanding $z/(e^z - 1)$ at zero, uniformly because $M_i \geq k \gg 1 + q$, gives

$$1 - \mathbb{E}_*(W_i | \mathcal{F}_i) = m(q) + O(1/M_i).$$

The last term in (8.4) lies in $[0, 1/M_i]$. Moreover $m'(q)$ is the variance of a tilted variable in $[0, 1]$ and is bounded. Using $|q(i+1) - q(i)| \leq 2/i$ therefore gives

$$|\mathbb{E}_*(Y_{i+1} - Y_i | \mathcal{F}_i) - [S(i+1) - S(i)]| \leq C(1/M_i + 1/i). \quad (8.7)$$

On the good event $M_i \geq i/4$, $1/M_i \leq 4/i$. On its complement, $M_i \geq k$ and (8.5) applies. Minkowski's inequality in L^2 now gives

$$\left\| \sum_{i=k}^{m-1} 1/M_i \right\|_2 \leq C \log n \quad (k \leq m \leq n). \quad (8.8)$$

Explicitly, the good part is bounded deterministically by $4 \sum 1/i$, and the norm of the bad part is at most $k^{-1} \sum_{i \geq k} Ce^{-ci/2}$.

The centered martingale has second moment $O(m)$ by orthogonality of its increments. Sum (8.7), use (8.8) and $|Y_k - S(k)| \leq k$, and note that replacing Y_i by M_i or $S(i)$ by $S(i+1)$ changes the difference by a bounded amount. We obtain

$$\mathbb{E}_*[(M_i - S(i+1))^2] \leq C(i + T^2 + k^2), \quad k \leq i < n. \quad (8.9)$$

This deliberately coarse estimate is sufficient at logarithmic-square scale.

8.3 Expected gain for one step

Freeze $q = q(i+1)$, $S = S(i+1)$, $M = M_i$, $l = L_i$, and $\Delta q = q(i+1) - q(i)$. Substituting the exact Doob formula into (8.2) gives

$$\begin{aligned} \log R_i &= \log \lambda_M(e^q) + \Delta q K_i - \Delta q l / M + \frac{q a j}{M(M+a)} \\ &\quad + \frac{h(q, M+a, j)}{M+a} - \frac{h(q(i), M, l)}{M}. \end{aligned} \quad (8.10)$$

The estimates (7.8), including the change of $x = j/M$ at fixed j , show that replacing the last two corrector terms by $[h(q, M, j) - h(q, M, l)]/M$ costs at most

$$C \left(\frac{Q^2}{M^2} + \frac{Q}{Mi} \right), \quad Q = 1 + q.$$

Also $qaj/[M(M+a)] \geq qaj/M^2 - q/M^2$. The Poisson identity then implies that the conditional expectation of the last term on the first line and all of the second line in (8.10) is at least

$$\frac{c}{M} - C \left(\frac{Q^2}{M^2} + \frac{Q}{Mi} \right). \quad (8.11)$$

We need a lower estimate as well as (7.3):

$$q/2 - C(1 + q/M) \leq c \leq q/2. \quad (8.12)$$

To verify it, sample a stationary uniform starting letter. The output letter J is then uniform, and

$$q(M-1)/(2M) - c = q \mathbb{E}_{\text{stat}}[(1-a)J/M].$$

Conditional on a forward increment d , exactly d starting letters wrap, and their outputs are $0, \dots, d-1$. Hence

$$\mathbb{E}_{\text{stat}}[(1-a)J/M \mid d] = d(d-1)/(2M^2) \leq W^2/2.$$

The un-tilted case of (7.11) gives $\mathbb{E}_{\text{stat}} W^2 \leq C/(1+q)^2$, proving (8.12).

For $q/M = o(1)$ uniformly, the exact eigenvalue has the lower expansion

$$\log \lambda_M(e^q) \geq \log(n\mu) + \log(M/S) + q/(2M) - Cq^2/M^2. \quad (8.13)$$

This follows from $x/(1-e^{-x}) = e^{x/2}(x/2)/\sinh(x/2)$ and the bounded Taylor remainder of $\log(\sinh(x/2)/(x/2))$ near zero. The calibration gives

$$\Delta q K_i \geq -K_i/S - C/i, \quad -\Delta q l/M \geq 0.$$

For the first estimate, compare $1/S(s)$ on $[i, i+1]$ with $1/S(i+1)$ using $|S'| \leq 1$, $S(s) \geq i/2$, and $K_i \leq i$. Combining these bounds with (8.11)–(8.13), using $K_i = M-2$ and dropping the favorable $2/S$, yields

$$\mathbb{E}_*(\log R_i \mid \mathcal{F}_i) - \log(n\mu/e) \geq \Psi(M/S) + \frac{q}{M} - C \left(\frac{1}{i} + \frac{1}{M} + \frac{Q^2}{M^2} + \frac{Q}{Mi} \right). \quad (8.14)$$

8.4 Summing the tracking errors and normalizing

On the good event $M_i \geq i/4$, the ratio $M_i/S(i+1)$ belongs to a fixed compact interval, for example $[1/5, 3]$ for large i . Since $\Psi(1) = \Psi'(1) = 0$,

$$\Psi(M_i/S) \geq -C(M_i - S)^2/i^2.$$

On the bad event $M_i \leq k$, while the deterministic support gives $M_i \leq i+1$, so $|\Psi(M_i/S)| \leq C + T$. Equations (8.5) and (8.9) imply

$$\begin{aligned} \sum_{i=k}^{n-1} \mathbb{E}_* \Psi(M_i/S(i+1)) &\geq -C \left(T + \frac{T^2 + k^2}{k} \right) - o(1) \\ &= -O(T + U^2 + k) = -o(T^2). \end{aligned} \quad (8.15)$$

For the positive main term, on good states

$$1/M_i \geq 1/S - C|M_i - S|/i^2.$$

Drop its nonnegative contribution on bad states, bound the missing q/S there by (8.5), and use Cauchy–Schwarz and (8.9). This gives

$$\begin{aligned}
\sum_{i=k}^{n-1} \mathbb{E}_* \frac{q(i+1)}{M_i} &\geq \sum_{i=k}^{n-1} \frac{q(i+1)}{S(i+1)} - CT \sum_{i=k}^{n-1} \frac{\sqrt{i} + T + k}{i^2} - o(1) \\
&\geq \sum_{i=k}^{n-1} \frac{q(i+1)}{S(i+1)} - O(T/\sqrt{k} + T^2/k + T) \\
&= \sum_{i=k}^{n-1} \frac{q(i+1)}{S(i+1)} - o(T^2).
\end{aligned} \tag{8.16}$$

The error terms on the right side of (8.14) sum in expectation to $O(T + T^2/k) = o(T^2)$ by the same good/bad split. Since q/S decreases, its right Riemann sum differs from its integral by at most $q(k)/S(k)$. Therefore

$$\sum_{i=k}^{n-1} \frac{q(i+1)}{S(i+1)} = \frac{1}{2}q(k)^2 + o(T^2). \tag{8.17}$$

The selected prefix has no multiplicity $k!$, but the missing logarithm is small at this scale. Specifically, subtract (6.19) from the desired normalization, or apply Stirling directly, to obtain

$$\begin{aligned}
&(k-1)q(k) + (n-k) \log(n\mu/e) - \log(n!\mu^n) \\
&= O(k \log k + k \log(1 + q(k)) + T) = O(T^2/U) = o(T^2).
\end{aligned} \tag{8.18}$$

Insert (8.14)–(8.18) into Jensen’s inequality (8.3). As $q(k)/T \rightarrow 1$,

$$\liminf_{n \rightarrow \infty} \frac{\log(a_n/(n!\mu^n))}{(\log n)^2} \geq \frac{1}{2}. \tag{8.19}$$

Together with (7.16), this proves the matching leading logarithmic correction

$$\log \frac{a_n}{n!\mu^n} = \left(\frac{1}{2} + o(1)\right) (\log n)^2, \quad \text{equivalently} \quad a_n = n!\mu^n \exp\left(\left(\frac{1}{2} + o(1)\right) (\log n)^2\right). \tag{8.20}$$

The error inside the exponential can still diverge. Thus (8.20) is a logarithmic asymptotic, not a relative-error coefficient equivalent and not an amplitude formula.

8.5 Retaining a quantitative remainder

The paired bounds prove more than the little- o statement in (8.20). All constants in the preceding estimates are independent of n after fixing a sufficiently large corrector constant C_0 . Their error budget is:

- both prefix discrepancies are $O(T^2/U)$;
- $q(k) = T + O(U)$ gives $q(k)^2 - T^2 = O(TU + U^2)$;
- the upper corrector and harmonic sums are $O(U^2 + T)$;
- the lower expected slack is $O(k + T + U^2)$;
- the lower main-term replacement and remaining errors are $O(T + U^2 + U)$;
- the right Riemann-sum discrepancy is $O(T/k)$, and the bad-state tails are exponentially small in k , times at most polynomial functions of T .

With $k = O(T^2/U^2)$, each item is $O(T^2/U)$; for example $TU/(T^2/U) = U^2/T \rightarrow 0$. Eventual negativity of the small-state ranges introduces no n -dependent error constant. Thus the upper and lower arguments establish exactly (1.6), proving the counting assertion of Theorem 1.2.

[write] 5 October 2026. The two places where this budget reaches T^2/U are the prefix discrepancies, which come from the cutoff $k \asymp T^2/U^2$ and the crude prefix bound $a_k \leq k!$. Part III removes both: a capped slope $\min(q/M, p)$ controls every small state (Section 29), so the cutoff can be $k = \lceil AT \rceil$, and a run decomposition of the prefixes (Section 30) replaces $k!$; keeping the exact normalization (28.4) together then makes every error $O(T)$.

9 Every fixed difference ascent parameter

Fix an integer $d \geq 0$. A d -ascent is an index with $x_{i+1} > x_i - d$. A d -ascent sequence starts at zero and obeys

$$0 \leq x_{i+1} \leq 1 + \text{dasc}(x_1, \dots, x_i). \quad (9.1)$$

This is the convention of Dukes–Sagan [6]; $d = 0$ is ordinary ascent and $d = 1$ is weak ascent. Write $a_n^{(d)}$, $H_n^{(d)}(v)$, and $K_n^{(d)}$ for its count, tilted count, and random statistic.

9.1 The exact state and a finite band comparison

The transition is

$$(K, L) \mapsto (K + \mathbf{1}_{\{j > L-d\}}, j), \quad 0 \leq j \leq K + 1. \quad (9.2)$$

If the indicator is zero, then $j \leq L - d \leq L \leq K$; if it is one, then $j \leq K + 1$, the new count. This proves $0 \leq L \leq K \leq n - 1$ for every $d \geq 0$, and hence $a_n^{(d)} \leq n!$. Every ordinary ascent is a d -ascent, so at every prefix an ordinary ascent sequence satisfies (9.1). Therefore

$$F_n^{\text{Fish}} \leq a_n^{(d)} \leq n!. \quad (9.3)$$

The inclusion is not statistic-preserving and is used only at unweighted mark one.

On an M -letter alphabet put

$$T_{d,v}(l, j) = v^{\mathbf{1}_{\{j > l-d\}}}, \quad C_d(v) = |d - 1| |v - 1| \max(v, v^{-1}).$$

For the weak-ascent eigenvector $f_l = v^{-l/M}$, at most $|d - 1|$ entries per row differ between $T_{d,v}$ and $T_{1,v}$. Since $f_j/f_l \leq \max(v, v^{-1})$, absolute rowwise comparison gives

$$T_{d,v}f \leq (\lambda_M(v) + C_d(v))f. \quad (9.4)$$

At $d = 0$ only the diagonal changes; at $d \geq 2$ a band below the weak-ascent threshold changes. Alphabet truncation only reduces its size, so the estimate also holds when $d > M$. Taking absolute differences makes it valid for $v < 1$.

By positivity and comparison with $\mathbf{1}$,

$$T_{d,v}^b \mathbf{1} \leq \max(v, v^{-1}) (\lambda_M(v) + C_d(v))^b \mathbf{1}. \quad (9.5)$$

For a fixed d and a compact positive mark range,

$$\lambda_M(v) + C_d(v) \leq M/g(v) + O_d(1).$$

This is a direct positive supersolution estimate; no perturbation theorem for eigenvalues or eigenvectors is assumed.

9.2 Applying the calibration and deriving the conclusions

A block of b actual transitions from (K, L) embeds in the alphabet of size $K + b + 2$, with the same weights $v^{K' - K}$. Equation (9.5) bounds its total by

$$\max(v, v^{-1})(\lambda_{K+b+2}(v) + C_d(v))^b,$$

whose eigenvalue majorant is $K/g(v) + O_d(b + 1)$. The hypotheses used in Section 4 have now all been verified: bounded count increments, support $K \leq n$, a prefix bound $k!$, uniformly conditioned positive comparison on compact marks, and this last block majorant. The same curve (4.1) gives the cancellation (4.2). Repeating (4.4)–(4.8), with constants depending additionally on the fixed d , proves

$$\limsup_{n \rightarrow \infty} \frac{1}{n} \log \frac{H_n^{(d)}(v)}{n!} \leq -\log \chi(v) \quad (v > 0). \quad (9.6)$$

At $v = 1$, (9.3) supplies the matching Fishburn lower bound, hence

$$\lim_{n \rightarrow \infty} (a_n^{(d)}/n!)^{1/n} = \mu.$$

Subtracting this unweighted limit from (9.6) gives exactly the moment-generating upper bound (5.5) for $K_n^{(d)}$. The two Chernoff signs prove exponential concentration at μ . Every state still has precisely $K + 2$ children, so

$$\frac{a_{n+1}^{(d)}}{a_n^{(d)}} = \mathbb{E}(K_n^{(d)} + 2), \quad \frac{a_{n+1}^{(d)}}{(n+1)a_n^{(d)}} \rightarrow \mu.$$

This proves all fixed- d assertions of Theorem 1.1.

Why fixed means fixed. The constants in the finite-band estimate depend on d . Nothing above is uniform for $d = d(n) \rightarrow \infty$. For example, when $d \geq n$, every transition of every length- n inversion sequence is a d -ascent. The constraint then permits all inversion sequences and the count is $n!$, whose factorial-normalized root is one. Subexponential factors for different fixed d also need not agree. Sections 6–8 establish the sharper logarithmic correction only for $d = 1$.

[write] **5 October 2026.** For fixed $d \geq 1$ the subexponential factors do differ. By Part III, (24.2), for fixed integers $d, d' \geq 1$,

$$\log \frac{a_n^{(d)}}{a_n^{(d')}} = \frac{d - d'}{2} (\log n)^2 + O_{d,d'}(\log n),$$

which tends to $\pm\infty$ when $d \neq d'$. The same holds with $d' = 0$ by Remark 33.1, where the factor is $C_F \sqrt{n}$. The logarithmic-square correction is therefore $\frac{d}{2}(\log n)^2$ for every fixed $d \geq 1$, not only for $d = 1$; the statement about $d = d(n) \rightarrow \infty$ is unchanged, and Part III claims no uniformity in d either.

10 Threshold inversion with integer bracketing

All logarithms are natural. For $X > 1$, let $L = \log X$ and $N = N(X)$. Since every nonempty object has at least two children, the counts strictly increase for $n \geq 1$; the duplicate $a_0 = a_1 = 1$ is irrelevant here. Consequently

$$\log a_{N-1} < L \leq \log a_N. \quad (10.1)$$

Set

$$f(x) = x \log(\mu x/e), \quad f'(x) = \log(\mu x). \quad (10.2)$$

The large positive solution of $f(x) = L$ is

$$x = x(L) = \frac{L}{W(\mu L/e)}, \quad x \sim L/\log L. \quad (10.3)$$

Indeed, putting $w = W(\mu L/e)$ gives $we^w = \mu L/e$, so $\mu x/e = e^w$ and $f(x) = xw = L$.

10.1 Consequence of the root theorem for every fixed d

Stirling and the root theorem give

$$\log a_n^{(d)} = f(n) + o(n).$$

For the threshold $N_d(X)$, the analogue of (10.1) first implies $N_d \sim L/\log L$. Since $f(N_d) - f(N_d - 1) = O(\log N_d)$, those same inequalities give $f(N_d) - L = o(N_d)$. The mean value theorem between N_d and $x(L)$ now has derivative asymptotic to $\log L$, whence

$$N_d(X) = \frac{L}{W(\mu L/e)} + o\left(\frac{L}{(\log L)^2}\right) \quad (10.4)$$

for each fixed d . This is the inverse accuracy justified by the common root theorem alone.

[write] 5 October 2026. Superseded for every fixed d : Part III, Corollary 24.2, gives $N_d(X) = x - \frac{d}{2} \log x + O_d(1)$ for $d \geq 1$, and Remark 33.1 gives $N_0(X) = x + O(1)$, with $x = L/W(\mu L/e)$ as here (Part III's threshold is $\min\{n \geq 1 : \cdot\}$, which differs from this one only at $X \leq 1$).

10.2 The signed weak-ascent inverse

Set $g(x) = f(x) + \frac{1}{2}(\log x)^2$. Theorem 1.2 and Stirling give

$$\log a_n = g(n) + O\left(\frac{(\log n)^2}{\log \log n}\right). \quad (10.5)$$

The Stirling contribution $\frac{1}{2} \log(2\pi n) + O(1/n)$ is smaller than this error. Since $g(N) - g(N-1) = O(\log N)$, both threshold inequalities (10.1) imply

$$g(N) - L = O\left(\frac{(\log N)^2}{\log \log N}\right).$$

The root theorem first gives $N \sim x \sim L/\log L$. Put

$$y = x - \frac{(\log x)^2}{2 \log(\mu x)}. \quad (10.6)$$

This displacement is $O(\log x)$. Taylor expansion of the explicitly smooth function f , with $f'(x) = \log(\mu x)$ and $f''(x) = 1/x$, gives

$$f(y) = L - \frac{1}{2}(\log x)^2 + O((\log x)^2/x).$$

Also $(\log y)^2 - (\log x)^2 = O((\log x)^2/x)$, so $g(y) - L = O((\log x)^2/x)$. Between N and y , the derivative g' is asymptotic to $\log x$. The mean value theorem therefore gives

$$N - y = O(\log x / \log \log x),$$

proving (1.7). Only the smooth main term was differentiated; no regularity of the unknown counting remainder was assumed.

For an explicit asymptotic bracketing formulation, the error statement supplies some constants C, n_0 valid for all $n \geq n_0$. Choose any sufficiently large constant A compared with C . For all sufficiently large L , the integers

$$n_- = \left\lfloor y - A \frac{\log x}{\log \log x} \right\rfloor, \quad n_+ = \left\lceil y + A \frac{\log x}{\log \log x} \right\rceil$$

satisfy $\log a_{n_-} < L \leq \log a_{n_+}$, hence $n_- < N(X) \leq n_+$. This follows from $g' \sim \log x$ and the two-sided error in (10.5); integer rounding changes the smooth value by only $O(\log x)$.

These existence constants are not numerically certified here. To validate a threshold for a particular finite X , use the exact recurrence and integer comparisons. The leading displacement is also $-\frac{1}{2} \log L + o(\log L)$, since $\log x \sim \log L$, but (1.7) retains the more informative smooth correction and a smaller error.

[write] **5 October 2026.** Superseded: Part III, Corollary 24.2 with $c = 1$, proves $N(X) = x - \frac{(\log x)^2}{2 \log(\mu x)} + O(1) = x - \frac{1}{2} \log x + O(1)$; the integer brackets above can then be taken of bounded width. The existence constants are still not certified (Part III, question 3 of Section 37).

11 Exact verification and exploratory numerical evidence

The source archive separates reproducible finite exact checks from floating-point diagnostics and from the analytic proofs. None of these finite checks proves a limiting assertion.

11.1 Fail-closed exact checks

The supplied Python checker uses integer and rational arithmetic only. It raises explicit exceptions on every failed condition and uses no Python `assert` statements, so its checks survive optimization with `python3 -O`. It verifies:

1. all 501 stored integer terms $a_0, \dots, a_{500}$ against a fresh positive-recurrence computation, including the 25 initial displayed OEIS values recorded in the source audit;
2. the polynomial derivative recurrence (2.6) against the full weak-ascent distribution through length 16;
3. direct enumeration against the state recurrence for selected fixed d , and the exact $K + 2$ extension identity;
4. exact rational finite-matrix identities and fixed-band supersolution inequalities, taking rational r and $v = r^{-M}$ so the eigenvector and eigenvalue are rational, with samples on both sides of $v = 1$ and shifts larger than the alphabet. The corrector identity is tested after dividing b, c, h by q , leaving exact rational expressions when r is rational.

The certificate records its complete expected grid and exact rational values. Missing cases, altered fractions, malformed indices, or changed small terms are rejected. A separate corruption runner deliberately changes these inputs and requires a nonzero verifier exit in both ordinary and optimized Python. Passing a corrupted input would fail the corruption test. The source archive is also extracted into a fresh directory for a clean replay of its checks and PDF build.

The rational certificates concern only the stated finite samples. They do not certify inequalities involving transcendental calibrations, the limiting constant μ , or the all- n proofs. Those conclusions rest on Sections 3–10 and the cited Fishburn theorem.

11.2 Floating diagnostics and their interpretation

A distinct optional high-precision script samples the frozen eigenvector, the fixed-band majorant, and the moving potential inequality. These are numerical sanity checks with explicit tolerances, not exact rational certificates. It also computes the following diagnostics from the exact integer terms. Let

$$r_n = \frac{a_n}{na_{n-1}}, \quad B_n = n(r_n/\mu - 1).$$

n	r_n	B_n
20	0.734801728900261	4.17400792494
50	0.671027431273686	5.18979407459
100	0.644076903552652	5.94640403239
200	0.628272669627415	6.69342350796
300	0.622367973464320	7.12628450003
400	0.619221951142305	7.43171294968
500	0.617249929461980	7.66771836584

The theorem now establishes the limiting value $\mu = 0.607927101854\dots$ independently of these data. Over the displayed range B_n grows slowly, suggesting a logarithmic correction to the scaled ratio. A factor $\exp(\frac{1}{2}(\log n)^2)$ would contribute a leading $\log n$ term to B_n under suitably controlled smooth asymptotics. The leading logarithmic-square coefficient is now proved by Sections 7–8; the table does not prove it. A rate such as $B_n \sim \log n$ would require additional control of consecutive errors and is not implied by our logarithmic remainder. Smaller logarithmic corrections and an amplitude likewise cannot be settled by this finite table.

[write] **5 October 2026.** A further diagnostic from the shipped terms `data/105-growth-weak_ascent_terms.txt` (binary64, recomputed at the write): the residual $R_n = \log(a_n/(n!\mu^n)) - \frac{1}{2}(\log n)^2$ is 3.356, 4.197, 4.725, 5.112, 5.419 at $n = 100, 200, 300, 400, 500$, and $R_n/\log n$ is 0.73, 0.79, 0.83, 0.85, 0.87. This is consistent with Part III's remainder $O(\log n)$ and with a possible term $c' \log n$ (open, Section 37.1); it proves neither, and it does not identify c' .

12 Further questions and a refined formal starting point

12.1 The prefactor remains the main asymptotic problem

The quadratic corrector improves the preliminary coefficient $3/4$ to $1/2$, and the exact path measure supplies a matching lower bound. Their quantitative remainder is still much larger than the potential next scales $\log n \log \log n$ or $\log n$. Determining those scales and their coefficients requires further analysis. In particular, the fixed prefix cutoff deliberately sacrifices lower-order precision in return for global control of every reachable state.

A full coefficient equivalent would require an additive $o(1)$ error after specifying all divergent terms in the logarithm, together with the constant term that determines the amplitude. The present $O((\log n)^2/\log \log n)$ remainder is far from that accuracy. More refined calibration of the initial boundary and higher correctors are possible directions; their effectiveness remains to be proved.

Other precise questions include a full limit for $n^{-1} \log(H_n(v)/n!)$ away from $v = 1$, a large-deviation principle, variance asymptotics and a central limit theorem, dependence of the subexponential factor on fixed d , and crossover regimes when d grows. The upper-pressure argument alone does not settle any of them.

[write] **5 October 2026.** Status after Parts II and III. The next scale: Part III proves the remainder $O(\log n)$, which excludes a next term of exact order $\log n \log \log n$; whether there is a term $c' \log n$ is Part III's first question. The dependence on fixed d is answered for every fixed $d \geq 1$ at the

logarithmic-square scale (coefficient $\frac{d}{2}$, remainder $O_d(\log n)$); for $d = 0$ the Fishburn asymptotic gives no such term. The large-deviation principle of Part II, Theorem 13.2, concerns the number of levels, not the ascent count K_n : the full tilted pressure, a large-deviation principle, variance asymptotics and a central limit theorem for K_n remain open, as do the crossover regimes in d . See Section 12.3.

12.2 Level marking and the solvable zero-level anchor

For an additional exact starting point, let ℓ mark adjacent equalities, and let $\mathcal{F}(t, u, v, \ell)$ count nonempty weak ascent sequences, where t marks length, u weak ascents, and v the last entry. Put $\mathcal{W}_\ell(t, u) = \mathcal{F}(t, u, 1, \ell)$. Appending a new letter to (k, l) assigns weight 1 below l , ℓu at l , and u above l . Finite geometric sums therefore give

$$\mathcal{F} = t + \frac{t}{1-v} [\mathcal{W}_\ell + (\ell u - 1 + (1-\ell)uv)\mathcal{F} - uv^2\mathcal{W}_\ell(t, uv)]. \quad (12.1)$$

The numerator is divisible by $1-v$ coefficientwise. This is a direct formal derivation consistent with the level-marked framework of Mansour [9].

The formal kernel root and substitution are

$$v_\ell(t, u) = \frac{1+t-\ell ut}{1+(1-\ell)ut}, \quad \phi_\ell(t, u) = uv_\ell(t, u),$$

so kernel cancellation gives

$$\mathcal{W}_\ell(t, u) = \frac{t(1-u)}{1+(1-\ell)ut} + uv_\ell(t, u)^2 \mathcal{W}_\ell(t, \phi_\ell(t, u)). \quad (12.2)$$

At $\ell = 1$, this is equivalent to (2.5) through $G(u, t) = u + u^2\mathcal{W}_1(t, u)$. Thus it does not remove the nonlinear endpoint problem.

At $\ell = 0$, $\phi_0 = u(1+t)/(1+ut)$ is a Möbius map. If $y = 1/u$, its iterates satisfy

$$y_{j+1} = \frac{y_j + t}{1+t}, \quad y_j = 1 + \frac{y_0 - 1}{(1+t)^j}.$$

The zero-level objects are precisely ordinary ascent sequences without adjacent equal letters. If $A_F(z)$ is the ordinary Fishburn series and $P(t) = 1 + \mathcal{W}_0(t, 1)$, deleting repeated adjacent letters preserves strict ascent counts, and reinserting positive runs gives

$$A_F(z) = P\left(\frac{z}{1-z}\right), \quad P(t) = A_F\left(\frac{t}{1+t}\right). \quad (12.3)$$

These are formal identities. Differentiating the level-marked equation around $\ell = 0$ offers a perturbative route, but passage to $\ell = 1$ would require new uniform remainder estimates. For positive level weight, a repeated level itself enlarges the future alphabet, so (12.3) cannot simply be extended unchanged.

[write] 5 October 2026, attribution. The identity (12.3) is not new, and this Part states it without credit. It is the binomial transform between Fishburn and primitive Fishburn objects, Jelínek's Fact 5.1 [8] (Report 107 cites Facts 5.1 and 5.2 of the same paper, and credits Fact 5.2, the ratio $e^{-\pi^2/6}$, to Drmota). Ascent sequences with no two equal adjacent letters (*primitive* ascent sequences), and runs of equal letters in ascent sequences, were studied earlier by Dukes, Kitaev, Rémel and Steingrímsson [5]. Part II proves the identity by the same run compression, in coefficient form (20.2)–(20.3), and uses it for the zero-level atom (Section 20). The series $P(t)$ here has the coefficients P_n of Part II ($P_0 = 1$); it is not the pressure $P(q)$ of Section 5.

Final scope. The factorial exponential constant, scaled ratio, and concentration are resolved here for each fixed d . Weak ascent sequences additionally have a sharp leading logarithmic-square correction, its quantitative remainder, and a signed Lambert inverse refinement. The full coefficient asymptotic remains an open part of this project.

12.3 Further questions and research

[write] (5 October 2026.) Under Vladimir’s standing rule of 4 October 2026, this subsection collects every claim of this Part that is stated without proof, with its source, the argument sketched for it and what is missing, and records how Parts II and III have re-scoped it. Nothing in this Part was found to be wrong.

1. *A relative-error equivalent and an amplitude* (“What remains open” after Theorem 1.2; Sections 8 and 12). Sketch offered: “more refined calibration of the initial boundary and higher correctors”; an equivalent needs an additive $o(1)$ error after all divergent terms and the constant term. Missing: everything below the $O(\log n)$ now proved by Part III (its question 2).
2. *The next scale and its coefficient* (Section 12: “ $\log n \log \log n$ or $\log n$ ”). Re-scoped: Part III proves the remainder $O(\log n)$, so a term of exact order $\log n \log \log n$ is excluded. Open: whether the residual is $c' \log n + O(1)$, and the value of c' (Part III, question 1). The residuals of the [write] note in Section 11 are evidence only.
3. *The full tilted pressure* $\lim_n n^{-1} \log(H_n(v)/n!)$ for $v \neq 1$ (Theorem 1.1 gives the upper bound (1.3), which “need not be an equality away from $v = 1$ ”; “a lower tilted bound at $q \neq 0$ has not been used or proved”). Missing: any lower bound at $v \neq 1$; the Fishburn inclusion used at $v = 1$ does not preserve the statistic.
4. *A large-deviation principle, variance asymptotics and a central limit theorem for the ascent count K_n* (Section 12). Not addressed by Parts II or III, whose probability law concerns the number of levels, a different statistic. Missing: the tilted pressure of item 3 and a fluctuation-scale analysis.
5. *Dependence on d* (Section 9, “Why fixed means fixed”, and Section 12). Answered for every fixed $d \geq 1$ at the logarithmic-square scale by Part III (and for $d = 0$ by Remark 33.1). Open: crossover regimes for $d = d(n) \rightarrow \infty$, between the fixed- d law and the factorial count $n!$ reached at $d \geq n$ (Part III, question 5).
6. *The Borel transform* (Section 2). The radius $\pi^2/6$ of $B(1, t)$ follows from the root theorem; its singular type at $t = \pi^2/6$ and an analytic continuation of the formal identity (2.8) are not established. Missing: an analytic, not only formal, treatment of (2.5).
7. *The perturbative route from the zero-level anchor* (Section 12). Sketch: differentiate the level-marked equation (12.2) around $\ell = 0$, where (12.3) solves it. Missing: uniform remainder estimates for passage to $\ell = 1$. Parts II and III reach every positive level weight by a probabilistic route instead; the generating-function route remains unexplored.
8. $B_n \sim \log n$ for the ratio diagnostic of Section 11, suggested by the table and by smooth heuristics for a factor $\exp(\frac{1}{2}(\log n)^2)$. Missing: control of consecutive differences of the remainder, which neither this Part nor Part III provides.
9. *Effective constants* for the inverse brackets of Section 10 (“not numerically certified”). Part III, question 3.

Part II

Weighted weak ascent levels and their probability law

Bundle Report 107, archive `A336070_Weighted_Levels_and_Large_Deviations_Source.zip`, delivered as `report107.tex` (18 pages), dated 2 October 2026. Delivered title “Weighted Weak Ascent Levels and Their Probability Law: Logarithmic square growth and large deviations”; author line “Research report 107 / Prepared for Vladimir”. Printed as delivered, with **[write]** notes. Delivered Section k is Section $k + 12$ here, and its statement or equation $k.j$ is $(k + 12).j$. The delivered “report 105” is Part I. Section 23 is added in the write.

Reading conventions of Part II	Elsewhere in the report
$H_n(w)$: w marks levels; $v = e^q$ an auxiliary ascent mark	Part I: $H_n(v)$, v marks weak ascents
V, V_i : circular increment; W_0 : Lambert	Part I: W, W_i and W
$\alpha(q) = S'$	Part I: $\alpha = 3q/(2S)$, and $m(q)$ for S'
P_n : primitive ascent sequences; F_n : Fishburn numbers	Part I: F_n^{Fish} ; P a pressure or kernel
$R = \pi^2/6$ in Section 20; $R = M/S$ elsewhere	Part I: R_i likelihood ratio
$b_n = (\log n)^2, U_n = \log \log n$	Part I: b block length; $U = \log T$

Abstract of this Part (as delivered). For a weak ascent sequence x , let $E(x)$ count equal adjacent pairs and let $H_n(w) = \sum_{|x|=n} w^{E(x)}$. For fixed positive w , we prove

$$\log H_n(w) = \log(n!) + n \log(6/\pi^2) + \frac{w}{2} (\log n)^2 + O_w \left(\frac{(\log n)^2}{\log \log n} \right),$$

with uniform constants on each compact positive weight interval. An exact positive weighted kernel, a quadratic corrector, and a tracked path change of measure give both bounds. The calibration and corrector are proved here, so this report has no dependency on report 105.

Under weight w , the number of levels divided by $(\log n)^2$ converges to $w/2$ in every fixed L^p , and satisfies a full large-deviation principle with rate $x \log(2x/w) - x + w/2$ for $x \geq 0$. We also give a quantitative mean estimate, the exact zero-level logarithmic rate, and a signed threshold-inverse correction. The zero-level enumeration uses the established primitive Fishburn ratio, due to Drmota and recorded by Jelínek; that ratio is not a new result. No total-variation Poisson approximation, central limit theorem, local limit theorem, variance equivalent, amplitude, or full transseries is asserted.

13 Definitions and principal results

A weak ascent of $x = (x_1, \dots, x_n)$ is an index $i < n$ with $x_i \leq x_{i+1}$. A weak ascent sequence starts at $x_1 = 0$ and obeys

$$0 \leq x_{i+1} \leq 1 + \text{wasc}(x_1, \dots, x_i), \quad 1 \leq i < n. \quad (13.1)$$

A *level* is an equal adjacent pair. Put

$$E(x) = \#\{i < n : x_i = x_{i+1}\}, \quad H_n(w) = \sum_{|x|=n} w^{E(x)}, \quad \mu = \frac{6}{\pi^2}.$$

The empty sequence has weight one, so $H_0(w) = H_1(w) = 1$. The present w marks levels only. The auxiliary mark $v = e^q$ appearing in the proof marks every weak ascent and is a different variable.

Theorem 13.1 (Weighted logarithmic asymptotic). *For every compact interval $\mathcal{W} = [w_-, w_+] \subset (0, \infty)$ there are $C_{\mathcal{W}}, n_{\mathcal{W}} > 0$ such that, for $w \in \mathcal{W}$ and $n \geq n_{\mathcal{W}}$,*

$$\left| \log H_n(w) - \log(n!) - n \log \mu - \frac{w}{2} (\log n)^2 \right| \leq C_{\mathcal{W}} \frac{(\log n)^2}{\log \log n}. \quad (13.2)$$

[write] **5 October 2026.** At $w = 1$, $H_n(1) = a_n$ and this is the counting assertion of Part I, Theorem 1.2, with the same remainder; the proof below is a second, self-contained route. Superseded in its remainder: Part III, Theorem 24.1, proves (13.2) with $C_{\mathcal{W}} \log n$ in place of $C_{\mathcal{W}} (\log n)^2 / \log \log n$, with the same compact uniformity.

For $w > 0$, let $\mathbb{P}_{n,w}$ assign x probability $w^{E(x)} / H_n(w)$, and write $E_n = E(x)$ and $b_n = (\log n)^2$. At $w = 1$ this is the uniform law.

Theorem 13.2 (Level law). *For each fixed $w > 0$ and $t \in \mathbb{R}$,*

$$\lim_{n \rightarrow \infty} \frac{1}{b_n} \log \mathbb{E}_{n,w} e^{tE_n} = \Lambda_w(t) := \frac{w}{2} (e^t - 1). \quad (13.3)$$

Moreover $E_n/b_n \rightarrow w/2$ in probability and in every fixed L^p , $1 \leq p < \infty$; all fixed positive moments converge to those of the constant $w/2$. Uniformly for w in a compact positive interval,

$$\mathbb{E}_{n,w} E_n = \frac{w}{2} b_n + O_{\mathcal{W}} \left(\frac{b_n}{\sqrt{\log \log n}} \right). \quad (13.4)$$

The variables E_n/b_n obey a full large-deviation principle on $\mathbb{R}$ at speed b_n , with good rate function

$$I_w(x) = \begin{cases} x \log(2x/w) - x + w/2, & x \geq 0, \\ +\infty, & x < 0, \end{cases} \quad (13.5)$$

Here $0 \log 0 = 0$. For each $\eta > 0$, some $c_{w,\eta} > 0$ gives, for all sufficiently large n ,

$$\mathbb{P}_{n,w}(|E_n/b_n - w/2| \geq \eta) \leq 2e^{-c_{w,\eta} b_n}. \quad (13.6)$$

[write] **5 October 2026.** The mean estimate (13.4) is superseded by Part III, Corollary 34.1: $\mathbb{E}_{n,w} E_n = \frac{w}{2} b_n + O_{\mathcal{W}}((\log n)^{3/2})$, with absolute central moments $O((\log n)^{3p/2})$ and variance $O((\log n)^3)$. The remainder of the theorem stands as stated; with Part III's remainder the error in (19.2) becomes $O(1/\log n)$.

Theorem 13.3 (Exact zero-level rate). *Let P_n be the number of primitive ordinary ascent sequences, meaning ordinary ascent sequences with no adjacent repetitions. Then $H_n(0) = P_n$, and for fixed $w > 0$,*

$$\mathbb{P}_{n,w}(E_n = 0) = \frac{P_n}{H_n(w)}, \quad \log \mathbb{P}_{n,w}(E_n = 0) = -\frac{w}{2} b_n + O_{\mathcal{W}} \left(\frac{b_n}{\log \log n} \right). \quad (13.7)$$

The error is compact-uniform for positive w . In particular the exact atom has logarithmic rate $-I_w(0) = -w/2$.

[write] **5 October 2026.** With Part III's remainder the error in (13.7) becomes $O_{\mathcal{W}}(\log n)$, compact-uniformly; neither manuscript states this. The deduction is [write] Remark 20.1 at the end of Section 20.

Theorem 13.4 (Threshold inverse). *For $w > 0$, define $N_w(X) = \min\{n \geq 0 : H_n(w) \geq X\}$. For $L = \log X$ and the principal Lambert function W_0 , put*

$$x = \frac{L}{W_0(\mu L/e)}.$$

As $X \rightarrow \infty$,

$$N_w(X) = x - \frac{w(\log x)^2}{2\log(\mu x)} + O_{\mathcal{W}}\left(\frac{\log x}{\log \log x}\right). \quad (13.8)$$

This holds uniformly for w in a fixed compact positive interval. For a positive integer w , $H_n(w)$ is an integer count obtained by coloring each level independently in w colors; for real $w > 0$ it is a weighted threshold.

[write] **5 October 2026.** Superseded: Part III, Corollary 24.2 with $c = w$, replaces the error $O_{\mathcal{W}}(\log x / \log \log x)$ by $O_{\mathcal{W}}(1)$, compact-uniformly (with the threshold $\min\{n \geq 1 : \cdot\}$, which differs from N_w only for $X \leq 1$). The non-claim “not a bounded-error inverse” at the end of Section 21 is superseded with it.

Scope. Theorem 13.1 is a logarithmic asymptotic, not a relative-error coefficient equivalent. The limiting log-MGF has the Poisson form, but its error is too large to prove total-variation Poisson approximation, mod-Poisson convergence, a central or local limit theorem, a sharp variance scale, or sharp tail prefactors. We do not differentiate its unknown remainder. Neither the theorem nor the probability law is uniform as w tends to zero with n . The exact atom in Theorem 13.3 uses a separate argument; an LDP alone would not supply its lower bound.

Literature and independence. Mansour [9] studies this same level statistic by exact generating functions. His Section 2.1 and Theorem 1 provide enumerative formulas and finite coefficients, not the probability limits proved here. Jelínek’s Fact 5.2 [8], credited there to Drmota, already gives the primitive/all Fishburn ratio $\exp(-\pi^2/6)$. We explicitly use and rederive that elementary consequence of the known Fishburn estimate in Section 20. A bounded primary-source check through 2 October 2026 found no competing level-law theorem; no exhaustive priority claim is intended. This is a separate continuation of report 105. Every calibration and corrector fact needed for the new weighted argument is established below, without importing that report or modifying it.

[write] **5 October 2026.** “Report 105” is Part I. This Part re-derives Part I’s calibration ((15.2) is (6.2)), frozen eigenvector, corrector and Poisson identity (Lemma 16.1 is (7.6)) and reachability bound; these repetitions are kept, because Part III cites them from here.

14 Exact state transfer and monotonicity

At length i , write K for the number of weak ascents, l for the last letter, and $M = K + 2$. The allowed next letter is $j \in \{0, \dots, M - 1\}$, and

$$a = \mathbf{1}_{\{j \geq l\}}, \quad e = \mathbf{1}_{\{j=l\}}, \quad (K, l) \mapsto (K + a, j), \quad \text{weight multiplier } w^e. \quad (14.1)$$

Induction gives $0 \leq l \leq K \leq i - 1$. Thus each sequence is an inversion sequence and there are at most $i!$ unweighted prefixes. Let $A_i(K, l; w)$ be the total weight in that state. Starting with $A_1(0, 0; w) = 1$, the exact positive recurrence adds

$$A_{i+1}(K + a, j; w) += w^{\mathbf{1}_{\{j=l\}}} A_i(K, l; w) \quad (0 \leq j \leq K + 1).$$

Summing all states yields $H_i(w)$, a polynomial with nonnegative integer coefficients. The first polynomials are

$$\begin{aligned} H_1(w) &= 1, & H_2(w) &= 1 + w, \\ H_3(w) &= 2 + 3w + w^2, & H_4(w) &= 5 + 11w + 6w^2 + w^3. \end{aligned}$$

Exactly one of the $K + 2$ children is a level, so

$$\frac{H_{i+1}(w)}{H_i(w)} = \mathbb{E}_{i,w}(K + 1 + w) \geq 1 + w > 1 \quad (i \geq 1). \quad (14.2)$$

This proves strict increase for every real $w > 0$ beyond the initial equality $H_0 = H_1$. In particular $H_i(w) \geq (1 + w)^{i-1}$, and every sufficiently large threshold is attained.

We will also need a deterministic lower bound on reachable M . Split a sequence into strictly decreasing runs, with a new run beginning at each weak ascent. The run beginning after the r th weak ascent starts with a letter at most r , so has at most $r + 1$ entries. Therefore

$$i \leq \frac{(K + 1)(K + 2)}{2}, \quad M \geq \sqrt{2i}. \quad (14.3)$$

15 Calibration and the frozen weighted kernel

15.1 The calibrated trajectory

Fix the terminal length n . Define

$$g(v) = \frac{\log v}{v - 1}, \quad \chi(v) = \int_v^\infty \frac{\log t}{t(t - 1)} dt, \quad (15.1)$$

using removable values at $v = 1$. Since $\chi(1) = \sum_{m \geq 1} m^{-2} = \pi^2/6$, the equations

$$s = n\mu\chi(e^{q(s)}), \quad S(s) = n\mu g(e^{q(s)}) \quad (15.2)$$

define $q(s) \geq 0$ and $S(s) > 0$ for $0 < s \leq n$. Equivalently,

$$\frac{s}{n\mu} = \int_{q(s)}^\infty \frac{u}{e^u - 1} du.$$

Differentiation gives

$$q' = -\frac{1}{S}, \quad S' = \alpha(q) := \frac{e^q}{e^q - 1} - \frac{1}{q}, \quad q(n) = 0, \quad S(0+) = 0. \quad (15.3)$$

The function $\alpha(q)$ is the mean of $x \in [0, 1]$ under a density proportional to e^{qx} . Hence $1/2 \leq \alpha \leq 1$, and its derivative, a variance, is bounded. It follows that

$$\frac{s}{2} \leq S(s) \leq s, \quad 0 \leq q(s) \leq 2 \log(n/s). \quad (15.4)$$

Write $q_i = q(i)$ and $S_i = S(i)$. With $\Delta q = q_{i+1} - q_i$,

$$\Delta q \leq -\frac{1}{S_{i+1}}, \quad |\Delta q| \leq \frac{2}{i}, \quad 0 \leq S_{i+1} - S_i \leq 1. \quad (15.5)$$

Moreover

$$(q/S)' = -\frac{1 + q\alpha(q)}{S^2} \leq 0, \quad \int_k^n \frac{q(s)}{S(s)} ds = \frac{1}{2} q_k^2. \quad (15.6)$$

Expanding $(e^u - 1)^{-1} = e^{-u} + O(e^{-2u})$ in the tail integral gives

$$\chi(e^q) = (q + 1)e^{-q} + O((q + 1)e^{-2q}) \quad (q \rightarrow \infty). \quad (15.7)$$

15.2 Exact Perron and Doob formulas

Freeze $q \geq 0$, $v = e^q$, and integer $M \geq 1$. For $0 \leq l, j < M$, put

$$T(l, j) = v^{\mathbf{1}_{\{j \geq l\}}}, \quad f_l = e^{-ql/M}.$$

At $q > 0$, write $r = e^{-q/M}$. Geometric summation gives

$$Tf = \lambda f, \quad \lambda = \frac{v-1}{1-r}. \quad (15.8)$$

At $q = 0$, $f = 1$ and $\lambda = M$. The equality weight changes just the diagonal:

$$T_w = T + (w-1)vI, \quad T_w f = \lambda_w f, \quad \lambda_w = \lambda + (w-1)v > 0. \quad (15.9)$$

Let $P(l, j) = T(l, j)f_j/(\lambda f_l)$ and $P_w(l, j) = T_w(l, j)f_j/(\lambda_w f_l)$. Then

$$P(l, j) = \frac{1-r}{1-r^M} r^{(j-l) \bmod M}. \quad (15.10)$$

Both are circulant stochastic kernels, with uniform stationary law. Define

$$t = \frac{v}{\lambda} = \frac{1-e^{-q/M}}{1-e^{-q}}, \quad D_w = 1 + (w-1)t, \quad \delta = \frac{(w-1)t}{D_w}. \quad (15.11)$$

The exact identity is

$$P_w = (1-\delta)P + \delta I. \quad (15.12)$$

When $w < 1$, $\delta < 0$: this is an algebraic identity, *not* a probability mixture. The actual off-diagonal entries are $P(l, j)/D_w$, and the diagonal entries are wt/D_w . They are strictly positive. At $q = 0$ all formulas use the exact limits

$$\lambda_w = M + w - 1, \quad t = 1/M, \quad \delta = \frac{w-1}{M+w-1}, \quad P_w(l, j) = \frac{w^{\mathbf{1}_{\{j=l\}}}}{M+w-1}. \quad (15.13)$$

Throughout the proof $w \in \mathcal{W} = [w_-, w_+] \subset (0, \infty)$ and constants indexed by $\mathcal{W}$ are uniform there. Put $Q = 1 + q$. Globally,

$$0 < t \leq \min(1, Q/M), \quad D_w \geq \min(1, w_-), \quad P_w(l, j) \leq \frac{\max(1, w_+)}{\min(1, w_-)} P(l, j). \quad (15.14)$$

Indeed $1 - e^{-q/M} \leq q/M$ and $q/(1 - e^{-q}) = q + q/(e^q - 1) \leq q + 1$; also $t \leq 1$ directly. For $M \geq C_{\mathcal{W}}Q$ with sufficiently large constant,

$$t = q/M + O(1/M + Q^2/M^2), \quad (15.15)$$

$$\log(\lambda_w/\lambda) = (w-1)q/M + O_{\mathcal{W}}(1/M + Q^2/M^2), \quad |\delta| \leq C_{\mathcal{W}}Q/M. \quad (15.16)$$

To see the uniformity at zero, let $Q_0(z) = z/(e^z - 1)$ and $\gamma(z) = (1 - e^{-z})/z$, both with removable values. Then $t = (q + Q_0(q))\gamma(q/M)/M$, with $0 \leq Q_0 \leq 1$ and $\gamma(z) = 1 + O(z)$. Taylor-expand $\log(1 + (w-1)t)$ only after $|w-1|t \leq 1/2$ is assured. The $O(1/M)$ error is necessary at $q = 0$. For small states, where such expansion may fail, the global one-sided estimate remains valid:

$$\log(\lambda_w/\lambda) \leq (w_+ - 1)_+ Q/M, \quad (15.17)$$

by $\log(1+x) \leq x$ for $w > 1$, and by its nonpositive sign for $w \leq 1$.

16 The exact quadratic corrector

16.1 Finite Poisson identity and stable formula

Initially take $q > 0$. Under $P(l, \cdot)$ let the next index be J , and write

$$b_l = \mathbb{E}_P[q \mathbf{1}_{\{J \geq l\}} J/M], \quad c = \frac{1}{M} \sum_{l=0}^{M-1} b_l, \quad B = \frac{r}{1-r}, \quad D_0 = 1 - r^M.$$

Geometric summation gives

$$b_l = \frac{q}{MD_0} \{l + B - (M + B)r^{M-l}\}, \quad (16.1)$$

$$c = \frac{q}{M^2 D_0} \{M(M-1)/2 + MB - (M+B)BD_0\}. \quad (16.2)$$

Define the polynomial, including its endpoint at $l = M$,

$$h_l = \left(\frac{c}{B} + \frac{q}{2MD_0 B} \right) l - \frac{ql^2}{2MD_0 B}. \quad (16.3)$$

Lemma 16.1. *On $0 \leq l < M$, $(P - I)h = c - b$. Also $h_0 = 0$.*

Proof. Set $y_l = h_l + c - b_l$. Direct subtraction in (16.1)–(16.3) gives $y_{l+1} - y_l = (y_l - h_l)/B$ for $l < M-1$. Since $\sum_l (c - b_l) = 0$, summing the differences shows the same equation at the cyclic boundary. Thus $y_l - ry_{l+1} = (1-r)h_l$ with cyclic indices. If C denotes forward cyclic shift, inversion of $I - rC$ gives

$$y = \frac{1-r}{1-r^M} \sum_{d=0}^{M-1} r^d C^d h = Ph.$$

This proves the claim. At $q = 0$, set $h = b = c = 0$; all identities extend continuously. $\square$

For estimates set $H(q, M, x) = h(q, M, Mx)$, allowing real M in derivatives. Algebra in (16.3) gives the stable representation

$$H = Ax(1-x) + J_0x, \quad A = \frac{q^2}{2(1-e^{-q})} \beta(q/M), \quad J_0 = Q_0(q) - Q_0(q/M), \quad \beta(z) = \frac{e^z - 1}{z}. \quad (16.4)$$

All values at zero are removable, and $H(0, M, x) = 0$. If $M \geq C_0 Q$ for a sufficiently large absolute C_0 , then uniformly in $0 \leq x \leq 1$,

$$|H| + |H_x| \leq CQ^2, \quad |H_q| \leq CQ, \quad |H_M| \leq CQ^3/M^2, \quad -1 \leq J_0 \leq 0. \quad (16.5)$$

Indeed β and its derivatives are bounded for $0 \leq q/M \leq 1$; the factor $q^2/[2(1-e^{-q})]$ is $O(Q^2)$ with derivative $O(Q)$, including its expansion $q/2 + O(q^2)$ at zero. The functions Q_0, Q'_0 are bounded on $[0, \infty)$ and Q_0 decreases from one to zero. Differentiating (16.4) now proves the bounds. The last inequality controls exactly the seam jump $H(1) - H(0)$.

16.2 Geometric moments and the corrector constant

Let $V = ((J - l) \bmod M)/M$. Its distribution under P is independent of l , with weights proportional to r^{MV} . In particular

$$\mathbb{E}_P V = \frac{1}{M(e^{q/M} - 1)} - \frac{1}{e^q - 1}. \quad (16.6)$$

Nonincreasing weights give $0 \leq \mathbb{E}_P V \leq 1/2$; the displayed formula gives $\mathbb{E}_P V \leq 1/q$ when $q > 0$. Hence $\mathbb{E}_P V = O(Q^{-1})$, including at zero. For every fixed integer $j \geq 0$ and constant $A_0 > 0$, enlarging C_0 in terms of A_0 gives

$$\mathbb{E}_P[V^j \exp(A_0 Q^2 V/M)] \leq C_j Q^{-j}. \quad (16.7)$$

For $q \geq 1$, arrange $A_0 Q^2/M \leq q/2$. The tilted geometric weights are bounded by $\exp(-qV/2)$, their normalization is $O(q/M)$, and the j th moment sum is $O(M/q^{j+1})$. These follow either by a geometric sum or comparison with its decreasing exponential integral, uniformly when M/q is bounded below. For $q \leq 1$, use $V \leq 1$ and bounded Q . By the positive entrywise bound (15.14), (16.7) holds under P_w too, with compact-uniform constants. This does not use (15.12) as a positive mixture.

Uniform stationarity yields

$$0 \leq c \leq \frac{q(M-1)}{2M} \leq \frac{q}{2}, \quad c \geq \frac{q}{2} - C(1 + q/M). \quad (16.8)$$

For the lower estimate, compare c with $q\mathbb{E}(J/M) = q(M-1)/(2M)$ under a uniform starting state. The deficit is $q\mathbb{E}[(1-a)J/M]$. Conditional on forward integer increment d , wrapping starts produce outputs $0, \dots, d-1$, so that conditional expectation equals $d(d-1)/(2M^2) \leq V^2/2$. Equation (16.7) gives the required bound. Stationarity is used to compute c , never as a claim about the evolving last letter.

16.3 Weighted first moment and exponential estimate

Integrating H_x around a forward circular arc, using the seam jump at most once, gives

$$|h_J - h_l| \leq CQ^2V + 1. \quad (16.9)$$

For $0 \leq \zeta \leq 1$, put

$$Z = q\mathbf{1}_{\{J \geq l\}}J/M + \zeta(h_J - h_l). \quad (16.10)$$

Lemma 16.1 gives $\mathbb{E}_P Z = (1 - \zeta)b_l + \zeta c \in [0, q]$. On the diagonal $Z = ql/M$, so the algebraic identity (15.12) gives exactly

$$\mathbb{E}_{P_w} Z = \mathbb{E}_P Z + \delta(ql/M - \mathbb{E}_P Z) = (1 - \zeta)b_l + \zeta c + O_{\mathcal{W}}(Q^2/M). \quad (16.11)$$

For the uncut case $\zeta = 1$, this reads $\mathbb{E}_{P_w} Z = c + \delta(ql/M - c)$. Also (16.9) and (16.7), for $j = 0, 2$, imply

$$\mathbb{E}_{P_w}[Z^2 e^{|Z|/M}] \leq C_{\mathcal{W}} Q^2. \quad (16.12)$$

Here $|Z| \leq CQ + CQ^2V$; the two moment estimates control each term after squaring. The exponential Taylor remainder and $\log(1+x) \leq x$ therefore prove

$$\log \mathbb{E}_{P_w} e^{Z/M} \leq \frac{(1 - \zeta)b_l + \zeta c}{M} + C_{\mathcal{W}} \frac{Q^2}{M^2}. \quad (16.13)$$

This is a genuine fluctuation bound, not merely a stationary average. At $q = 0$, $Z = 0$ exactly.

17 The compact uniform upper bound

17.1 Moving cutoff and regular states

Write $T = \log n$ and $U = \log T$. Set

$$D_* = \frac{3}{2} + (w_+ - 1)_+, \quad B_* = \max(100, 72D_*^2), \quad k = \left\lceil B_* T^2 / U^2 \right\rceil. \quad (17.1)$$

We henceforth take n sufficiently large that $U > 0$ and $k < n$. The dependence of B_* on the weight interval is important for small states. Let $\zeta(r) = 0$ for $r \leq 1/2$, $\zeta(r) = 4r - 2$ for $1/2 \leq r \leq 3/4$, and $\zeta(r) = 1$ for $r \geq 3/4$. Define

$$F_i(K, l) = \exp(q_i K - q_i l/M + \zeta(M/S_i)h(q_i, M, l)/M), \quad M = K + 2. \quad (17.2)$$

At the terminal time, $F_n = 1$ exactly. For $i \geq k$, abbreviate $q = q_{i+1}$, $S = S_{i+1}$, $Q = 1 + q$, $R = M/S$, and $\Delta q = q_{i+1} - q_i$. First suppose $R \geq 0.49$. Then $M \geq 0.245(i + 1)$, while $i/Q \rightarrow \infty$ uniformly for $i \geq k$. All corrector estimates apply throughout the parameter interpolations below.

Replacing the actual next-state correction $\zeta((M + a)/S)h(q, M + a, j)/(M + a)$ by $\zeta(M/S)h(q, M, j)/M$ costs $O_{\mathcal{W}}(Q^2/i^2)$. To verify this, differentiating $H(q, M, j/M)/M$ at fixed j gives $O(Q^2/M^2 + Q^3/M^3)$, and the cutoff change costs $O(Q^2/(SM))$. Replacing the current correction by its value at q, S costs the same order: use $|\Delta q| \leq 2/i$, $H_q = O(Q)$, and cutoff change at most $4M/(S_i S)$. The corners of ζ cause no problem because only its Lipschitz bound is used. The remaining eigenvector-denominator difference satisfies

$$-\frac{qj}{M+a} + \frac{qj}{M} = \frac{qaj}{M(M+a)} \leq \frac{qaj}{M^2}. \quad (17.3)$$

Factoring the exact weighted eigenvector therefore gives the pointwise estimate

$$\sum_j w^{\mathbf{1}_{\{j=l\}}} \frac{F_{i+1}(K+a, j)}{F_i(K, l)} \leq \lambda_w \exp(\Delta q K - \Delta q l/M + C_{\mathcal{W}} Q^2/i^2) \mathbb{E}_{P_w} e^{Z/M}, \quad (17.4)$$

with Z from (16.10) and $\zeta = \zeta(M/S)$. The inequality $2 \sinh(z/2) \geq z$ yields, for all $q \geq 0$,

$$\lambda \leq \frac{M}{g(e^q)} \exp(q/(2M)). \quad (17.5)$$

Put $\Psi(R) = \log R - R + 1$. For $R \geq 3/4$ the cutoff equals one. The leading contributions in (17.4) are $q/(2M)$ from (17.5), $(w-1)q/M$ from (15.16), and $c/M \leq q/(2M)$ from (16.13). Their sum is at most wq/M , even if $w < 1/2$. The calibration contribution is exactly

$$\log(M/g) - K/S - \log(n\mu/e) = \Psi(R) + 2/S,$$

and $-\Delta q l/M \leq 2/i$. Thus

$$\log \left\{ \frac{\sum_j w^{\mathbf{1}_{\{j=l\}}} F_{i+1}/F_i}{n\mu/e} \right\} \leq \Psi(R) + \frac{wq}{SR} + C_{\mathcal{W}}(Q^2/i^2 + 1/i). \quad (17.6)$$

For $0.49 \leq R < 3/4$, use $(1-\zeta)bl + \zeta c \leq q$ instead. The bound is

$$\Psi(R) + C_{\mathcal{W}}(Q/i + Q^2/i^2 + 1/i). \quad (17.7)$$

Since $\Psi(R) \leq \Psi(3/4) < 0$, this is negative throughout the intermediate range for all sufficiently large n , uniformly on $\mathcal{W}$. For the full-corrector range, let $\beta_* = wq/S = o(1)$. When $0 \leq \beta_* \leq 1/8$,

$$\sup_{R \geq 3/4} \{\Psi(R) + \beta_*/R\} \leq \beta_* + 2\beta_*^2. \quad (17.8)$$

Indeed the derivative has numerator $R - R^2 - \beta_*$. The maximum is at $r_+ = (1 + \sqrt{1 - 4\beta_*})/2 \in (3/4, 1]$; the maximum value has derivative $1/r_+ \leq 1 + 4\beta_*$. Integrate from zero. The added quadratic term is $O_{\mathcal{W}}(Q^2/i^2)$.

17.2 All small states

Suppose $R < 0.49$. For sufficiently large i , both correction cutoffs vanish:

$$M/S_i \leq 0.49(1 + 2/i) < 1/2, \quad (M + a)/S \leq 0.49 + 2/(i + 1) < 1/2. \quad (17.9)$$

No derivative estimate or expansion in q/M will be used here. Apply the crude discrepancy bound $qaj/M^2 \leq q/M$, the exact eigenvector, (15.17), and (17.5). This gives globally in the small range

$$\log \left\{ \frac{\sum_j w^{1_{\{j=l\}}} F_{i+1}/F_i}{n\mu/e} \right\} \leq \Psi(R) + \frac{D_*Q}{SR} + 6/i. \quad (17.10)$$

Set $\alpha_* = D_*Q/S$ and $m = \sqrt{2i}/S$. Reachability (14.3) gives $R \geq m$; also $\alpha_* \leq 6D_*T/i = o(1)$ uniformly for $i \geq k$. If $[m, 0.49]$ is nonempty, the function $h_{\alpha_*}(r) = \Psi(r) + \alpha_*/r$ has its maximum there at an endpoint for large n . Its derivative has numerator $r - r^2 - \alpha_*$; the smaller critical point is a minimum and the larger tends to one, outside this interval. At the reachable endpoint,

$$h_{\alpha_*}(m) \leq -\frac{1}{2} \log i + 1 + \log(2\sqrt{2}) + \frac{3D_*T}{\sqrt{2i}}. \quad (17.11)$$

Here use $S \geq i/2$ and $Q \leq 3T$, discarding the favorable $-m$. The right side decreases with i . By (17.1), at $i = k$ it is at most

$$-U + \log U + O_{\mathcal{W}}(1) + U/4 = -3U/4 + \log U + O_{\mathcal{W}}(1) \rightarrow -\infty.$$

At the other endpoint, $h_{\alpha_*}(0.49) = \Psi(0.49) + \alpha_*/0.49$ tends uniformly to a fixed negative number. Both endpoint bounds absorb $6/i$ in (17.10). If $m > 0.49$ the interval is empty. This settles every reachable small state, including M much smaller than q . Combining the three ranges with (17.8) proves the global one-step inequality

$$\sum_j w^{1_{\{j=l\}}} F_{i+1}(K + a, j) \leq F_i(K, l) \frac{n\mu}{e} \exp \left\{ \frac{wq_{i+1}}{S_{i+1}} + C_{\mathcal{W}} \left(\frac{Q_{i+1}^2}{i^2} + \frac{1}{i} \right) \right\}. \quad (17.12)$$

17.3 Prefix and telescope

At length k , a nonzero cutoff forces $M > S_k/2 \geq k/4$, so the absolute corrector exponent is $O(Q_k^2/k)$. There are at most $k!$ prefixes, each with at most $k - 1$ levels, and $K \leq k - 1$. Thus

$$\sum_{|x|=k} w^{E(x)} F_k(K, l) \leq k! \exp \{ kq_k + k \log \max(1, w_+) + CQ_k^2/k \}. \quad (17.13)$$

The corrector estimates are valid when the cutoff is nonzero; when it is zero there is no correction to bound. Positivity and $F_n = 1$ permit backward iteration of (17.12). Since q/S decreases,

$$\sum_{i=k}^{n-1} \frac{q_{i+1}}{S_{i+1}} \leq \frac{1}{2} q_k^2, \quad \sum_{i=k}^{n-1} \frac{Q_{i+1}^2}{i^2} = O(T^2/k), \quad \sum_{i=k}^{n-1} \frac{1}{i} = O(T).$$

Consequently

$$\log H_n(w) \leq \log(k!) + kq_k + (n - k) \log(n\mu/e) + \frac{w}{2} q_k^2 + O_{\mathcal{W}}(k + Q_k^2/k + T + T^2/k). \quad (17.14)$$

As $k = n^{o(1)}$, (15.7) implies $q_k \rightarrow \infty$, and taking logarithms in the calibration gives

$$q_k = T - \log k + \log(q_k + 1) + \log \mu + O(e^{-q_k}) = T + O_{\mathcal{W}}(U).$$

Stirling's formula and $k/(n\mu) = \chi(e^{qk})$ give the useful exact cancellation

$$\begin{aligned} \log(k!) + kq_k + (n-k)\log(n\mu/e) - \log(n!) - n\log\mu \\ = k\log[e^{qk}\chi(e^{qk})] + \frac{1}{2}\log(k/n) + O(1/k) = O_{\mathcal{W}}(kU + T). \end{aligned} \quad (17.15)$$

The term $wq_k^2/2$ is $wT^2/2 + O_{\mathcal{W}}(TU + U^2)$. Since $k = O_{\mathcal{W}}(T^2/U^2)$ and $TU = O(T^2/U)$ eventually, every remaining error in (17.14) is $O_{\mathcal{W}}(T^2/U)$. This proves the upper half of Theorem 13.1.

18 The matching weighted lower bound

18.1 A positive path law and its exact likelihood

For this lower bound choose $k = \lceil 100T^2/U^2 \rceil$; any fixed positive multiplier would suffice. Begin with the single legal prefix of k zeros. Its state is

[write] **5 October 2026.** Part III, Section 31, repeats this section almost word for word with the cutoff $k = \lceil AT \rceil$; its closing paragraph replaces (18.21) and the final error count by the exact normalization (28.4), giving the lower bound with error $O_{\mathcal{W}}(\log n)$.

$K_k = k - 1$, $L_k = 0$, $M_k = k + 1$ and its weight is w^{k-1} . At each time $i = k, \dots, n - 1$, sample j from P_w , with parameters $q = q_{i+1}$ and $M = M_i$. Denote this path law by $\mathbb{P}_w^*$, with expectation $\mathbb{E}_w^*$. The weak-ascent count never decreases, so every path has

$$M_i \geq k + 1, \quad \frac{M_i}{1 + q_{i+1}} \geq \frac{k}{1 + 2T} \longrightarrow \infty. \quad (18.1)$$

All corrector and eigenvalue expansions therefore apply on every sampled path, uniformly for $w \in \mathcal{W}$ once n is large enough. Use the uncut potential

$$F_i(K, l) = \exp\{q_i K - q_i l / (K + 2) + h(q_i, K + 2, l) / (K + 2)\}.$$

For a step let

$$R_i^w = \frac{w^e F_{i+1}(K + a, j)}{F_i(K, l) P_w(l, j)}. \quad (18.2)$$

The product telescopes, and $F_n = 1$, so the total weighted mass of completions of this fixed prefix is exactly

$$w^{k-1} F_k(k - 1, 0) \mathbb{E}_w^* \exp\left(\sum_{i=k}^{n-1} \log R_i^w\right).$$

It is a positive sub-sum of $H_n(w)$ even for $w < 1$. Jensen gives

$$\log H_n(w) \geq (k - 1) \log w + (k - 1) q_k + \sum_{i=k}^{n-1} \mathbb{E}_w^* \log R_i^w. \quad (18.3)$$

There is no $k!$ factor in this single-prefix lower bound.

18.2 Tracking the sampled state

Put $X_i = L_i/M_i$, $Y_i = K_i - X_i$, and $V_i = ((j - L_i) \bmod M_i)/M_i$. A deterministic calculation gives

$$Y_{i+1} - Y_i = 1 - V_i + aX_{i+1}/M_i \in [0, 2]. \quad (18.4)$$

Indeed $j/M_i - X_i = V_i - (1 - a)$, while $j/M_i = X_{i+1}(1 + a/M_i)$. This includes equality steps, for which $V_i = 0$ and $a = 1$. Since the identity part of (15.12) has $V = 0$,

$$\mathbb{E}_{P_w} V = (1 - \delta) \mathbb{E}_P V, \quad |\mathbb{E}_{P_w} V - \mathbb{E}_P V| \leq C_{\mathcal{W}}/M. \quad (18.5)$$

For the second inequality both factors matter: $|\delta| = O_{\mathcal{W}}(Q/M)$ and $\mathbb{E}_P V = O(Q^{-1})$. At $q = 0$, $\mathbb{E}_{P_w} V = (M-1)/(2(M+w-1))$, still differing from the unweighted mean by $O_{\mathcal{W}}(1/M)$. Equation (16.6) also gives

$$1 - \mathbb{E}_P V = \alpha(q) + \frac{1 - Q_0(q/M)}{q} = \alpha(q) + O(1/M), \quad (18.6)$$

where the quotient has value $1/(2M)$ at zero. The last term in (18.4) lies in $[0, 1/M_i]$. Boundedness of α' and (15.5) now give

$$\mathbb{E}_w^*[Y_{i+1} - Y_i \mid \mathcal{F}_i] \geq \frac{1}{2} - C_{\mathcal{W}}/M_i \geq \frac{1}{3}, \quad (18.7)$$

$$|\mathbb{E}_w^*[Y_{i+1} - Y_i \mid \mathcal{F}_i] - (S_{i+1} - S_i)| \leq C_{\mathcal{W}}(1/M_i + 1/i). \quad (18.8)$$

The threshold for the second inequality in (18.7) may depend on $\mathcal{W}$. Subtract conditional means in (18.4) to obtain a martingale with increments bounded in absolute value by two. Since $Y_k = k - 1$, the event $M_i < i/8$ forces its displacement below

$$i/8 - (k-1) - (i-k)/3 = -5i/24 - 2k/3 + 1.$$

Azuma's inequality proves, for absolute $c, C > 0$ after a sufficiently large compact-uniform threshold,

$$\mathbb{P}_w^*(M_i < i/8) \leq Ce^{-ci} \quad (k \leq i < n). \quad (18.9)$$

At $i = k$ the event is empty. No independence is assumed. On good states $M_i \geq i/8$, $1/M_i \leq 8/i$; on bad states use $M_i \geq k$ and (18.9). The L^2 triangle inequality yields

$$\left\| \sum_{i=k}^{m-1} 1/M_i \right\|_2 \leq 8 \sum_{i=k}^{m-1} 1/i + \frac{1}{k} \sum_{i=k}^{m-1} Ce^{-ci/2} \leq CT. \quad (18.10)$$

The martingale variance is $O(m)$, and $|Y_k - S_k| \leq k$. Sum (18.8), use (18.10), and note $M_i - Y_i = 2 + X_i$. We obtain

$$\mathbb{E}_w^*[(M_i - S_{i+1})^2] \leq C_{\mathcal{W}}(i + T^2 + k^2). \quad (18.11)$$

18.3 Conditional one-step gain

Freeze $q = q_{i+1}$, $S = S_{i+1}$, $M = M_i$, $l = L_i$, and $\Delta q = q_{i+1} - q_i$. Substituting the Doob kernel into (18.2) cancels w^e exactly and gives

$$\begin{aligned} \log R_i^w &= \log \lambda_w + \Delta q K_i - \Delta q l/M + \frac{qaj}{M(M+a)} \\ &\quad + \frac{h(q, M+a, j)}{M+a} - \frac{h(q_i, M, l)}{M}. \end{aligned} \quad (18.12)$$

Using (16.5), replacement of the two corrector terms by $[h(q, M, j) - h(q, M, l)]/M$ costs at most

$$C(Q^2/M^2 + Q/(Mi)). \quad (18.13)$$

For the next-state change, differentiating at fixed j includes both H_x and H_M , giving $O(Q^2/M^2 + Q^3/M^3)$; the current-time change costs $O(Q|\Delta q|/M)$. Equation (18.1) justifies the whole interpolation. Also $qaj/[M(M+a)] \geq qaj/M^2 - q/M^2$. Take conditional expectation and use the exact weighted corrector identity (16.11) at $\zeta = 1$, then (16.8). The last three terms of (18.12) are at least

$$q/(2M) - C_{\mathcal{W}}(1/M + Q^2/M^2 + Q/(Mi)). \quad (18.14)$$

The eigenvalue satisfies exactly

$$\lambda = n\mu \frac{M}{S} \frac{q/M}{1 - e^{-q/M}}.$$

The identity $\log[z/(1 - e^{-z})] = z/2 + \log[(z/2)/\sinh(z/2)]$ implies, in the present range,

$$\log \lambda \geq \log(n\mu) + \log(M/S) + q/(2M) - Cq^2/M^2.$$

Together with (15.16),

$$\log \lambda_w \geq \log(n\mu) + \log(M/S) + (w - 1/2)q/M - C_W(1/M + Q^2/M^2). \quad (18.15)$$

A negative intermediate coefficient when $w < 1/2$ causes no inequality reversal. Since $1/S(s)$ changes by $O(1/i^2)$ on $[i, i + 1]$ and $K_i \leq i$, calibration gives

$$\Delta q K_i \geq -K_i/S - C/i, \quad -\Delta q l/M \geq 0.$$

Using $K_i = M - 2$ and dropping the favorable $2/S$, combine these inequalities to obtain

$$\begin{aligned} \mathbb{E}_w^*[\log R_i^w \mid \mathcal{F}_i] - \log(n\mu/e) \\ \geq \Psi(M_i/S_{i+1}) + \frac{wq_{i+1}}{M_i} - C_W \left(\frac{1}{i} + \frac{1}{M_i} + \frac{Q_{i+1}^2}{M_i^2} + \frac{Q_{i+1}}{M_i i} \right). \end{aligned} \quad (18.16)$$

Thus the net leading gain is exactly wq/M .

18.4 Summation and prefix normalization

On good states, $M_i/S_{i+1} \in [1/16, 2]$; the double zero of Ψ at one gives $\Psi(M_i/S_{i+1}) \geq -C(M_i - S_{i+1})^2/i^2$. On bad states, $M_i \geq k$ and $S_{i+1} \leq n$, so $|\Psi| \leq C + T$. Equations (18.9) and (18.11) imply

$$\sum_{i=k}^{n-1} \mathbb{E}_w^* \Psi(M_i/S_{i+1}) \geq -C_W(T + T^2/k + k) - o(1). \quad (18.17)$$

On good states, $1/M_i \geq 1/S_{i+1} - C|M_i - S_{i+1}|/i^2$. Drop the nonnegative bad-state part of q_{i+1}/M_i , retaining the exponentially small loss of q_{i+1}/S_{i+1} . Cauchy–Schwarz and (18.11), with $q \leq 2T$, yield

$$\sum_{i=k}^{n-1} \mathbb{E}_w^* \frac{q_{i+1}}{M_i} \geq \sum_{i=k}^{n-1} \frac{q_{i+1}}{S_{i+1}} - C_W(T/\sqrt{k} + T^2/k + T) - o(1). \quad (18.18)$$

The explicit errors in (18.16) sum in expectation to $O_W(T + T^2/k)$: on good states use $M_i \geq i/8$, and on bad states use $M_i \geq k$ and (18.9). Every omitted bad-state sum is an exponentially small factor e^{-ck} times a polynomial in T, k . The decreasing function q/S has right Riemann sum differing from its integral by at most $q_k/S_k = O(T/k)$. Thus

$$\sum_{i=k}^{n-1} \frac{q_{i+1}}{S_{i+1}} = \frac{1}{2}q_k^2 + O(T/k). \quad (18.19)$$

Combining (18.16)–(18.19) gives

$$\sum_{i=k}^{n-1} \mathbb{E}_w^* \log R_i^w \geq (n - k) \log(n\mu/e) + \frac{w}{2}q_k^2 - C_W(k + T + T^2/k + T/\sqrt{k}). \quad (18.20)$$

Again (15.7) gives $q_k = T + O(U)$ and $q_k^2 = T^2 + O(TU + U^2)$. For $v = e^{q_k}$, Stirling and $k = n\mu\chi(v)$ give

$$\begin{aligned} & (k-1)q_k + (n-k)\log(n\mu/e) - \log(n!\mu^n) \\ &= k[\log(v\chi(v)) - \log k + 1] - q_k - \frac{1}{2}\log(2\pi n) + O(1/n). \end{aligned} \quad (18.21)$$

Its absolute size is $O(k\log k + k\log(1+q_k) + T) = O(T^2/U)$. The weighted prefix adds $(k-1)\log w = O_{\mathcal{W}}(k)$. Substitute (18.20)–(18.21) into (18.3). All remaining errors are bounded by

$$O_{\mathcal{W}}\{k\log k + k\log(1+q_k) + k + T + T^2/k + T/\sqrt{k} + TU + U^2\} = O_{\mathcal{W}}(T^2/U).$$

This proves the matching lower half, and completes Theorem 13.1.

19 The probability law

Write $b_n = (\log n)^2$ and $U_n = \log \log n$. For each real t , the finite sum defining H_n gives the exact identity

$$\mathbb{E}_{n,w} e^{tE_n} = \frac{H_n(we^t)}{H_n(w)}. \quad (19.1)$$

Theorem 13.1, used at the two positive weights, yields

$$\Lambda_{n,w}(t) := \frac{1}{b_n} \log \mathbb{E}_{n,w} e^{tE_n} = \frac{w}{2}(e^t - 1) + O_{\mathcal{W},J}(1/U_n), \quad (19.2)$$

uniformly when $w \in \mathcal{W}$ and t lies in a fixed compact real interval J . Indeed we^t then stays in a fixed compact positive interval. The tilt is fixed before taking $n \rightarrow \infty$; (19.2) is not a fluctuation-scale local expansion.

19.1 Concentration and moments

Let $X_n = E_n/b_n$. For $\eta > 0$ and a small fixed $t > 0$, Chernoff's inequality gives

$$\mathbb{P}_{n,w}(X_n \geq w/2 + \eta) \leq \exp\{b_n[\Lambda_{n,w}(t) - t(w/2 + \eta)]\} \leq e^{-c_{w,\eta}b_n}$$

for large n , because the limiting exponent has derivative $-\eta$ at zero. A small fixed $t < 0$ gives the same conclusion for $X_n \leq w/2 - \eta$. If that threshold is negative the event is empty; when it is zero the negative-tilt argument still works. This proves (13.6).

The mean is obtained without differentiating the counting remainder. The function $\Lambda_{n,w}$ is convex and differentiable, with $\Lambda'_{n,w}(0) = \mathbb{E}_{n,w} E_n/b_n$ and $\Lambda_{n,w}(0) = 0$. For $h > 0$, its secant slopes give

$$-\frac{\Lambda_{n,w}(-h)}{h} \leq \Lambda'_{n,w}(0) \leq \frac{\Lambda_{n,w}(h)}{h}. \quad (19.3)$$

Letting first $n \rightarrow \infty$ and then $h \downarrow 0$ proves $\mathbb{E}_{n,w} E_n \sim (w/2)b_n$. More quantitatively, use the uniform error in (19.2) on a fixed enlarged compact interval and put $h = U_n^{-1/2}$. The secant errors are $O_{\mathcal{W}}(h + 1/(hU_n)) = O_{\mathcal{W}}(U_n^{-1/2})$. This proves (13.4) without any derivative bound on the remainder.

For moment convergence, fix $t > 0$. Equation (19.2) gives $\Lambda_{n,w}(t) \leq C$ eventually. For $y \geq 2C/t$,

$$\mathbb{P}_{n,w}(X_n \geq y) \leq \exp(-tb_n y/2). \quad (19.4)$$

Since $b_n \geq 1$ eventually, this is a uniform exponential tail. Tail integration gives uniform integrability of X_n^p for every fixed $p > 0$ (indeed of every higher fixed moment). Together with

concentration, this gives convergence of every positive moment and L^p convergence for $p \geq 1$. In particular $\text{Var}_{n,w}(E_n) = o((\log n)^4)$; no equivalent or sharper variance order follows here. **[write] 5 October 2026.** Part III, (34.3), sharpens this to the upper bound $\text{Var}_{n,w}(E_n) = O((\log n)^3)$; the variance order is still unknown (Section 23).

19.2 A full large deviation principle

For completeness we prove the LDP by elementary tilting. For any closed $F \subset \mathbb{R}$ and open $G \subset \mathbb{R}$, the claim is

$$\limsup_{n \rightarrow \infty} b_n^{-1} \log \mathbb{P}_{n,w}(X_n \in F) \leq - \inf_{x \in F} I_w(x), \quad (19.5)$$

$$\liminf_{n \rightarrow \infty} b_n^{-1} \log \mathbb{P}_{n,w}(X_n \in G) \geq - \inf_{x \in G} I_w(x). \quad (19.6)$$

Use $\log 0 = -\infty$ and the usual empty-set convention. The Legendre supremum of $\Lambda_w(t) = w(e^t - 1)/2$ is

$$\sup_{t \in \mathbb{R}} \{tx - \Lambda_w(t)\} = I_w(x).$$

For $x > 0$ its maximizer satisfies $e^t = 2x/w$; for $x = 0$ the supremum is $w/2$, approached as $t \rightarrow -\infty$; and for $x < 0$ it is infinite. The rate is lower semicontinuous and tends to infinity as $x \rightarrow \infty$, so it is good.

Upper bound. For $t > 0$ and $R > 0$,

$$\limsup_n b_n^{-1} \log \mathbb{P}_{n,w}(X_n \geq R) \leq \Lambda_w(t) - tR.$$

Fix $t = 1$ and send $R \rightarrow \infty$ to obtain exponential tightness. Negative X_n are impossible. Let $C \subset [0, \infty)$ be compact and $a < \inf_C I_w$. For each $x \in C$ choose a finite t_x with $t_x x - \Lambda_w(t_x) > a$. This is possible even at zero since $a < I_w(0)$. Continuity gives a neighborhood V_x on which $t_x y - \Lambda_w(t_x) > a$. Markov's inequality, valid for either sign of t_x , and (19.2) give

$$\limsup_n b_n^{-1} \log \mathbb{P}_{n,w}(X_n \in V_x) \leq -a.$$

A finite subcover gives the compact upper bound. Let a increase to $\inf_C I_w$; combine with exponential tightness to obtain (19.5) for every closed set.

Lower bound at positive points. Fix $x > 0$ and $t = \log(2x/w)$. Tilt $\mathbb{P}_{n,w}$ by e^{tE_n} and divide by its expectation. The new law is exactly $Q_n = \mathbb{P}_{n,we^t} = \mathbb{P}_{n,2x}$. The concentration just proved implies $Q_n(|X_n - x| < \varepsilon) \rightarrow 1$, exponentially fast. Undoing the tilt,

$$\begin{aligned} \mathbb{P}_{n,w}(|X_n - x| < \varepsilon) &= \mathbb{E}_{Q_n} \left[e^{b_n \Lambda_{n,w}(t) - t b_n X_n} \mathbf{1}_{\{|X_n - x| < \varepsilon\}} \right] \\ &\geq e^{b_n [\Lambda_{n,w}(t) - tx - |t|\varepsilon]} Q_n(|X_n - x| < \varepsilon). \end{aligned}$$

The logarithm of the final probability divided by b_n tends to zero. Thus the liminf is at least $-I_w(x) - |t|\varepsilon$. If G contains x , use arbitrarily small ε whose interval is contained in G , then optimize over positive $x \in G$.

The endpoint zero. If an open G contains zero, it contains positive points approaching zero. Right-continuity of I_w at zero therefore supplies the same lower bound there. If G has no nonnegative point, the lower claim is vacuous. This finishes the full LDP and Theorem 13.2. This endpoint step concerns neighborhoods of zero; it does not determine the exact atom. The latter follows in the next section.

20 The exact zero level event

20.1 Identification and the established Fishburn input

An ordinary ascent sequence uses the same defining bound as (13.1), with strict ascents instead of weak ascents. If a weak ascent sequence has no levels, its weak ascents are strict in every prefix. It is therefore exactly a primitive ordinary ascent sequence. This proves

$$H_n(0) = P_n. \quad (20.1)$$

Here evaluating at zero means taking the constant term of the polynomial $H_n(w)$; each zero-level word contributes one. Let F_n be the ordinary Fishburn number, with $F_0 = 1$. Compress each maximal constant run in an ordinary ascent sequence to one entry. Strict-ascent counts before each run are unchanged, so the compressed word remains legal and primitive. Conversely, replacing each entry of a primitive word by any positive number of equal copies preserves all bounds. These operations are inverse, giving

$$F_n = \sum_{m=1}^n \binom{n-1}{m-1} P_m \quad (n \geq 1). \quad (20.2)$$

[write] **5 October 2026.** This identity, and its series form below, is Jelínek's Fact 5.1 [8]; Part I states it as (12.3) without credit.

In formal ordinary series, $F(z) = P(z/(1-z))$ and $P(t) = F(t/(1+t))$. These compositions are coefficientwise legitimate because the inner series has zero constant term, even though the outer series has radius zero. Coefficient extraction yields

$$P_n = \sum_{j=0}^{n-1} (-1)^j \binom{n-1}{j} F_{n-j}, \quad n \geq 1, \quad P_0 = 1. \quad (20.3)$$

The upper index is $n-1$, not n . The same substitution counts primitive Fishburn matrices: replace every occupied entry in a zero-one Fishburn matrix by a positive integer. Thus the number of primitive matrices of size n agrees with P_n by the same inverse substitution, without requiring a statistic-preserving bijection. There is no analogous run-compression claim for the general positive-weight weak-ascent model: inserted equalities create additional weak ascents and change subsequent allowed alphabets.

The only external coefficient asymptotic used in this section is the established ordinary Fishburn estimate [8, 7],

$$F_m = C_F m! \mu^m \sqrt{m} (1 + O(1/m)), \quad C_F = 12\sqrt{3} \pi^{-5/2} e^{\pi^2/12}. \quad (20.4)$$

The primitive/all limiting ratio below is already Jelínek's Fact 5.2, credited to Drmota [8]; the transform proof is prior work too. We include its summability argument to make the endpoint application explicit, and not as a novelty claim.

20.2 Summable domination and quantitative ratio

Write $R = 1/\mu = \pi^2/6$ and $F_m = C_F m! \mu^m \sqrt{m} b_m$. Then $b_m = 1 + O(1/m)$ and every $b_m > 0$. There are constants $0 < c \leq C < \infty$ with $c \leq b_m \leq C$ for all $m \geq 1$, by absorbing finitely many small m into the asymptotic bounds. Exact factorial cancellation gives, for $0 \leq j \leq n-1$,

$$\binom{n-1}{j} \frac{F_{n-j}}{F_n} = \frac{R^j}{j!} (1 - j/n)^{3/2} \frac{b_{n-j}}{b_n}. \quad (20.5)$$

Extend this term by zero for $j \geq n$. It is bounded absolutely by $(C/c)R^j/j!$, a summable sequence. For each fixed j , the factor after $R^j/j!$ tends to one. Dominated convergence in (20.3) gives $P_n/F_n \rightarrow e^{-R}$, with the full near- n range controlled. For a quantitative bound choose D such that $|b_m - 1| \leq D/m$ for all $m \geq 1$. With $m = n - j$ and $d = (m/n)^{3/2}$,

$$\begin{aligned} |db_m/b_n - 1| &\leq |d - 1| + d|b_m - b_n|/b_n \\ &\leq \frac{3j}{2n} + \frac{D}{c}d(1/m + 1/n) \leq \frac{3j/2 + 2D/c}{n}. \end{aligned} \quad (20.6)$$

Here the derivative of $x^{3/2}$ is bounded by $3/2$ on $[0, 1]$, and $d/m = \sqrt{m}/n^{3/2} \leq 1/n$. Summing (20.6) against $R^j/j!$ costs $O(1/n)$; the missing tail obeys

$$\sum_{j \geq n} \frac{R^j}{j!} \leq \frac{1}{n} \sum_{j \geq 0} \frac{jR^j}{j!} = Re^R/n.$$

Consequently

$$\frac{P_n}{F_n} = e^{-R} + O(1/n), \quad P_n = C_P n! \mu^n \sqrt{n} (1 + O(1/n)), \quad C_P = 12\sqrt{3}\pi^{-5/2}e^{-\pi^2/12}. \quad (20.7)$$

20.3 Application at positive weight

For fixed $w > 0$, the numerator of $\mathbb{P}_{n,w}(E_n = 0)$ is P_n , independent of w . Thus

$$\log \mathbb{P}_{n,w}(E_n = 0) = \log P_n - \log H_n(w).$$

Equation (20.7) supplies $\log P_n = \log(n!) + n \log \mu + \frac{1}{2} \log n + \log C_P + O(1/n)$. Subtract Theorem 13.1; the half-logarithm and constant are absorbed by its error. This proves Theorem 13.3, including compact-uniformity. The combination with the weighted theorem is the present endpoint conclusion. The primitive ratio is prior work. No relative-error equivalent for $\mathbb{P}_{n,w}(E_n = 0)$ follows from the present denominator estimate. At $w = 0$ the induced law is concentrated on primitive objects and $E_n = 0$ identically; this is a separate endpoint model, not a uniform extension to n -dependent weights approaching zero.

Remark 20.1 ([**write**], 5 October 2026). With Part III's Theorem 24.1, for every compact $\mathcal{W} \subset (0, \infty)$,

$$\log \mathbb{P}_{n,w}(E_n = 0) = -\frac{w}{2}(\log n)^2 + O_{\mathcal{W}}(\log n) \quad (w \in \mathcal{W}).$$

Proof. By (20.1) the numerator is P_n , so $\log \mathbb{P}_{n,w}(E_n = 0) = \log P_n - \log H_n(w)$. By (20.7), $\log P_n = \log(n!) + n \log \mu + \frac{1}{2} \log n + \log C_P + O(1/n)$, with no dependence on w . By (24.1), $\log H_n(w) = \log(n!) + n \log \mu + \frac{w}{2}(\log n)^2 + O_{\mathcal{W}}(\log n)$, uniformly for $w \in \mathcal{W}$. Subtracting, the factorial and exponential terms cancel and $\frac{1}{2} \log n + \log C_P + O(1/n) = O(\log n)$ is absorbed. $\square$

In the scale of Theorem 13.3, the error is $O(\sqrt{b_n})$ instead of $O(b_n/\log \log n)$. It is still a logarithmic statement: since the $O_{\mathcal{W}}(\log n)$ of Part III is not identified, no relative-error equivalent for the atom follows.

21 The threshold inverse

Equation (14.2) proves that $N_w(X)$ is well-defined and that no eventual-monotonicity assumption is needed, even for real weights. The initial equality $H_0 = H_1 = 1$ is irrelevant as $X \rightarrow \infty$. By Stirling and Theorem 13.1, for integers $n \rightarrow \infty$,

$$\log H_n(w) = f(n) + \frac{w}{2}(\log n)^2 + O_{\mathcal{W}}\left(\frac{(\log n)^2}{\log \log n}\right), \quad f(t) = t \log(\mu t/e). \quad (21.1)$$

Let $L = \log X$, and let $x = L/W_0(\mu L/e)$, so $f(x) = L$. For sufficiently large x , define

$$z = x - \frac{w(\log x)^2}{2 \log(\mu x)}, \quad A_w(t) = f(t) + \frac{w}{2}(\log t)^2.$$

Uniformly on a fixed compact $\mathcal{W}$, $z - x = O_{\mathcal{W}}(\log x)$. Taylor's theorem, using $f'(t) = \log(\mu t)$ and $f''(t) = 1/t$, gives

$$A_w(z) = L + O_{\mathcal{W}}((\log x)^2/x). \quad (21.2)$$

Indeed the first-order change of f cancels $w(\log x)^2/2$, while the quadratic remainder of f and the change in the log-square term are both $O_{\mathcal{W}}((\log x)^2/x)$. Put $d_x = \log x / \log \log x$. For a sufficiently large fixed $C_{\mathcal{W}}$, form the integer brackets

$$m_- = \lfloor z - C_{\mathcal{W}}d_x \rfloor, \quad m_+ = \lceil z + C_{\mathcal{W}}d_x \rceil.$$

On this range $A'_w(t) = \log(\mu t) + w \log t/t \sim \log x$ uniformly. Therefore

$$A_w(m_{\pm}) - L = \pm C_{\mathcal{W}} \frac{(\log x)^2}{\log \log x} (1 + o(1)) + O_{\mathcal{W}}(\log x),$$

where rounding contributes only $O(\log x)$. Choose $C_{\mathcal{W}}$ larger than the uniform error constant in (21.1). Equations (21.1)–(21.2) then ensure, for all sufficiently large X ,

$$\log H_{m_-}(w) < L < \log H_{m_+}(w).$$

Exact monotonicity gives $m_- < N_w(X) \leq m_+$, proving (13.8) with its signed negative correction. This is an integer-bracket argument, not a formal differentiation of a sequence or an assumed coefficient ratio. For positive integer w , independently coloring each level realizes every summand $w^{E(x)}$ as an exact number of colored words. For arbitrary real $w > 0$, the same proof concerns a weighted sum. The inverse error diverges; it is not a bounded-error inverse or a certified finite numerical error bar.

22 Reproducibility and limits of finite verification

The source package contains the complete article, its final PDF, a deterministic build script, exact-rational fixtures and verifiers, a corruption campaign, a file manifest, and validation records. No external report is required to build or follow the proof. Source hashes identify the fixed mathematical inputs and immutable report-105 baseline used for provenance; the package contains only the report and its reproducibility materials.

The exact checks choose rational $r \in (0, 1)$ and rational $w > 0$, then set $v = r^{-M}$. The relation $q = -M \log r$ is symbolic: q is generally not rational. Instead the checks use the rational quantities b/q , c/q , and h/q from (16.1)–(16.3). This makes the eigenvector, row-stochasticity, signed-mixture, and divided Poisson identities exact over the rational numbers. The $q = 0$ endpoint is checked separately with its actual limiting kernel and the vanishing undivided corrector. No floating-point residual is labelled an exact check.

Independent polynomial recurrence and exhaustive inversion-sequence enumeration test the finite level polynomials. Ordinary and primitive ascent recurrences test the two binomial transforms and the no-level identification. Deliberate fixture corruptions must be rejected by explicit guards in both ordinary Python and optimized Python; correctness gates do not rely on assertions that optimization removes. These finite tests protect formulas, signs, indexing, and implementation. They do not prove any of the uniform analytic estimates, limits, or large-deviation statements, whose proofs are in Sections 15–21.

What remains unproved. The present results do not identify an amplitude or a relative-error equivalent for $H_n(w)$; they do not identify smaller log-scale coefficients, a full transseries, or an asymptotic uniform in vanishing or unbounded weights. Equality with Poisson leading log-MGF and large-deviation scales is not a Poisson approximation theorem. A central or local limit, variance equivalent, positive-level atom asymptotic, and sharp tail prefactor would require additional control.

23 Further questions and research

[write] (5 October 2026.) This section was added in the write, under Vladimir’s standing rule of 4 October 2026. It collects every claim of Part II stated without proof or explicitly left open (Sections 13 and 22), with the re-scoping by Part III. Nothing in this Part was found to be wrong.

1. *Poisson approximation.* The limiting log-MGF $\frac{w}{2}(e^t - 1)$ at speed b_n has the Poisson form, which suggests that E_n is close to a Poisson variable with mean $\frac{w}{2}b_n$. Total-variation Poisson approximation and mod-Poisson convergence are not proved (“its error is too large”). Missing: a fixed-tilt remainder $o(1)$ in $\log \mathbb{E}_{n,w} e^{tE_n}$, not only $o(b_n)$; Part III gives $O(\log n)$, still too large.
2. *Fluctuations.* A central or local limit theorem and the variance order. Re-scoped: Part III proves $\text{Var}_{n,w}(E_n) = O((\log n)^3)$ (an upper bound, not an order); the Poisson form would suggest, without implying, order $(\log n)^2$ (Part III, question 4).
3. *Positive-level atoms and tail prefactors.* Asymptotics of $\mathbb{P}_{n,w}(E_n = k)$ for $k \geq 1$, and sharp prefactors in the large-deviation tails. The zero atom needed a separate argument (Theorem 13.3); no analogue for $k \geq 1$ is given.
4. *An amplitude, a relative-error equivalent, smaller log-scale coefficients and a transseries for $H_n(w)$.* As for Part I, items 1–2; Part III, questions 1–2.
5. *Uniformity as $w \rightarrow 0$ or $w \rightarrow \infty$ with n* (Section 13, “Scope”; Section 20 for the separate endpoint $w = 0$). Part III, question 5.
6. *A relative-error equivalent for the zero atom $\mathbb{P}_{n,w}(E_n = 0)$,* which would need the amplitude of $H_n(w)$ (item 4); Remark 20.1 gives the error $O(\log n)$ in the logarithm.
7. *Certified inverse constants* (Section 21). Part III, question 3.

Part III

Sharp logarithmic remainders and bounded error inverses

Bundle Report 108, archive `Ascent_Sequences_Sharp_Remainders_and_Bounded_Inverses_Source.zip`, delivered as `report108.tex` (24 pages), dated 2 October 2026. Delivered title “Sharp Logarithmic Remainders and Bounded Error Inverses: Weighted weak ascents and fixed difference ascents”; author line “Research report 108 / Prepared for Vladimir”. Printed as delivered, with **[write]** notes, except its Sections 3 and 4, which repeat Part II’s Sections 15 and 16 word for word and are printed there. Delivered Section k is Section $k + 23$ here, and its statement or equation $k.j$ is $(k + 23).j$; references to equations of its Sections 3–4 point to Part II.

Reading conventions of Part III	Elsewhere in the report
$H_n(w)$: w marks levels	Part I: $H_n(v)$, v marks weak ascents
$B_n = H_n(w)$ or $a_n^{(d)}$ (a count)	Part I: $B_n = n(r_n/\mu - 1)$ (a ratio diagnostic)
$c \in \{w, d\}$: the log-square coefficient	Parts I, II: c the corrector mean (also here)
A : cutoff constant, $k = \lceil AT \rceil$	Parts I, II: A in the stable corrector; $k \asymp T^2/U^2$
K : the number of d -ascents in the difference model	Part I: $K_n^{(d)}$
F_i capped, $\tilde{F}_i$ uncut potentials	Part I: $\tilde{V}_i, U_i$; Part II: F_i

Abstract of this Part (as delivered). For level-weighted weak ascent sequences and every fixed positive integer difference-ascent parameter, we prove factorial growth with logarithmic-square corrections and an $O(\log n)$ remainder. The coefficients are $w/2$ and $d/2$, respectively, and the weak case at weight one is OEIS A336070. An all-state capped transfer bound and a sharper prefix count reduce the remainder to logarithmic order. A positive path change of measure supplies the lower bound; the fixed-difference argument explicitly controls its shifted lower boundary. The resulting threshold inverses have bounded additive errors after a signed logarithmic correction to the Lambert-W inverse. Exact finite checks accompany the analytic proofs. These results leave polynomial-factor count uncertainty; they do not determine amplitudes, relative equivalents, or exact integer thresholds.

24 Main results and their meaning

The weak ascent count is OEIS A336070. Its factorial scale hides a substantial subexponential correction. This report sharpens that correction to a polynomial multiplicative uncertainty and, as a consequence, determines a threshold inverse up to a bounded additive error.

A weak ascent sequence is a nonnegative integer word $x = (x_1, \dots, x_n)$ with $x_1 = 0$ and

$$x_{i+1} \leq 1 + \#\{t < i : x_{t+1} \geq x_t\}.$$

Let $E(x) = \#\{i < n : x_{i+1} = x_i\}$ and

$$H_n(w) = \sum_{|x|=n} w^{E(x)}, \quad \mu = 6/\pi^2.$$

Here $w > 0$ weights equal adjacent pairs, called *levels*. The empty word has weight one. Independently, for an integer $d \geq 1$, a *difference d ascent* is an index with $x_{i+1} > x_i - d$; a

difference ascent sequence satisfies the same upper bound with the number of these d -ascents. Denote its unweighted count by $a_n^{(d)}$. In particular $a_n^{(1)} = H_n(1)$.

Theorem 24.1 (Sharp logarithmic remainders). *For every compact $\mathcal{W} = [w_-, w_+] \subset (0, \infty)$,*

$$\log H_n(w) = \log(n!) + n \log \mu + \frac{w}{2}(\log n)^2 + O_{\mathcal{W}}(\log n), \quad (24.1)$$

uniformly for $w \in \mathcal{W}$. For every fixed integer $d \geq 1$,

$$\log a_n^{(d)} = \log(n!) + n \log \mu + \frac{d}{2}(\log n)^2 + O_d(\log n). \quad (24.2)$$

The constants and the starting threshold for n can depend on the indicated compact interval or fixed integer.

[write] **5 October 2026.** This theorem strengthens Part I, Theorem 1.2 (the case $w = 1$, equivalently $d = 1$), and Part II, Theorem 13.1, whose remainders were $O((\log n)^2 / \log \log n)$. It answers, for every fixed $d \geq 1$ and at the logarithmic-square scale, Part I's question on the dependence of the subexponential factor on fixed d (Section 12). The case $d = 0$ of (24.2), which this Part's proof excludes, follows from the Fishburn asymptotic: [write] Remark 33.1.

Corollary 24.2 (Bounded error threshold inverses). *For $B_n = H_n(w)$ or $a_n^{(d)}$, set $c = w$ or d , respectively, and*

$$N(Y) = \min\{n \geq 1 : B_n \geq Y\}, \quad L = \log Y, \quad x = \frac{L}{W_0(\mu L/e)},$$

where W_0 is the principal real Lambert function. Then

$$N(Y) = x - \frac{c(\log x)^2}{2 \log(\mu x)} + O(1) = x - \frac{c}{2} \log x + O(1). \quad (24.3)$$

The weighted error is uniform on each fixed positive compact interval; the difference-ascent error may depend on fixed d .

[write] **5 October 2026.** This corollary supersedes the inverses (1.7) and (10.4) of Part I and Theorem 13.4 of Part II, and the statements there that those inverses are not bounded-error inverses.

The two models have different meanings. Substituting an integer $w = d$ does not turn levels into difference ascents. For example $H_3(2) = 12$, whereas $a_3^{(2)} = 6$. The coincidence of the two correction coefficients reflects related transfer calculations, not an equality of the families.

Exponentiating (24.1) says that $H_n(w)$ lies between $n! \mu^n \exp\{(w/2) \log^2 n\}$ times n^{-C} and the same expression times n^C , eventually. This is a polynomial-factor uncertainty. It is not a relative-error equivalent and it does not identify a power of n , an amplitude, a full transseries, or a finite certified error constant. The bounded inverse is substantially more precise than its leading Lambert term, but is not eventual exact integer recovery.

Scope and method. The proof is self-contained. A frozen geometric kernel and an exact quadratic correction are combined with a calibration curve. A capped slope controls every small state, including transitions across the cap. A run decomposition removes the crude factorial prefix loss. These two improvements allow a prefix of length $k \asymp \log n$, and retaining the exact normalization cancellation makes every accumulated error $O(\log n)$. For fixed d , a shifted row replaces the eigenvector formula and a separate boundary argument recovers its full coefficient. No numerical fit supplies any analytic step.

The ordinary-ascent endpoint $d = 0$ is the classical Fishburn family and is separate; the zero prefix used below would not have $k - 1$ ascents there. For real $d > 0$, integrality of the letters makes the effective parameter $\lceil d \rceil$. No uniformity for $d = d(n)$, or for weights approaching zero or infinity, is claimed. No OEIS identifiers for unrestricted $d = 2, 3$ are asserted.

[write] **5 October 2026.** The conclusions of Theorem 24.1 and Corollary 24.2 nevertheless hold at $d = 0$, with coefficient 0, by the Fishburn asymptotic instead of this proof: [write] Remark 33.1.

25 Exact states and recurrences

At length i , let K be the relevant ascent count and l the last letter, and put $M = K + 2$. A legal extension chooses $j \in \{0, \dots, M - 1\}$. For weak ascents it updates

$$a = \mathbf{1}_{\{j \geq l\}}, \quad e = \mathbf{1}_{\{j = l\}}, \quad (K, l) \mapsto (K + a, j), \quad \text{multiplier } w^e. \quad (25.1)$$

In the difference model, $a = \mathbf{1}_{\{j > l - d\}}$ and the multiplier is one. Induction gives $0 \leq l \leq K \leq i - 1$: a non-ascent decreases the letter, while an ascent permits at most the new K . In particular every letter x_i is at most $i - 1$.

Starting with $A_1(0, 0; w) = 1$, the exact weighted recurrence is

$$A_{i+1}(K + a, j; w) = w^e A_i(K, l; w), \quad 0 \leq j \leq K + 1.$$

Summing states gives $H_i(w)$. For example

$$H_1 = 1, \quad H_2 = 1 + w, \quad H_3 = 2 + 3w + w^2, \quad H_4 = 5 + 11w + 6w^2 + w^3.$$

If $C_d(n, k, l)$ counts difference-ascent states, the equivalent incoming form is

$$C_d(n + 1, k, j) = \sum_{l=j+d}^k C_d(n, k, l) + \sum_{l=0}^{\min(k-1, j+d-1)} C_d(n, k - 1, l). \quad (25.2)$$

Equation (25.2) is for $n \geq 1$ and $0 \leq j \leq k \leq n$. Define states outside their support to be zero; empty sums also vanish. The first sum is a non-ascent; the second is an ascent. This formula also gives the $d = 0$ benchmark, although our asymptotic proof excludes that endpoint.

Exactly one weak extension is a level, so

$$H_{i+1}(w) = \sum_{K, l} (K + 1 + w) A_i(K, l; w) \geq (1 + w) H_i(w) > H_i(w). \quad (25.3)$$

Similarly $a_{i+1}^{(d)} = \sum_{K, l} (K + 2) C_d(i, K, l) \geq 2a_i^{(d)}$. These strict inequalities for $i \geq 1$ justify first-crossing inverses.

Split a sequence at each ascent into runs of non-ascents. For $d \geq 1$ each such run is strictly decreasing. The run beginning after the r th ascent starts at most at r , so has at most $r + 1$ entries. Thus both models satisfy

$$i \leq (K + 1)(K + 2)/2, \quad M \geq \sqrt{2i}. \quad (25.4)$$

26 Calibration and the frozen weighted kernel

[write] **5 October 2026.** Report 108's Section 3, with this title and its two subsections "The calibrated trajectory" and "Exact Perron and Doob formulas", is word for word Report 107's Section 3, printed as Section 15 of Part II. It is not printed twice. Its equations and statements are cited in this Part under their Part II numbers, for example (15.2), (15.12) and (15.17).

27 The exact quadratic corrector

[write] **5 October 2026.** Report 108's Section 4, with this title and its three subsections "Finite Poisson identity and stable formula", "Geometric moments and the corrector constant" and "Weighted first moment and exponential estimate", is word for word Report 107's Section 4, printed as Section 16 of Part II, including Lemma 16.1. It is not printed twice; this Part cites its equations under their Part II numbers, for example (16.5), (16.10) and (16.13).

28 A logarithmic cutoff and exact normalization

Write $T = \log n$. Choose a sufficiently large fixed A , depending only on $\mathcal{W}$ or on fixed d , and put

$$k = \lceil AT \rceil. \quad (28.1)$$

The size requirements on A will be made explicit by inequalities below. Local kernel and corrector constants are selected first; A is then fixed before n tends to infinity. In particular, all relevant local estimates hold when Q/M is sufficiently small and fixed. They do not require $Q/M \rightarrow 0$: the exponential-moment proof only needs its extra tilt to be at most $q/2$, and the logarithm expansion only needs $|(w-1)t| \leq 1/2$.

For $i \geq k$ and $T \geq 2$, $Q_i/i \leq 3/A$. A global calibration bound is

$$e^q \chi(e^q) = \int_0^\infty e^{-u} \{q + u + Q_0(q + u)\} du \leq q + 2. \quad (28.2)$$

This identity uses $t/(e^t - 1) = e^{-t}(t + Q_0(t))$, and is valid at $q = 0$. The positive tail expansion (15.7) also gives

$$q_k = T + \log(\mu/A) + O_A(T^{-1}), \quad q_k^2 = T^2 + O_A(T). \quad (28.3)$$

Here is a noncircular derivation. Since $k/n \rightarrow 0$, $q_k \rightarrow \infty$. Taking logarithms of the tail yields

$$q_k = T - \log k + \log(q_k + 1) + \log \mu + O(e^{-q_k}).$$

Together with $q_k \leq 2T$, this first gives $T - \log T + O_A(1) \leq q_k \leq T + O_A(1)$, hence $q_k/T \rightarrow 1$. Rewrite the identity as

$$q_k - T = \log(\mu/A) + \log((q_k + 1)/T) - \log(k/(AT)) + O(e^{-q_k}).$$

It gives a bounded difference and then (28.3) by substitution. The ceiling contributes only $O_A(T^{-1})$.

The deterministic zero-prefix normalization is especially important. With $v = e^{q_k}$, Stirling and $k = n\mu\chi(v)$ give

$$\begin{aligned} D_n &:= (k-1)q_k + (n-k)\log(n\mu/e) - \log(n!\mu^n) \\ &= k[\log(v\chi(v)) - \log k + 1] - q_k - \frac{1}{2}\log(2\pi n) + O(n^{-1}). \end{aligned} \quad (28.4)$$

By (28.3) and the tail expansion, $v\chi(v)/k = A^{-1} + O_A(T^{-1})$, so $D_n = O_A(T)$. The terms $\log(v\chi(v))$ and $\log k$ must be kept together. Estimating them separately would lose an avoidable $T \log T$.

29 The capped weighted upper bound

29.1 Constants and the capped potential

Put $\Psi(R) = \log R - R + 1$, $\eta_0 = -\Psi(0.49) > 0$, $\eta_1 = -\Psi(3/4) > 0$, and $b = (w_+ - 1)_+$. Choose a fixed $p > 0$ with

$$p \leq 1, \quad (3/2 + b)p \leq \eta_0/4.$$

Let $\zeta(R)$ be zero for $R \leq 1/2$, linear with slope four on $[1/2, 3/4]$, and one for $R \geq 3/4$. Define

$$F_i(K, l) = \exp\{q_i K - \min(q_i/M, p)l + \zeta(M/S_i)h(q_i, M, l)/M\}, \quad M = K + 2. \quad (29.1)$$

Where $\zeta = 0$, define the whole corrector term to be zero, without invoking local derivative estimates. At time n , $F_n = 1$ exactly.

Call a state uncapped if $q_i/M \leq p$, assigning equality to this case. Along any path q_i decreases and M_i increases; therefore every successor of an uncapped state is uncapped. A capped state has

$$M < 2T/p, \quad M/S_i < 4/(Ap), \quad (M + a)/S_{i+1} \leq 4/(Ap) + 2/(i + 1). \quad (29.2)$$

Take $A \geq 32/p$, so both correctors vanish on every transition from a capped state, eventually.

Select local constants C_0, C_1 for the fixed region $M \geq C_0 Q$ first. After p is fixed, put

$$B_{W,p} = \max(1, w_+) \frac{e^{p+1}(1 + e)}{1 - e^{-p}}.$$

Choose $A \geq \max(16C_0, 32/p, 32w_+, 6eB_{W,p})$, enlarged further so the intermediate-state error below is at most $\eta_1/2$. Each requirement depends only on fixed constants. Subsequent increases in n make all $1/i$ and $1/M$ residuals small uniformly for $i \geq k$. There is no circular choice.

29.2 Regular uncapped states

Fix $i \geq k$, and abbreviate $q = q_{i+1}$, $Q = 1 + q$, $S = S_{i+1}$, $R = M/S$, and $\Delta q = q - q_i$. If $R \geq 0.49$, then $M \geq 0.245(i + 1)$, so sufficiently large A makes the cap inactive and ensures $M \geq C_0(1 + q_i)$ on all interpolation segments.

Replace the next corrector by $\zeta(M/S)h(q, M, j)/M$, and the current corrector by $\zeta(M/S)h(q, M, l)/M$. Each replacement costs $O_W(Q^2/i^2)$. For detail, differentiating $H(q, M, j/M)/M$ at fixed j gives

$$\frac{H_M}{M} - \frac{(j/M)H_x + H}{M^2} = O(Q^2/M^2 + Q^3/M^3).$$

The next cutoff change costs $O(Q^2/(SM))$; the time change costs $O(Q|\Delta q|/M)$; and the current cutoff change is bounded by its Lipschitz constant times $M|S - S_i|/(S_i S)$. Thus its corners cause no gap. Also

$$-\frac{qj}{M+a} + \frac{qj}{M} = \frac{qaj}{M(M+a)} \leq \frac{qaj}{M^2}.$$

Factoring the frozen eigenvector gives

$$\sum_j w^{\mathbf{1}_{\{j=l\}}} \frac{F_{i+1}(K+a, j)}{F_i(K, l)} \leq \lambda_w \exp\{\Delta q K - \Delta q l/M + C_W Q^2/i^2\} \cdot \mathbb{E}_{P_w} \exp(Z/M), \quad (29.3)$$

with Z from (16.10) and $\zeta = \zeta(R)$. Since $2 \sinh(z/2) \geq z$, the global bound

$$\lambda \leq \frac{M}{g(e^q)} \exp(q/(2M)) \quad (29.4)$$

holds without a small-ratio assumption. For $R \geq 3/4$, the eigenvalue correction, the weight correction, and the corrector mean contribute, respectively,

$$q/(2M), \quad (w-1)q/M + O_{\mathcal{W}}(1/M + Q^2/M^2), \quad c/M \leq q/(2M).$$

Their sum is bounded by wq/M plus the stated errors, also when $w < 1$. Calibration gives

$$\log(M/g) - K/S - \log(n\mu/e) = \Psi(R) + 2/S, \quad -\Delta q l/M \leq 2/i.$$

Consequently

$$\log \frac{\sum_j w^{\mathbf{1}_{\{j=l\}}} F_{i+1}/F_i}{n\mu/e} \leq \Psi(R) + \frac{wq}{SR} + C_{\mathcal{W}}(Q^2/i^2 + 1/i). \quad (29.5)$$

For $0.49 \leq R < 3/4$, use $(1-\zeta)b_l + \zeta c \leq q$ instead. The bound is

$$\Psi(R) + C_{\mathcal{W}}(Q/i + Q^2/i^2 + 1/i) < 0.$$

Indeed $\Psi(R) \leq -\eta_1$ and $Q/i \leq 3/A$: a sufficiently small fixed ratio suffices, rather than a ratio claimed to vanish.

For $\beta = wq/S \leq 1/8$, elementary maximization gives

$$\sup_{R \geq 3/4} \{\Psi(R) + \beta/R\} \leq \beta + 2\beta^2. \quad (29.6)$$

The interior maximum is at $R_+ = (1 + \sqrt{1-4\beta})/2$. The derivative of its value with respect to β is $1/R_+ \leq 1 + 4\beta$; integrate from zero. The endpoint is no larger for this interval of β . Thus (29.5) is at most $wq/S + C_{\mathcal{W}}(Q^2/i^2 + 1/i)$.

29.3 Small uncapped states

If $R < 0.49$, eventually both cutoffs vanish because

$$M/S_i \leq 0.49(1 + 2/i) < 1/2, \quad (M+a)/S \leq 0.49 + 2/(i+1) < 1/2.$$

All successors remain uncapped. The denominator discrepancy is at most q/M . Use only the exact eigenvector, (29.4), and the global weight bound (15.17) to obtain

$$\begin{aligned} \log \frac{\sum_j w^{\mathbf{1}_{\{j=l\}}} F_{i+1}/F_i}{n\mu/e} &\leq \Psi(R) + \frac{3q}{2M} + b \frac{Q}{M} + \frac{6}{i} \\ &\leq -\eta_0 + (3/2 + b)p + b/M + 6/i. \end{aligned} \quad (29.7)$$

Since $M \geq \sqrt{2i}$, the last expression is at most $-\eta_0/2$ eventually. No corrector estimate is used here.

29.4 Every capped state and every cap crossing

For a capped current state, both correctors vanish by (29.2). Freeze instead $f_l^{\text{cap}} = e^{-pl}$. Direct summation gives

$$\frac{(T_q f^{\text{cap}})_l}{f_l^{\text{cap}}} = \sum_{j < l} e^{p(l-j)} + e^q \sum_{j \geq l} e^{-p(j-l)} \leq \frac{e^{pM} + e^q}{1 - e^{-p}}.$$

The weighted row is at most $\max(1, w_+)$ times this expression. Because $pM < q_i \leq q + 2/i$, its numerator is at most $e^q(1 + e^{2/i})$.

The actual next slope is $p_{\text{new}} = \min(q/(M+a), p)$. When the successor stays capped, its ratio to the frozen vector is one. If it becomes uncapped, its log ratio is

$$(p - p_{\text{new}})j \leq p(M+a) - q < q_i - q + p \leq p + 2/i. \quad (29.8)$$

This covers a crossing caused by the change of time, by the increment of M , or by both. At cap equality the expression is continuous and the same bound holds. The remaining temporal factor is $\exp(\Delta q K) \leq 1$. Hence for $i \geq 2$,

$$\sum_j w^{\mathbf{1}_{\{j=l\}}} \frac{F_{i+1}(K+a, j)}{F_i(K, l)} \leq B_{\mathcal{W}, p} e^q. \quad (29.9)$$

By (28.2) and $(i+1) = n\mu\chi(e^q)$, its normalized value is at most

$$\frac{eB_{\mathcal{W}, p}(q+2)}{i+1} \leq \frac{3eB_{\mathcal{W}, p}}{A} \leq \frac{1}{2}.$$

This is a pointwise bound for every capped reachable state. No spectral expansion is made in that range.

All states have now been covered, including both cutoff corners and every cap equality. Combining the bounds yields

$$\sum_j w^{\mathbf{1}_{\{j=l\}}} F_{i+1}(K+a, j) \leq F_i(K, l) \frac{n\mu}{e} \exp \left\{ \frac{wq_{i+1}}{S_{i+1}} + C_{\mathcal{W}} \left(\frac{Q_{i+1}^2}{i^2} + \frac{1}{i} \right) \right\}. \quad (29.10)$$

30 The sharper prefix entropy and upper telescope

A length- k weak prefix with r strict descents has $K = k - 1 - r$ and entries in $\{0, \dots, k-1\}$. Cut it at its descents into $r+1$ nondecreasing runs. Discarding all legality and boundary restrictions, and permitting empty runs, gives the upper bound

$$N_{k,r} \leq [z^k](1-z)^{-k(r+1)} = \binom{k(r+2)-1}{k} \leq [e(r+2)]^k. \quad (30.1)$$

The maximal-run decomposition is unique; the coefficient already sums over run lengths, so no missing independent choice of cut locations occurs.

By (28.3), $q_k \geq ak$ eventually, where $a = 1/(2A)$. The function $1 + \log(r+2) - (a/2)r$ has a finite upper bound C_a on $r \geq 0$. Therefore

$$\sum_{r=0}^{k-1} e^{-q_k r} [e(r+2)]^k \leq e^{kC_a} \sum_{r \geq 0} e^{-kar/2} \leq \frac{e^{kC_a}}{1 - e^{-a/2}}. \quad (30.2)$$

The level weight is at most $\max(1, w_+)^{k-1}$. The last-letter part of the capped potential is nonpositive. A nonzero prefix corrector forces $M > S_k/2 \geq k/4$, so the local bounds apply, and its exponent is $O_{\mathcal{W}}(Q_k^2/k)$. Thus

$$\log \sum_{|x|=k} w^{E(x)} F_k(K, l) \leq (k-1)q_k + O_{\mathcal{W}}(k + Q_k^2/k). \quad (30.3)$$

Unlike a crude $k!$ bound, this estimate retains the strong penalty for many descents.

Positivity and $F_n = 1$ allow (29.10) to telescope backward. The decreasing function q/S satisfies

$$\sum_{i=k}^{n-1} \frac{q_{i+1}}{S_{i+1}} \leq \frac{1}{2} q_k^2, \quad \sum_{i=k}^{n-1} \left(\frac{Q_{i+1}^2}{i^2} + \frac{1}{i} \right) = O(T^2/k + T).$$

Combining with (30.3) gives

$$\begin{aligned} \log H_n(w) &\leq (k-1)q_k + (n-k) \log(n\mu/e) + \frac{w}{2} q_k^2 \\ &\quad + O_{\mathcal{W}}(k + Q_k^2/k + T + T^2/k) \\ &= \log(n!) + n \log \mu + \frac{w}{2} T^2 + O_{\mathcal{W}}(T), \end{aligned} \quad (30.4)$$

using (28.3) and the cancellation (28.4).

31 The matching weighted lower bound

31.1 A positive path law and its exact likelihood

Use the same logarithmic cutoff $k = \lceil AT \rceil$, increasing the fixed A if needed for the local kernel bounds. Begin with the single legal prefix of k zeros. Its state is

[write] 5 October 2026. This section repeats Part II, Section 18, word for word except for the cutoff (here $k = \lceil AT \rceil$ instead of $\lceil 100T^2/U^2 \rceil$), the lower bound $A/3$ in (31.1) and the sentences after it, the name $\tilde{F}_i$ of the uncut potential (there F_i), and the closing paragraph, which here uses the exact normalization (28.4). It is printed in full because Section 32 cites its equations.

$K_k = k-1$, $L_k = 0$, $M_k = k+1$ and its weight is w^{k-1} . At each time $i = k, \dots, n-1$, sample j from P_w , with parameters $q = q_{i+1}$ and $M = M_i$. Denote this path law by $\mathbb{P}_w^*$, with expectation $\mathbb{E}_w^*$. The weak-ascent count never decreases, so every path has

$$M_i \geq k+1, \quad \frac{M_i}{1+q_{i+1}} \geq \frac{k}{1+2T} \geq A/3 \quad (T \geq 1). \quad (31.1)$$

Choose $A/3$ at least as large as every fixed local threshold. All corrector and eigenvalue expansions then apply on every sampled path and on each short interpolation segment, uniformly for $w \in \mathcal{W}$. The ratio need not diverge. The later drift requirement $C_{\mathcal{W}}/M_i \rightarrow 0$ follows separately because $k \rightarrow \infty$. Use the uncut potential

$$\tilde{F}_i(K, l) = \exp\{q_i K - q_i l / (K+2) + h(q_i, K+2, l) / (K+2)\}.$$

For a step let

$$R_i^w = \frac{w^e \tilde{F}_{i+1}(K+a, j)}{\tilde{F}_i(K, l) P_w(l, j)}. \quad (31.2)$$

The product telescopes, and $\tilde{F}_n = 1$, so the total weighted mass of completions of this fixed prefix is exactly

$$w^{k-1} \tilde{F}_k(k-1, 0) \mathbb{E}_w^* \exp \left(\sum_{i=k}^{n-1} \log R_i^w \right).$$

It is a positive sub-sum of $H_n(w)$ even for $w < 1$. Jensen gives

$$\log H_n(w) \geq (k-1) \log w + (k-1)q_k + \sum_{i=k}^{n-1} \mathbb{E}_w^* \log R_i^w. \quad (31.3)$$

There is no $k!$ factor in this single-prefix lower bound.

31.2 Tracking the sampled state

Put $X_i = L_i/M_i$, $Y_i = K_i - X_i$, and $V_i = ((j - L_i) \bmod M_i)/M_i$. A deterministic calculation gives

$$Y_{i+1} - Y_i = 1 - V_i + aX_{i+1}/M_i \in [0, 2]. \quad (31.4)$$

Indeed $j/M_i - X_i = V_i - (1 - a)$, while $j/M_i = X_{i+1}(1 + a/M_i)$. This includes equality steps, for which $V_i = 0$ and $a = 1$. Since the identity part of (15.12) has $V = 0$,

$$\mathbb{E}_{P_w} V = (1 - \delta) \mathbb{E}_P V, \quad |\mathbb{E}_{P_w} V - \mathbb{E}_P V| \leq C_{\mathcal{W}}/M. \quad (31.5)$$

For the second inequality both factors matter: $|\delta| = O_{\mathcal{W}}(Q/M)$ and $\mathbb{E}_P V = O(Q^{-1})$. At $q = 0$, $\mathbb{E}_{P_w} V = (M - 1)/(2(M + w - 1))$, still differing from the unweighted mean by $O_{\mathcal{W}}(1/M)$. Equation (16.6) also gives

$$1 - \mathbb{E}_P V = \alpha(q) + \frac{1 - Q_0(q/M)}{q} = \alpha(q) + O(1/M), \quad (31.6)$$

where the quotient has value $1/(2M)$ at zero. The last term in (31.4) lies in $[0, 1/M_i]$. Boundedness of α' and (15.5) now give

$$\mathbb{E}_w^*[Y_{i+1} - Y_i \mid \mathcal{F}_i] \geq \frac{1}{2} - C_{\mathcal{W}}/M_i \geq \frac{1}{3}, \quad (31.7)$$

$$|\mathbb{E}_w^*[Y_{i+1} - Y_i \mid \mathcal{F}_i] - (S_{i+1} - S_i)| \leq C_{\mathcal{W}}(1/M_i + 1/i). \quad (31.8)$$

The threshold for the second inequality in (31.7) may depend on $\mathcal{W}$. Subtract conditional means in (31.4) to obtain a martingale with increments bounded in absolute value by two. Since $Y_k = k - 1$, the event $M_i < i/8$ forces its displacement below

$$i/8 - (k - 1) - (i - k)/3 = -5i/24 - 2k/3 + 1.$$

Azuma's inequality proves, for absolute $c, C > 0$ after a sufficiently large compact-uniform threshold,

$$\mathbb{P}_w^*(M_i < i/8) \leq Ce^{-ci} \quad (k \leq i < n). \quad (31.9)$$

At $i = k$ the event is empty. No independence is assumed. On good states $M_i \geq i/8$, $1/M_i \leq 8/i$; on bad states use $M_i \geq k$ and (31.9). The L^2 triangle inequality yields

$$\left\| \sum_{i=k}^{m-1} 1/M_i \right\|_2 \leq 8 \sum_{i=k}^{m-1} 1/i + \frac{1}{k} \sum_{i=k}^{m-1} Ce^{-ci/2} \leq CT. \quad (31.10)$$

The martingale variance is $O(m)$, and $|Y_k - S_k| \leq k$. Sum (31.8), use (31.10), and note $M_i - Y_i = 2 + X_i$. We obtain

$$\mathbb{E}_w^*[(M_i - S_{i+1})^2] \leq C_{\mathcal{W}}(i + T^2 + k^2). \quad (31.11)$$

31.3 Conditional one-step gain

Freeze $q = q_{i+1}$, $S = S_{i+1}$, $M = M_i$, $l = L_i$, and $\Delta q = q_{i+1} - q_i$. Substituting the Doob kernel into (31.2) cancels w^e exactly and gives

$$\begin{aligned} \log R_i^w &= \log \lambda_w + \Delta q K_i - \Delta q l/M + \frac{qaj}{M(M+a)} \\ &\quad + \frac{h(q, M+a, j)}{M+a} - \frac{h(q_i, M, l)}{M}. \end{aligned} \quad (31.12)$$

Using (16.5), replacement of the two corrector terms by $[h(q, M, j) - h(q, M, l)]/M$ costs at most

$$C(Q^2/M^2 + Q/(Mi)). \quad (31.13)$$

For the next-state change, differentiating at fixed j includes both H_x and H_M , giving $O(Q^2/M^2 + Q^3/M^3)$; the current-time change costs $O(Q|\Delta q|/M)$. Equation (31.1) justifies the whole interpolation. Also $qaj/[M(M+a)] \geq qaj/M^2 - q/M^2$. Take conditional expectation and use the exact weighted corrector identity (16.11) at $\zeta = 1$, then (16.8). The last three terms of (31.12) are at least

$$q/(2M) - C_W(1/M + Q^2/M^2 + Q/(Mi)). \quad (31.14)$$

The eigenvalue satisfies exactly

$$\lambda = n\mu \frac{M}{S} \frac{q/M}{1 - e^{-q/M}}.$$

The identity $\log[z/(1 - e^{-z})] = z/2 + \log[(z/2)/\sinh(z/2)]$ implies, in the present range,

$$\log \lambda \geq \log(n\mu) + \log(M/S) + q/(2M) - Cq^2/M^2.$$

Together with (15.16),

$$\log \lambda_w \geq \log(n\mu) + \log(M/S) + (w - 1/2)q/M - C_W(1/M + Q^2/M^2). \quad (31.15)$$

A negative intermediate coefficient when $w < 1/2$ causes no inequality reversal. Since $1/S(s)$ changes by $O(1/i^2)$ on $[i, i+1]$ and $K_i \leq i$, calibration gives

$$\Delta q K_i \geq -K_i/S - C/i, \quad -\Delta q l/M \geq 0.$$

Using $K_i = M - 2$ and dropping the favorable $2/S$, combine these inequalities to obtain

$$\begin{aligned} &\mathbb{E}_w^*[\log R_i^w \mid \mathcal{F}_i] - \log(n\mu/e) \\ &\geq \Psi(M_i/S_{i+1}) + \frac{wq_{i+1}}{M_i} - C_W \left(\frac{1}{i} + \frac{1}{M_i} + \frac{Q_{i+1}^2}{M_i^2} + \frac{Q_{i+1}}{M_i i} \right). \end{aligned} \quad (31.16)$$

Thus the net leading gain is exactly wq/M .

31.4 Summation and prefix normalization

On good states, $M_i/S_{i+1} \in [1/16, 2]$; the double zero of Ψ at one gives $\Psi(M_i/S_{i+1}) \geq -C(M_i - S_{i+1})^2/i^2$. On bad states, $M_i \geq k$ and $S_{i+1} \leq n$, so $|\Psi| \leq C + T$. Equations (31.9) and (31.11) imply

$$\sum_{i=k}^{n-1} \mathbb{E}_w^* \Psi(M_i/S_{i+1}) \geq -C_W(T + T^2/k + k) - o(1). \quad (31.17)$$

On good states, $1/M_i \geq 1/S_{i+1} - C|M_i - S_{i+1}|/i^2$. Drop the nonnegative bad-state part of q_{i+1}/M_i , retaining the exponentially small loss of q_{i+1}/S_{i+1} . Cauchy–Schwarz and (31.11), with $q \leq 2T$, yield

$$\sum_{i=k}^{n-1} \mathbb{E}_w^* \frac{q_{i+1}}{M_i} \geq \sum_{i=k}^{n-1} \frac{q_{i+1}}{S_{i+1}} - C_{\mathcal{W}}(T/\sqrt{k} + T^2/k + T) - o(1). \quad (31.18)$$

The explicit errors in (31.16) sum in expectation to $O_{\mathcal{W}}(T + T^2/k)$: on good states use $M_i \geq i/8$, and on bad states use $M_i \geq k$ and (31.9). Every omitted bad-state sum is an exponentially small factor e^{-ck} times a polynomial in T, k . The decreasing function q/S has right Riemann sum differing from its integral by at most $q_k/S_k = O(T/k)$. Thus

$$\sum_{i=k}^{n-1} \frac{q_{i+1}}{S_{i+1}} = \frac{1}{2}q_k^2 + O(T/k). \quad (31.19)$$

Combining (31.16)–(31.19) gives

$$\sum_{i=k}^{n-1} \mathbb{E}_w^* \log R_i^w \geq (n-k) \log(n\mu/e) + \frac{w}{2}q_k^2 - C_{\mathcal{W}}(k + T + T^2/k + T/\sqrt{k}). \quad (31.20)$$

The prefix contributes $(k-1) \log w = O_{\mathcal{W}}(k)$. At $k = \lceil AT \rceil$, every error in (31.20) is $O_{\mathcal{W}}(T)$. Apply (28.3) and (28.4) to (31.3) to obtain

$$\log H_n(w) \geq \log(n!) + n \log \mu + \frac{w}{2}T^2 - O_{\mathcal{W}}(T).$$

Together with (30.4), this proves (24.1). The lower potential is uncut and is distinct from the capped upper potential. All its transitions retain positive sampling probability, including when $w < 1$; the signed identity for P_w has been used only algebraically.

32 Fixed difference ascents and the shifted row

Fix an integer $d \geq 1$ for the remainder of this section. Write $s_d = d - 1$ and

$$l' = \max(l - s_d, 0), \quad m = l - l' = \min(s_d, l).$$

For integer letters, the ascent indicator is precisely $a = \mathbf{1}_{\{j \geq l'\}}$. This shift is $d - 1$, not d . The clipping at zero will matter in the lower bound.

32.1 Exact normalization and corrector cancellation

For $f_l = e^{-ql/M}$ and $T_d(l, j) = e^{q\mathbf{1}_{\{j \geq l'\}}}$, the weak row identity gives exactly

$$(T_d f)_l = \lambda f_{l'} = \lambda e^{mq/M} f_l, \quad P_d(l, j) = \frac{T_d(l, j) f_j}{\lambda e^{mq/M} f_l} = P(l', j). \quad (32.1)$$

This row normalizer depends on l near zero. We do not call f a Perron eigenvector for T_d . The normalized kernel need not be circulant or have a uniform stationary law.

Under $P_d(l, \cdot)$ the forward fraction $V = ((J - l') \bmod M)/M$ has the same truncated geometric law as in Section 16. Apply the weak Poisson identity at l' to obtain

$$\mathbb{E}_{P_d}\{qaJ/M + h_J - h_l\} = c + h_{l'} - h_l. \quad (32.2)$$

The shift is nonwrapping, so the derivative bound implies

$$|h_{l'} - h_l| \leq C_d Q^2 / M \quad (M \geq C_0 Q). \quad (32.3)$$

For $0 \leq \zeta \leq 1$, put $Z_d = qaJ/M + \zeta(h_J - h_l)$. Its exact mean is

$$\mathbb{E}_{P_d} Z_d = (1 - \zeta)b_{l'} + \zeta c + \zeta(h_{l'} - h_l).$$

The circular arc from l' to J , followed by (32.3), gives $|Z_d| \leq C_d(Q + Q^2V + Q^2/M)$. On enlarging a fixed $C_{0,d}$, the geometric moment proof applies on $M \geq C_{0,d}Q$ and gives

$$\mathbb{E}_{P_d}[Z_d^2 e^{|Z_d|/M}] \leq C_d Q^2.$$

The exponential Taylor remainder therefore proves

$$\log \mathbb{E}_{P_d} e^{Z_d/M} \leq \frac{(1 - \zeta)b_{l'} + \zeta c}{M} + C_d Q^2 / M^2. \quad (32.4)$$

All these estimates hold on a sufficiently small fixed interval of q/M , not merely as that ratio tends to zero. At $q = 0$, P_d is uniform and the undivided corrector is exactly zero.

32.2 The fixed difference capped upper bound

Use the same calibration and choose a fixed $p > 0$ such that

$$(d + 1/2)p < -\Psi(0.49)/2.$$

Then choose a sufficiently large fixed $A = A_d$, and $k = \lceil AT \rceil$. Use the capped potential (29.1), with the difference-ascent K . The monotonicity of q_i/M_i and all cutoff comparisons remain exact in this model.

For $R = M/S_{i+1} \geq 0.49$, choose $A > 10/p$ and large enough for $M \geq C_{0,d}(1 + q_i)$. The cap is inactive. The moving-corrector replacements have the same $O_d(Q^2/i^2)$ costs calculated in Section 29. Factoring (32.1), then using (32.4), gives

$$\begin{aligned} \log \frac{\sum_j F_{i+1}(K + a, j)/F_i(K, l)}{n\mu/e} &\leq \Psi(R) + (m + 1/2)q/M \\ &\quad + \frac{(1 - \zeta)b_{l'} + \zeta c}{M} + C_d(Q^2/i^2 + 1/i). \end{aligned} \quad (32.5)$$

For $R \geq 3/4$, $c \leq q/2$ and $m \leq d - 1$ make the coefficient at most d . For $0.49 \leq R < 3/4$, use $b_{l'} \leq q$, giving $d + 1/2$. The negative margin of Ψ absorbs the latter term for sufficiently large fixed A , since $Q/i \leq 3/A$. For $R \geq 3/4$, apply (29.6) with $\beta = dq/S \leq 4d/A \leq 1/8$ to get the bound

$$\frac{dq}{S} + C_d(Q^2/i^2 + 1/i). \quad (32.6)$$

When an uncapped state has $R < 0.49$, both correctors vanish and all its successors remain uncapped. The uncorrected exact row and the denominator discrepancy bounded by q/M give

$$\log \frac{\sum_j F_{i+1}/F_i}{n\mu/e} \leq \Psi(R) + (d + 1/2)q/M + 6/i \leq \Psi(0.49) + (d + 1/2)p + 6/i < 0 \quad (32.7)$$

eventually. This uses no local corrector estimate in a small state.

For a capped current state, take $A > 16/p$, so both endpoint correctors vanish as in (29.2). The exact clipped threshold gives

$$\begin{aligned} \frac{(T_d f^{\text{cap}})_l}{f_l^{\text{cap}}} &= \sum_{j < l'} e^{p(l-j)} + e^q \sum_{j \geq l'} e^{p(l-j)} \\ &\leq \frac{e^{pM} + e^q e^{p(d-1)}}{1 - e^{-p}}. \end{aligned} \quad (32.8)$$

If $l' = 0$ the first sum is empty. The second bound uses $l - l' \leq d - 1$ and remains valid even if $d > M$. The actual next slope costs at most $e^{p+2/i}$ by (29.8). Also $e^{pM} < e^{q_i} \leq e^{q+2/i}$ and $\exp(\Delta q K) \leq 1$. One may therefore take

$$B_{d,p} = \frac{e^{p+2}(e^2 + e^{p(d-1)})}{1 - e^{-p}}$$

to obtain a row bound $B_{d,p}e^q$. By (28.2), the normalized row is at most $4eB_{d,p}/A$ for $T \geq 1$. Choose $A \geq 8eB_{d,p}$, making it at most $1/2$. The bound covers all capped successors and all capped-to-uncapped transitions pointwise.

Select p after the local constants and A after p , the cap constant, and all fixed intermediate-state margins. Also enlarge A to satisfy the lower-chain thresholds below. These are finitely many compatible conditions depending only on d . The combined result, for every reachable state and every $i \geq k$, is

$$\sum_j F_{i+1}(K+a, j) \leq F_i(K, l) \frac{n\mu}{e} \exp \left\{ \frac{dq_{i+1}}{S_{i+1}} + C_d \left(\frac{Q_{i+1}^2}{i^2} + \frac{1}{i} \right) \right\}. \quad (32.9)$$

32.3 The shifted prefix encoding

If a length- k difference-ascent word has r non- d -ascents, then $K = k - 1 - r$. Define

$$y_i = x_i + (d-1)i \quad (1 \leq i \leq k).$$

This transformation is injective, $0 \leq y_i \leq dk - 1$, and

$$y_{i+1} \geq y_i \iff x_{i+1} > x_i - d.$$

The r non-ascents are exactly the strict descents of y . Split at those descents, discard the inter-run restrictions, and permit empty runs. Over an alphabet of size dk this proves

$$N_{k,r}^{(d)} \leq \binom{dk(r+1) + k - 1}{k} \leq [e(d(r+1) + 1)]^k \leq [e(d+1)(r+2)]^k. \quad (32.10)$$

The canonical ordered block tuple contains the boundary locations, so again no additional binomial factor is missing.

With $q_k \geq k/(2A)$, the geometric-penalty argument (30.2) gives $\sum_r N_{k,r}^{(d)} e^{-q_k r} = \exp(O_d(k))$. The prefix slope is nonpositive; a nonzero corrector has $M > k/4$ and costs $O_d(Q_k^2/k)$. Thus

$$\sum_{|x|=k} F_k(K, l) \leq \exp\{(k-1)q_k + O_d(T)\}.$$

Telescope (32.9) and apply (28.3)–(28.4). The accumulated local error is $O_d(T^2/k+T) = O_d(T)$, proving

$$\log a_n^{(d)} \leq \log(n!) + n \log \mu + \frac{d}{2} T^2 + O_d(T). \quad (32.11)$$

32.4 A positive lower path law and its tracker

Start from the legal prefix of k zeros. Its state is $(K_k, L_k) = (k-1, 0)$, with $M_k = k+1$. At time i , sample j under P_d frozen at $q = q_{i+1}$ and $M = M_i$, then make the true update. Every legal continuation has positive probability. Since $M_i \geq k+1$, sufficiently large fixed A guarantees all local hypotheses along every path and every interpolation segment.

Use the uncut lower potential

$$\tilde{F}_i(K, l) = \exp\{q_i K - q_i l/M + h(q_i, M, l)/M\}, \quad M = K + 2,$$

and the likelihood ratio

$$R_i^{(d)} = \frac{\tilde{F}_{i+1}(K+a, j)}{\tilde{F}_i(K, l)P_d(l, j)}.$$

The exact pathwise telescope and Jensen give

$$\log a_n^{(d)} \geq (k-1)q_k + \sum_{i=k}^{n-1} \mathbb{E}_d^* \log R_i^{(d)}. \quad (32.12)$$

There is no factorial prefix multiplier.

Put $X_i = L_i/M_i$ and $Y_i = K_i - X_i$. With $V = ((j-l') \bmod M_i)/M_i$, direct calculation gives

$$Y_{i+1} - Y_i = 1 - V + aX_{i+1}/M_i + m/M_i \in [0, 2]. \quad (32.13)$$

Indeed $j/M_i - l'/M_i = V - (1-a)$ and $j/M_i = X_{i+1}(1+a/M_i)$. The shift term is positive. Nonincreasing geometric weights give $\mathbb{E}V \leq 1/2$, so the conditional drift is at least $1/2$. Subtracting conditional means yields martingale increments bounded by two. From $Y_k = k-1$, Azuma's inequality proves

$$\mathbb{P}_d^*(M_i < i/4) \leq Ce^{-ci}. \quad (32.14)$$

For $i = k$ the event is empty. For $i > k$, that event forces a negative martingale displacement of at least $i/4 + k/2 - 1$ in magnitude. Consequently, for every fixed $u > 0$,

$$\mathbb{E}_d^* M_i^{-u} \leq (4/i)^u + C(k+1)^{-u} e^{-ci} \leq C_u i^{-u}. \quad (32.15)$$

The exact geometric mean from (16.6) gives

$$1 - \mathbb{E}V = \alpha(q) + \frac{1 - Q_0(q/M_i)}{q} = \alpha(q) + O(1/M_i).$$

The quotient at zero is $1/(2M_i)$; boundedness of $[1 - Q_0(z)]/z$ on a fixed small interval proves uniformity. Add the last two terms of (32.13) and the one-step calibration change to obtain

$$|\mathbb{E}_d^*[Y_{i+1} - Y_i \mid \mathcal{F}_i] - (S_{i+1} - S_i)| \leq C_d(1/M_i + 1/i).$$

By (32.15), $\|\sum_{j=k}^{i-1} 1/M_j\|_2 \leq C \sum_{j=k}^{i-1} 1/j \leq CT$. The martingale variance is $O(i)$, the initial mismatch is $O(k)$, and $M_i - Y_i = 2 + X_i$ is bounded. Hence

$$\mathbb{E}_d^*[(M_i - S_{i+1})^2] \leq C_d(i + T^2 + k^2). \quad (32.16)$$

These are statements about the specified auxiliary path law, with no claim of stationarity of the actual last letter.

32.5 The exact lower gain and the clipped shift

Substituting (32.1) into the likelihood ratio gives exactly

$$\begin{aligned} \log R_i^{(d)} &= \log \lambda + \Delta q K_i - \Delta q l/M + mq/M + \frac{qaj}{M(M+a)} \\ &\quad + \frac{h(q, M+a, j)}{M+a} - \frac{h(q_i, M, l)}{M}. \end{aligned} \quad (32.17)$$

The state-dependent mq/M has positive sign and cannot be omitted. Replacing the corrector difference by $(h_J - h_l)/M$ costs $O(Q^2/M^2 + Q/(Mi))$, as calculated explicitly for the weighted case. Also $qaj/[M(M+a)] \geq qaj/M^2 - q/M^2$. Use (32.2)–(32.3) and $c \geq q/2 - C(1 + q/M)$.

The same local eigenvalue inequality gives

$$\log \lambda \geq \log(n\mu) + \log(M/S) + q/(2M) - Cq^2/M^2,$$

while $\Delta q K_i \geq -K_i/S - C/i$ and $-\Delta q l/M \geq 0$. Combining these identities yields

$$\begin{aligned} \mathbb{E}_d^*[\log R_i^{(d)} \mid \mathcal{F}_i] - \log(n\mu/e) &\geq \Psi(M_i/S_{i+1}) + \frac{dq_{i+1}}{M_i} - \frac{(s_d - m)q_{i+1}}{M_i} \\ &\quad - C_d \left(\frac{1}{i} + \frac{1}{M_i} + \frac{Q_{i+1}^2}{M_i^2} + \frac{Q_{i+1}}{M_i i} \right). \end{aligned} \quad (32.18)$$

The sole deficit in the full-shift gain is $(s_d - m)q/M$, supported on $l < s_d$.

32.6 Summing the lower boundary deficit

For every $q \geq 0$ the largest row probability of the geometric law is bounded by

$$\frac{1 - e^{-q/M}}{1 - e^{-q}} \leq \frac{1 + q}{M}, \quad (32.19)$$

with value $1/M$ at zero. Use $1 - e^{-q/M} \leq q/M$ and $q/(1 - e^{-q}) \leq q + 1$. Let $B_i = \mathbf{1}_{\{L_i < s_d\}}$. If $s_d = 0$ there is no deficit. Otherwise $k > s_d$ eventually, and conditioning on the entire previous history gives, for $i \geq k + 1$,

$$\mathbb{P}_d^*(L_i < s_d \mid \mathcal{F}_{i-1}) \leq s_d(1 + q_i)/M_{i-1}.$$

The event and the denominator are not independent. Instead use the exact monotonicity $M_i \geq M_{i-1}$ *before* taking expectation:

$$\mathbb{E}_d^* \left[\frac{q_{i+1}}{M_i} B_i \mid \mathcal{F}_{i-1} \right] \leq s_d q_{i+1} (1 + q_i) M_{i-1}^{-2}. \quad (32.20)$$

By (32.15), $q_i \leq 2T$, and summation of i^{-2} , the total for $i \geq k + 1$ is $O_d(T^2/k)$. The initial state is at the boundary when $d > 1$ and contributes explicitly

$$(s_d - m_k)q_{k+1}/M_k = s_d q_{k+1}/(k + 1) = O_d(T/k).$$

Therefore

$$\sum_{i=k}^{n-1} \mathbb{E}_d^* \frac{(s_d - m_i)q_{i+1}}{M_i} = O_d(T^2/k). \quad (32.21)$$

This proof requires neither stationarity nor independence at the lower boundary.

32.7 Finishing the fixed difference lower bound

On good states $M_i \geq i/4$, the ratio M_i/S_{i+1} lies in $[1/8, 2]$, where $\Psi(M_i/S_{i+1}) \geq -C(M_i - S_{i+1})^2/i^2$. On the complementary event use $M_i \geq k+1$ and $|\Psi| \leq C+T$. Equations (32.14) and (32.16) give

$$\sum_i \mathbb{E}_d^* \Psi(M_i/S_{i+1}) \geq -C_d(T + T^2/k + k) - o(1).$$

On good states $1/M_i \geq 1/S_{i+1} - C|M_i - S_{i+1}|/i^2$. Drop the nonnegative bad-state contribution and retain the exponentially small loss of q_{i+1}/S_{i+1} . Cauchy-Schwarz then yields

$$\sum_i \mathbb{E}_d^* \frac{q_{i+1}}{M_i} \geq \sum_i \frac{q_{i+1}}{S_{i+1}} - C_d(T/\sqrt{k} + T^2/k + T) - o(1).$$

The explicit local errors in (32.18) sum to $O_d(T + T^2/k)$ by (32.15). Bad-state sums contain e^{-ck} times a polynomial in T, k , with no extra factor n . The right Riemann sum is $q_k^2/2 + O(T/k)$. Combining with (32.21),

$$\begin{aligned} \sum_i \mathbb{E}_d^* \log R_i^{(d)} &\geq (n-k) \log(n\mu/e) + \frac{d}{2} q_k^2 \\ &\quad - C_d(k + T + T^2/k + T/\sqrt{k}). \end{aligned} \quad (32.22)$$

At $k = \lceil AT \rceil$ the error is $O_d(T)$. Insert this in (32.12), then use the sharp calibration and exact normalization from Section 28. This proves

$$\log a_n^{(d)} \geq \log(n!) + n \log \mu + \frac{d}{2} T^2 - O_d(T).$$

Together with (32.11), it completes Theorem 24.1.

33 Bounded error threshold inversion

Let B_n be one of the two positive increasing quantities in Corollary 24.2, with correction coefficient c . Define the smooth functions

$$f(t) = t \log(\mu t/e), \quad G_c(t) = f(t) + \frac{c}{2} (\log t)^2.$$

Stirling and Theorem 24.1 imply

$$\log B_n = G_c(n) + O(\log n). \quad (33.1)$$

The half-logarithmic Stirling term is included in this error. All constants are compact-uniform in the weighted case.

Write $L = \log Y$ and $x = L/W_0(\mu L/e)$. If $u = W_0(\mu L/e)$, then $ue^u = \mu L/e$, so $\mu x/e = e^u$ and $f(x) = L$ exactly. The leading growth gives $N(Y) \sim x$: evaluate (33.1) near $(1 \pm \varepsilon)x$, where the main-term gap is of order $\varepsilon x \log x$, dominating all log-square and logarithmic errors. Monotonicity then brackets the first crossing, uniformly over any fixed weight interval.

At $N = N(Y)$, the neighboring integers satisfy

$$\log B_{N-1} < L \leq \log B_N.$$

Using (33.1) at both indices and $G_c(N) - G_c(N-1) = O(\log N)$ gives

$$G_c(N) - L = O(\log N). \quad (33.2)$$

This uses neither regularity nor monotonicity of the unknown remainder.

Put

$$y = x - \frac{c(\log x)^2}{2\log(\mu x)}.$$

Then $y - x = O(\log x)$, $f'(x) = \log(\mu x)$, and $f''(t) = 1/t$. Taylor expansion of the known smooth terms alone gives

$$f(y) = L - \frac{c}{2}(\log x)^2 + O((\log x)^2/x), \quad (\log y)^2 - (\log x)^2 = O((\log x)^2/x).$$

Therefore $G_c(y) - L = O((\log x)^2/x)$. Both N and y are asymptotic to x , and along their joining segment

$$G'_c(t) = \log(\mu t) + c\log(t)/t \sim \log x.$$

The mean-value theorem applied to (33.2) proves $N - y = O(1)$. Finally

$$\frac{(\log x)^2}{\log(\mu x)} = \log x - \log \mu + O(1/\log x),$$

which proves both forms of (24.3).

What the bounded error means. The correction is negative: the log-square enhancement makes the threshold occur earlier than the leading factorial inverse predicts. The integer rounding uncertainty is already included in $O(1)$. The proof supplies an eventually bounded distance, with unspecified constants, and does not supply a limiting constant or a computable finite bracket. A next coefficient of order $\log n$ in the counting logarithm would affect the inverse at constant order.

Remark 33.1 ([write], 5 October 2026; the endpoint $d = 0$). The case $d = 0$, ordinary ascent sequences counted by the Fishburn numbers $a_n^{(0)} = F_n^{\text{Fish}}$, is excluded from the proof of Theorem 24.1, but both of its conclusions hold there, with coefficient $c = d = 0$:

$$\log a_n^{(0)} = \log(n!) + n \log \mu + O(\log n), \quad N(Y) = x + O(1).$$

Proof. By the Fishburn asymptotic (5.2) (Hwang–Jin [7]), $\log a_n^{(0)} = \log(n!) + n \log \mu + \frac{1}{2} \log n + \log C_F + O(1/n)$, which is (24.2) at $d = 0$. For the inverse, every state has exactly $K + 2 \geq 2$ children for every $d \geq 0$ (Part I, Section 9), so $a_{i+1}^{(0)} \geq 2a_i^{(0)}$ and the first crossing is well defined. The proof of Corollary 24.2 in this section uses only (33.1), which now holds with $c = 0$, and this monotonicity; with $c = 0$, $G_0 = f$, $y = x$, and the mean-value step gives $N - x = O(1)$. $\square$

This is a deduction from a known theorem, not a new result on Fishburn numbers. It shows that the dichotomy between $d = 0$ and $d \geq 1$ is in the coefficient of $(\log n)^2$, and not in the form of the statements.

34 Sharper moment bounds for the number of levels

Sample a weak ascent sequence with probability $w^{E(x)}/H_n(w)$, and let E_n count its levels. Put $T = \log n$ and $m_w = (w/2)T^2$.

Corollary 34.1 (Quantitative level moments). *Uniformly for w in a fixed compact $\mathcal{W} \subset (0, \infty)$,*

$$\mathbb{E}_{n,w} E_n = m_w + O_{\mathcal{W}}(T^{3/2}), \tag{34.1}$$

$$\mathbb{E}_{n,w} |E_n - m_w|^p = O_{\mathcal{W},p}(T^{3p/2}) \quad (p > 0 \text{ fixed}), \tag{34.2}$$

$$\text{Var}_{n,w}(E_n) = O_{\mathcal{W}}(T^3). \tag{34.3}$$

The same absolute moment bound holds when centered at the actual mean. These are upper bounds, not fluctuation-scale equivalents or a central limit theorem.

[write] **5 October 2026.** This supersedes the mean estimate (13.4) of Part II, whose error was $O(b_n/\sqrt{\log \log n})$, and its remark that the variance is $o((\log n)^4)$.

Proof. Fix $\delta > 0$ and apply the compact-uniform theorem on $\mathcal{W}^* = [w_-e^{-\delta}, w_+e^{\delta}]$. The exact identity

$$\mathbb{E}_{n,w} e^{tE_n} = H_n(we^t)/H_n(w)$$

gives, for $|t| \leq \delta$ and $Z = E_n - m_w$,

$$\log \mathbb{E}_{n,w} e^{tZ} = \frac{w}{2} T^2 (e^t - 1 - t) + O_{\mathcal{W}^*}(T) \leq C_0 T + C_1 T^2 t^2. \quad (34.4)$$

To bound the mean without differentiating that error, use convexity of the exact finite log-MGF. Its derivative at zero lies between its secant slopes at $\pm h$. With $h = T^{-1/2}$, expansion of the explicit leading term gives $m_w + O_{\mathcal{W}}(T^2 h + T/h) = m_w + O_{\mathcal{W}}(T^{3/2})$, proving (34.1).

Chernoff's inequality for both signs of t gives, with $b = 2C_1\delta$,

$$\mathbb{P}_{n,w}(|Z| \geq u) \leq \min\{1, 2 \exp(C_0 T - u^2/(4C_1 T^2))\}, \quad 0 \leq u \leq bT^2, \quad (34.5)$$

$$\mathbb{P}_{n,w}(|Z| \geq u) \leq \min\{1, 2 \exp(C_0 T - \delta u/2)\}, \quad u \geq bT^2. \quad (34.6)$$

For the first estimate take $t = u/(2C_1 T^2)$; once it reaches δ , use $t = \delta$. Choose a fixed $B > 0$ with $B^2 \geq 8C_0 C_1$, and split

$$\mathbb{E}|Z|^p = p \int_0^\infty u^{p-1} \mathbb{P}(|Z| > u) du$$

at $BT^{3/2}$ and bT^2 . The initial interval costs $O(T^{3p/2})$. On the middle interval (34.5) is at most $2 \exp(-u^2/(8C_1 T^2))$, whose integral over all positive u is $O_p(T^p)$. On the last interval (34.6) is at most $2 \exp(-\delta u/4)$ for large n , and its integral is exponentially small in T^2 times a polynomial. This proves (34.2). For $p = 2$, variance is at most $\mathbb{E}(E_n - m_w)^2$. Centering at the actual mean follows from (34.1) and $|a + b|^p \leq C_p(|a|^p + |b|^p)$. $\square$

In particular $E_n/T^2 \rightarrow w/2$ in every fixed positive absolute moment, and the mean error is improved by the sharp remainder. The variance upper bound does not assert that the variance has order T^3 . The fixed-tilt log-MGF still has leading term $(w/2)T^2(e^t - 1)$, but this fact alone does not prove Poisson approximation, mod-Poisson convergence, a central or local limit theorem, or sharp tail prefactors.

35 Exact benchmarks and primary literature

Table 1 records selected exact values from (25.2). They agree with Dukes–Sagan's unrestricted Table 3 [6], and its $d = 1$ column agrees with OEIS A336070 [10]. The $d = 0$ column is provided to check indexing, not as an application of the present positive- d proof.

[write] **5 October 2026.** The $d = 1$ column is A336070 and agrees with the 501 terms of Part I (data/105-growth-weak_ascent_terms.txt); the $d = 0$ column is the Fishburn numbers (OEIS A022493). The whole table was recomputed independently at placement, and every entry agrees.

Bényi–Claesson–Dukes [2] develop weak ascent sequences and related bijections. Mansour [9] treats the level statistic through exact multivariate generating functions. These enumerative results are context for the weighted family; no probability or logarithmic-remainder conclusion in this report is deduced from a numerical coefficient fit.

Table 1: Exact unrestricted difference-ascent counts

n	$d = 0$	$d = 1$	$d = 2$	$d = 3$
0	1	1	1	1
1	1	1	1	1
2	2	2	2	2
3	5	6	6	6
4	15	23	24	24
5	53	106	118	120
6	217	567	682	714
7	1014	3440	4506	4896
8	5335	23286	33376	37854
9	31240	173704	273200	324792
10	201608	1414102	2444274	3055320

[write] 5 October 2026. The Mansour citation was not checked against the publication at placement or at this write.

Dukes–Sagan [6] define difference ascents by precisely the convention used here. Their Problem 8.7 asks for the unrestricted generating function for general d . Their Section 6 recursion instead concerns the d -increasing subclass, whose $d = 2$ count at $n = 4$ is 22, not the unrestricted 24. Difference ascents also differ from the separately named p -ascent model that changes the allowance above the ascent count.

Zang–Zhou [11] give related bijections. Cerbai–Claesson–Sagan [3, 4] study a modified encoding and its self-modified subclass. Enumeration formulas for self-modified or pattern-avoiding restrictions must not be silently substituted for unrestricted $a_n^{(d)}$. The ordinary Fishburn family has a separate asymptotic literature, including Hwang–Jin [7].

The bounded primary-source review supporting this report, through 2 October 2026, did not locate an unrestricted asymptotic theorem of the two sharpened forms in Theorem 24.1. This is not an exhaustive priority certificate. Exact formulas and bijections remain attributed to their sources. No OEIS identifier for unrestricted $d = 2$ or $d = 3$ was verified; failed searches are not evidence that no entry exists.

36 Reproducibility and the boundary of finite checks

The package includes this complete TeX article, its final PDF, deterministic build scripts, exact fixtures and checkers, deliberate corruption tests, an integrity manifest, and validation records. The article requires no earlier report to follow its proof or to rebuild its PDF. Source-provenance hashes identify the checked input versions without making those earlier packages build dependencies.

The exact algebra chooses rational $r \in (0, 1)$, rational $w > 0$, and integer M , setting $v = r^{-M}$. The real parameter $q = -M \log r$ is kept symbolic; it is not declared rational. The finite corrector identities are checked using the rational values b/q , c/q , and h/q from (16.1)–(16.3). The $q = 0$ endpoint is checked separately using its actual limiting kernel and vanishing undivided corrector. The weighted signed identity at $w < 1$ and the shifted row normalizer are checked against their positive entries.

Exact state recurrences are compared with independent inversion-sequence enumeration, including the shift convention and parameters larger than the state alphabet. Prefix encodings are checked on finite words, and rational geometric capped-row inequalities and crossing algebra are exercised without confusing those finite instances with a proof for all real parameters. The

analytic proof of uniformity, the cap, the prefix cancellation, and the inverse is in the body of the article.

The guards use explicit exceptions rather than removable Python assertions. The validation campaign runs under both ordinary Python and `python -O`, and requires deliberate fixture corruptions to be rejected. The final archive is extracted into a fresh directory, its manifest is verified, and the checks and deterministic PDF build are replayed there. The regenerated PDF must agree byte for byte with the packaged final PDF in the documented environment. Every final page is rendered and visually checked for layout problems. These safeguards test implementation, formulas, signs, and packaging; they are not formal proof-assistant verification of the analytic theorem.

37 Further questions justified by the result

The logarithmic remainder isolates several concrete next problems without resolving them.

1. Determine whether the residual $\log B_n - \log(n!) - n \log \mu - (c/2) \log^2 n$ has a coefficient times $\log n$ plus a bounded remainder. The present bounds allow this possibility but do not imply it.
2. Identify an amplitude and a relative-error equivalent, if such a regular expansion exists. This requires sharper prefix matching and more detailed control of the evolving state, beyond an $O(\log n)$ loss.
3. Find effective constants and starting thresholds for a certified finite inverse bracket. Bounded-error asymptotics alone cannot decide an exact integer crossing.
4. For levels, determine the variance scale and establish, if valid, a central or local limit theorem. The moment bounds here are too weak to select that scale.
5. Understand transitions when w tends to zero or d grows with n . The constants in the present proof depend on a fixed positive compact weight interval or fixed d , so no such transition follows by substitution.

A full transseries would require substantially more information than the logarithmic pressure expansion proved here. The present result supplies a rigorous starting point for these refinements, rather than evidence that any particular one must hold.

37.1 Further questions and research

[write] (5 October 2026.) Questions 1–5 above are this Part’s own list and stay as printed. Under Vladimir’s standing rule of 4 October 2026 the write adds the remaining open statements of this Part. Nothing in this Part was found to be wrong.

6. *The unrestricted generating function for general d* (Dukes–Sagan, Problem 8.7 [6]; Section 35). Open. Part I’s catalytic equation (2.5) (Auli–Elizalde) covers $d = 1$, and the Fishburn series $d = 0$; the recurrence (25.2) is all this Part provides for $d \geq 2$.
7. *OEIS identification* of the unrestricted $d = 2$ and $d = 3$ columns of Table 1. This Part verified none and notes that failed searches are not evidence of absence; not searched again at the write.
8. *Why the coefficients $w/2$ and $d/2$ coincide*. Section 24 says the coincidence “reflects related transfer calculations” (in the proofs, the shift $l' = l - (d - 1)$ contributes mq/M with $m \leq d - 1$, as the level weight contributes $(w - 1)q/M$), and that the families differ ($H_3(2) = 12$, $a_3^{(2)} = 6$). A structural or combinatorial explanation is not given.

9. Real $d > 0$. This Part's remark that the effective parameter is $\lceil d \rceil$ is correct: for integers, $y > -d$ if and only if $y \geq 1 - \lceil d \rceil$, that is, $y > -\lceil d \rceil$ (checked at the write). No question remains there; uniformity in d is item 5 above.

References

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- [9] T. Mansour, *Statistics in Weak Ascent Sequences*, Mathematics **14**(8) (2026), article 1378. doi:10.3390/math14081378. Section 2.1 and Theorem 1 concern levels and their exact generating function (Parts II and III). [Cited by all three Parts. **[write]** Not checked against the publication at placement or at this write (5 October 2026): its existence, its bibliographic data and the content attributed to it are unverified here.]
- [10] OEIS Foundation Inc., *A336070*, weak ascent sequences (number of weak ascent sequences of length n ; also a vincular-pattern inversion-sequence count).

<https://oeis.org/A336070>. Source check: 2 October 2026 (Part I); accessed 2 October 2026 (Part III). OEIS data are available under CC BY-SA 4.0.

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